Question #1 (Topic: Exam A)*Topic 1*

A retail company has a generative AI (GenAI) product recommendation application that uses Amazon Bedrock. The application suggests products to customers based on browsing history and demographics. The company needs to implement fairness evaluation across multiple demographic groups to detect and measure bias in recommendations between two prompt approaches. The company wants to collect and monitor fairness metrics in real time. The company must receive an alert if the fairness metrics show a discrepancy of more than 15% between demographic groups. The company must receive weekly reports that compare the performance of the two prompt approaches.
Which solution will meet these requirements with the LEAST custom development effort?

A. Configure an Amazon CloudWatch dashboard to display default metrics from Amazon Bedrock API calls. Create custom metrics based on model outputs. Set up Amazon EventBridge rules to invoke AWS lambda functions that perform post-processing analysis on model responses and publish custom fairness metrics.
B. Create the two prompt variants in Amazon Bedrock Prompt Management. Use Amazon Bedrock Flows to deploy the prompt variants with defined traffic allocation. Configure Amazon Bedrock guardrails that have content filters to monitor demographic fairness. Set up Amazon CloudWatch alarms on the GuardrailContentSource dimension that use InvocationsIntervened metrics to detect recommendation discrepancy threshold violations.
C. Set up Amazon SageMaker Clarify to analyze model outputs. Publish fairness metrics to Amazon CloudWatch. Create CloudWatch composite alarms that combine SageMaker Clarify bias metrics with Amazon Bedrock latency metrics to provide a comprehensive fairness evaluation dashboard.
D. Create an Amazon Bedrock model evaluation job to compare fairness between the two prompt variants. Enable model invocation logging in Amazon CloudWatch. Set up CloudWatch alarms for InvocationsIntervened metrics with a dimension for each demographic group.

Correct Answer: C

Aspect	Option B (My Original Pick)	Option C (Correct)

Fairness measurement	❌ Guardrails detect *harmful content*, not *bias/fairness across demographics* — this is a proxy at best	✅ SageMaker Clarify is **purpose-built** for bias detection with actual statistical fairness metrics (DPL, KL divergence, demographic parity)
15% discrepancy detection	❌ InvocationsIntervened measures guardrail triggers, not recommendation disparity between demographic groups	✅ Clarify computes actual fairness metrics that can directly measure % discrepancy between demographic groups
Real-time monitoring	✅ Guardrails are inline	✅ Publishing Clarify metrics to CloudWatch enables continuous monitoring and alerting
Comprehensive evaluation	❌ Content filtering is one-dimensional	✅ Composite alarms combine bias metrics + latency (if the system is systematically slower for certain groups, that's also a fairness concern)

Question #2 (Topic: Exam A)*Topic 1*

A finance company is developing an AI assistant to help clients plan investments and manage their portfolios. The company identifies several high-risk conversation patterns such as requests for specific stock recommendations or guaranteed returns. High-risk conversation patterns could lead to regulatory violations if the company cannot implement appropriate controls.

The company must ensure that the AI assistant does not provide inappropriate financial advice, generate content about competitors, or make claims that are not factually grounded in the company's approved financial guidance. The company wants to use Amazon Bedrock Guardrails to implement a solution.

Which combination of steps will meet these requirements? (Choose three.)

A. Add the high-risk conversation patterns to a denied topics guardrail.
B. Configure a content filter guardrail to filter prompts that contain the high-risk conversation patterns.
C. Configure a content filter guardrail to filter prompts that contain competitor names.
D. Add the names of competitors as custom word filters. Set the input and output actions to block.
E. Set a low grounding score threshold.

F. Set a high grounding score threshold.

Correct Answer: ADF

Question #3 (Topic: Exam A)*Topic 1*

A company has deployed an AI assistant as a React application that uses AWS Amplify, an AWS AppSync GraphQL API, and Amazon Bedrock Knowledge Bases. The application uses the GraphQL API to call the Amazon Bedrock RetrieveAndGenerate API for knowledge base interactions. The company configures an AWS Lambda resolver to use the RequestResponse invocation type. Application users report frequent timeouts and slow response times. Users report these problems more frequently for complex questions that require longer processing.
The company needs a solution to fix these performance issues and enhance the user experience. Which solution will meet these requirements?

A. Use AWS Amplify AI Kit to implement streaming responses from the GraphQL API and to optimize client-side rendering.
B. Increase the timeout value of the Lambda resolver. Implement retry logic with exponential backoff.
C. Update the application to send an API request to an Amazon SQS queue. Update the AWS AppSync resolver to poll and process the queue.
D. Change the RetrieveAndGenerate API to the InvokeModelWithResponseStream API. Update the application to use an Amazon API Gateway WebSocket API to support the streaming response.

Correct Answer: A

Question #4 (Topic: Exam A)*Topic 1*

An ecommerce company operates a global product recommendation system that needs to switch between multiple foundation models (FM) in Amazon Bedrock based on regulations, cost optimization, and performance requirements. The company must apply custom controls based on proprietary business logic, including dynamic cost thresholds, AWS Region-specific compliance rules, and real-time A/B testing across multiple FMs. The system must be able to switch between FMs without deploying new code. The system must route user requests based on complex rules including user tier, transaction value, regulatory zone, and real-time cost metrics that change

hourly and require immediate propagation across thousands of concurrent requests.
Which solution will meet these requirements?

A. Deploy an AWS Lambda function that uses environment variables to store routing rules and Amazon Bedrock FM IDs. Use the Lambda console to update the environment variables when business requirements change. Configure an Amazon API Gateway REST API to read request parameters to make routing decisions.
B. Deploy Amazon API Gateway REST API request transformation templates to implement routing logic based on request attributes. Store Amazon Bedrock FM endpoints as REST API stage variables. Update the variables when the system switches between models.
C. Configure an AWS Lambda function to fetch routing configurations from the AWS AppConfig Agent for each user request. Run business logic in the Lambda function to select the appropriate FM for each request. Expose the FM through a single Amazon API Gateway REST API endpoint.
D. Use AWS Lambda authorizers for an Amazon API Gateway REST API to evaluate routing rules that are stored in AWS AppConfig. Return authorization contexts based on business logic. Route requests to model-specific Lambda functions for each Amazon Bedrock FM.

Correct Answer: C

Question #5 (Topic: Exam A)*Topic 1*

A company is developing an internal generative AI (GenAI) assistant that uses Amazon Bedrock to summarize corporate documents for multiple business units. The GenAI assistant must generate responses in a consistent format that includes a document summary, classification of business risks, and terms that are flagged for review. The GenAI assistant must adapt the tone of responses for each user's business unit, such as legal, human resources, or finance. The GenAI assistant must block hate speech, inappropriate topics, and sensitive information such as personal health information.
The company needs a solution to centrally manage prompt variants across business units and teams. The company wants to minimize ongoing orchestration efforts and maintenance for post-processing logic. The company also wants to have the ability to adjust content moderation criteria for the GenAI assistant over time.
Which solution will meet these requirements with the LEAST maintenance overhead?

A. Use Amazon Bedrock Prompt Management to configure reusable templates and business unit-specific prompt variants. Apply Amazon Bedrock guardrails that have category filters and sensitive term lists to block prohibited content.

B. Use Amazon Bedrock Prompt Management to define base templates. Enforce business unit-specific tone by using system prompt variables. Configure Amazon Bedrock guardrails to apply audience-based threshold tuning. Manage the guardrails by using an internal administration API.
C. Use Amazon Bedrock with business unit-based instruction injection in API calls. Store response formatting rules in Amazon DynamoDB. Use AWS Step functions to validate responses. Use Amazon Comprehend to apply content filters after the GenAI assistant generates responses.
D. Use Amazon Bedrock with custom prompt templates that are stored in Amazon DynamoDB. Create one AWS Lambda function to select business unit-specific prompts. Create a second Lambda function to call Amazon Comprehend to filter prohibited content from responses.

Correct Answer: A

Question #6 (Topic: Exam A)*Topic 1*

A financial services company is building a customer support application that retrieves relevant financial regulation documents from a database based on semantic similarities to user queries. The application must integrate with Amazon Bedrock to generate responses. The application must be able to search documents that are in English, Spanish, and Portuguese. The application must filter documents by metadata such as publication date, regulatory agency, and document type. The database stores approximately 10 million document embeddings. To minimize operational overhead, the company wants a solution that minimizes management and maintenance effort. The application must provide low-latency responses for real-time customer interactions.
Which solution will meet these requirements?

A. Use Amazon OpenSearch Serverless to provide vector search capabilities and metadata filtering. Connect to Amazon Bedrock Knowledge Bases to enable Retrieval Augmented Generation (RAG) capabilities that use an Anthropic Claude foundation model (FM).
B. Deploy an Amazon Aurora PostgreSQL database with the pgvector extension. Define tables to store embeddings and metadata. Use SQL queries to perform similarity searches. Send retrieved documents to Amazon Bedrock to generate responses.
C. Use Amazon S3 Vectors to configure a vector index and non-filterable metadata fields. Integrate S3 Vectors with Amazon Bedrock to enable Retrieval Augmented Generation (RAG) capabilities.
D. Set up an Amazon Neptune Analytics graph database. Configure a vector index that has appropriate dimensionality to store document embeddings. Use Amazon Bedrock to perform graph-based retrieval and to generate responses.

Correct Answer: A

Question #7 (Topic: Exam A)*Topic 1*

A medical company is building a generative AI (GenAI) application that uses RAG to provide evidence-based medical information. The application uses Amazon OpenSearch Service to retrieve vector embeddings. Users report that searches frequently miss results that contain exact medical terms and acronyms and return too many semantically similar but irrelevant documents. The company needs to improve retrieval quality and maintain low end user latency, even as the document collection grows to millions of documents.
Which solution will meet these requirements with the LEAST operational overhead?

A. Configure hybrid search by combining vector similarity with keyword matching to improve semantic understanding and exact term and acronym matching.
B. Increase the dimensions of the vector embeddings from 384 to 1536. Use a post-processing AWS Lambda function to filter out irrelevant results after retrieval.
C. Replace OpenSearch Service with Amazon Kendra. Use query expansion to handle medical acronyms and terminology variants during pre-processing.
D. Implement a two-stage retrieval architecture in which initial vector search results are re-ranked by an ML model that is hosted on Amazon SageMaker AI.

Correct Answer: A

Question #8 (Topic: Exam A)*Topic 1*

A company runs a generative AI (GenAI)-powered summarization application in an application AWS account that uses Amazon Bedrock. The application architecture includes an Amazon API Gateway REST API that forwards requests to AWS Lambda functions that are attached to private VPC subnets. The application summarizes sensitive customer records that the company stores in a governed data lake in a centralized data storage account. The company has enabled Amazon S3, Amazon Athena, and AWS Glue in the data storage account.
The company must ensure that calls that the application makes to Amazon Bedrock use only private connectivity between the company's application VPC and Amazon Bedrock. The company's data lake must provide fine-grained column-level access across the company's AWS accounts.
Which solution will meet these requirements?

A. In the application account, create interface VPC endpoints for Amazon Bedrock runtimes. Run Lambda functions in private subnets. Use IAM conditions on inference and data-plane policies to allow calls only to approved endpoints and roles. In the data storage account, use AWS Lake Formation LF-tag-based access control to create table and column-level cross-account grants.

B. Run Lambda functions in private subnets. Configure a NAT gateway to provide access to Amazon Bedrock and the data lake. Use S3 bucket policies and ACLs to manage permissions. Export AWS CloudTrail logs to Amazon S3 to perform weekly reviews.
C. Create a gateway endpoint only for Amazon S3 in the application account. Invoke Amazon Bedrock through public endpoints. Use database-level grants in AWS Lake Formation to manage data access. Stream AWS CloudTrail logs to Amazon CloudWatch Logs. Do not set up metric filters or alarms.
D. Use VPC endpoints to provide access to Amazon Bedrock and Amazon S3 in the application account. Use only IAM path-based policies to manage data lake access. Send AWS CloudTrail logs to Amazon CloudWatch Logs. Periodically create dashboards and allow public fallback for cross-Region reads to reduce setup time.

Correct Answer: A

Question #9 (Topic: Exam A)*Topic 1*

A media company must use Amazon Bedrock to implement a robust governance process for AI-generated content. The company needs to manage hundreds of prompt templates. Multiple teams use the templates across multiple AWS Regions to generate content. The solution must provide version control with approval workflows that include notifications for pending reviews. The solution must also provide detailed audit trails that document prompt activities and consistent prompt parameterization to enforce quality standards.
Which solution will meet these requirements?

A. Configure Amazon Bedrock Studio prompt templates. Use Amazon CloudWatch to create dashboards that display prompt usage metrics. Store the approval status of content in Amazon DynamoDB. Use AWS Lambda functions to enforce approvals.
B. Use Amazon Bedrock Prompt Management to implement version control. Configure AWS CloudTrail for audit logging. Use IAM policies to control approval permissions. Create parameterized prompt templates by specifying variables.
C. Use AWS Step Functions to create an approval workflow. Store prompts as documents in Amazon S3. Use tags to implement version control. Use Amazon EventBridge to send notifications.
D. Deploy Amazon SageMaker Canvas with prompt templates that are stored in Amazon S3. Use AWS CloudFormation to implement version control. Use AWS Config to enforce approval policies.

Correct Answer: B

Requirement	How Option B Addresses It
Version control for hundreds of templates	Amazon Bedrock **Prompt Management** natively supports creating, organizing, and versioning prompt templates — purpose-built for this exact use case
Multi-team, multi-Region usage	Bedrock Prompt Management is a managed service accessible across regions; IAM policies ensure consistent access control across teams
Approval workflows with permissions	**IAM policies** control who can create, update, publish, or approve prompt versions — enforcing role-based approval gates (e.g., only reviewers can promote a draft to production)
Detailed audit trails	**AWS CloudTrail** captures all Bedrock API calls including CreatePrompt, UpdatePrompt, GetPrompt, and CreatePromptVersion — providing comprehensive documentation of all prompt activities
Consistent parameterization	Bedrock Prompt Management natively supports **parameterized templates with variables** (e.g., {{tone}}, {{audience}}, {{content_type}}), enforcing quality standards across teams

Question #10 (Topic: Exam A)*Topic 1*

A company is developing a customer support application that uses Amazon Bedrock foundation models (FMs) to provide real-time AI assistance to the company's employees. The application must display AI-generated responses character by character as the responses are generated. The application needs to support thousands of concurrent users with minimal latency. The responses typically take 15 to 45 seconds to finish.
Which solution will meet these requirements?

A. Configure an Amazon API Gateway WebSocket API with an AWS Lambda integration. Configure the WebSocket API to invoke the Amazon Bedrock InvokeModelWithResponseStream API and stream partial responses through WebSocket connections.

B. Configure an Amazon API Gateway REST API with an AWS Lambda integration. Configure the REST API to invoke the Amazon Bedrock standard InvokeModel API and implement frontend client-side polling every 100 ms for complete response chunks.
C. Implement direct frontend client connections to Amazon Bedrock by using IAM user credentials and the InvokeModelWithResponseStream API without any intermediate gateway or proxy layer.
D. Configure an Amazon API Gateway HTTP API with an AWS Lambda integration. Configure the HTTP API to cache complete responses in an Amazon DynamoDB table and serve the responses through multiple paginated GET requests to frontend clients.

Correct Answer: A

Question #11 (Topic: Exam A)*Topic 1*

A company is using Amazon Bedrock to design an application to help researchers apply for grants. The application is based on an Amazon Nova Pro foundation model (FM). The application contains four required inputs and must provide responses in a consistent text format. The company wants to receive a notification in Amazon Bedrock if a response contains bullying language. However, the company does not want to block all flagged responses.
The company creates an Amazon Bedrock flow that takes an input prompt and sends it to the Amazon Nova Pro FM. The Amazon Nova Pro FM provides a response.
Which additional steps must the company take to meet these requirements? (Choose two.)

A. Use Amazon Bedrock Prompt Management to specify the required inputs as variables. Select an Amazon Nova Pro FM. Specify the output format for the response. Add the prompt to the prompts node of the flow.
B. Create an Amazon Bedrock guardrail that applies the hate content filter. Set the filter response to block. Add the guardrail to the prompts node of the flow.
C. Create an Amazon Bedrock prompt router. Specify an Amazon Nova Pro FM. Add the required inputs as variables to the input node of the flow. Add the prompt router to the prompts node. Add the output format to the output node.
D. Create an Amazon Bedrock guardrail that applies the insults content filter. Set the filter response to detect. Add the guardrail to the prompts node of the flow.
E. Create an Amazon Bedrock application inference profile that specifies an Amazon Nova Pro FM. Specify the output format for the response in the description. Include a tag for each of the input variables. Add the profile to the prompts node of the flow.

Correct Answer: AD

Question #12 (Topic: Exam A)*Topic 1*

A healthcare company is using Amazon Bedrock to build a Retrieval Augmented Generation (RAG) application that helps practitioners make clinical decisions. The application must achieve high accuracy for patient information retrievals, identify hallucinations in generated content, and reduce human review costs.
Which solution will meet these requirements?

A. Use Amazon Comprehend to analyze and classify RAG responses and to extract medical entities and relationships. Use AWS Step Functions to orchestrate automated evaluations. Configure Amazon CloudWatch metrics to track entity recognition confidence scores. Configure CloudWatch to send an alert when accuracy falls below specified thresholds.
B. Implement automated large language model (LLM)-based evaluations that use a specialized model that is fine-tuned for medical content to assess all responses. Deploy AWS Lambda functions to parallelize evaluations. Publish results to Amazon CloudWatch metrics that track relevance and factual accuracy.
C. Configure Amazon CloudWatch Synthetics to generate test queries that have known answers on a regular schedule, and track model success rates. Set up dashboards that compare synthetic test results against expected outcomes.
D. Deploy a hybrid evaluation system that uses an automated LLM-as-a-judge evaluation to initially screen responses and targeted human reviews for edge cases. Use Amazon SageMaker Feature Store to maintain evaluation datasets. Use a built-in Amazon Bedrock evaluation to track retrieval precision and hallucination rates.

Correct Answer: D

Question #13 (Topic: Exam A)*Topic 1*

Company configures a landing zone in AWS Control Tower. The company handles sensitive data that must remain within the European Union. The company must use only the eu-central-1 Region. The company uses SCPs to enforce data residency policies. GenAI developers at the company are assigned IAM roles that have full permissions for Amazon Bedrock.

The company must ensure that GenAI developers can use the Amazon Nova Pro model through Amazon Bedrock only by using cross-Region inference (CRI) and only in eu-central-1. The company enables model access for the GenAI developer IAM roles in Amazon Bedrock. However, when a GenAI developer attempts to invoke the model through the Amazon Bedrock Chat/Text playground, the GenAI developer receives the following error.
User: arn:aws:sts::123456789012:assumed-role/AssumedDevRole/DevUserName
Action: bedrock:InvokeModelWithResponseStream
On resource(s): arn:aws:bedrock:eu-west-3::foundation-model/amazon.nova-pro-v1:0
Context: a service control policy explicitly denies the action
The company needs a solution to resolve the error. The solution must retain the company's existing governance controls and must provide precise access control. The solution must comply with the company's existing data residency policies.
Which combination of solutions will meet these requirements? (Choose two.)

A. Add an AdministratorAccess policy to the GenAI developer IAM role.
B. Extend the existing SCPs to enable CRI for the eu.amazon.nova-pro-v1:0 inference profile.
C. Enable Amazon Bedrock model access for Amazon Nova Pro in the eu-west-3 Region.
D. Validate that the GenAI developer IAM roles have permissions to invoke Amazon Nova Pro through the eu.amazon.nova-pro.v1:0 inference profile on all European Union AWS Regions that can serve the model.
E. Extend the existing SCP to enable CRI for the eu.* inference profile.

Correct Answer: BD

Question #14 (Topic: Exam A)*Topic 1*

A financial services company is developing a customer service AI assistant by using Amazon Bedrock. The AI assistant must not discuss investment advice with users. The AI assistant must block harmful content, mask personally identifiable information (PII), and maintain audit trails for compliance reporting. The AI assistant must apply content filtering to both user inputs and model responses based on content sensitivity.
The company requires an Amazon Bedrock guardrail configuration that will effectively enforce policies with minimal false positives. The solution must provide multiple handling strategies for

multiple types of sensitive content.
Which solution will meet these requirements?

A. Configure a single guardrail and set content filters to high for all categories. Set up denied topics for investment advice and include sample phrases to block. Set up sensitive information filters that apply the block action for all PII entities. Apply the guardrail to all model inference calls.
B. Configure multiple guardrails by using tiered policies. Create one guardrail and set content filters to high. Configure the guardrail to block PII for public interactions. Configure a second guardrail and set content filters to medium. Configure the second guardrail to mask PII for internal use. Configure multiple topic-specific guardrails to block investment advice and set up contextual grounding checks.
C. Configure a guardrail and set content filters to medium for harmful content. Set up denied topics for investment advice and include clear definitions and sample phrases to block. Configure sensitive information filters to mask PII in responses and to block financial information in inputs. Enable both input and output evaluations that use custom blocked messages for audits.
D. Create a separate guardrail for each use case. Create one guardrail that applies a harmful content filter. Create a guardrail to apply topic filters for investment advice. Create a guardrail to apply sensitive information filters to block PII. Use AWS Step Functions to chain the guardrails together sequentially. Use conditional logic based on content classification.

Correct Answer: C

Question #15 (Topic: Exam A)*Topic 1*

An ecommerce company is developing a generative AI (GenAI) solution that uses Amazon Bedrock with Anthropic Claude to recommend products to customers. Customers report that some of the recommended products are not available for sale on the website or are not relevant to the customer. Customers also report that the solutions takes a long time to generate some recommendations.
The company investigates the issues and finds that most interactions between customers and the product recommendation solution are unique. The company confirms that the solutions recommends products that are not in the company's product catalog. The company must resolve these issues.
Which solution will meet this requirement?

A. Increase grounding within Amazon Bedrock Guardrails. Enable Automated Reasoning checks. Set up provisioned throughput.

B. Use prompt engineering to restrict the model responses to relevant products. Use streaming techniques such as the InvokeModelWithResponseStream action to reduce perceived latency for the customers.
C. Create an Amazon Bedrock knowledge base. Implement Retrieval Augmented Generation (RAG). Set the PerformanceConfigLatency parameter to optimized.
D. Store product catalog data in Amazon OpenSearch Service. Validate the model's product recommendations against the product catalog. Use Amazon DynamoDB to implement response caching.

Correct Answer: C

Question #16 (Topic: Exam A)*Topic 1*

A company is using AWS Lambda and REST APIs to build a reasoning agent to automate support workflows. The system must preserve memory across interactions, share the relevant agent state, and support event-driven invocation and synchronous invocation. The system must also enforce access control and session-based permissions.
Which combination of steps provides the MOST scalable solution? (Choose two.)

A. Use Amazon Bedrock AgentCore to manage memory and session-aware reasoning. Deploy the agent with built-in identity support, event handling, and observability.
B. Register the Lambda functions and the REST APIs as actions by using Amazon API Gateway and Amazon EventBridge. Enable Amazon Bedrock AgentCore to invoke the Lambda functions and the REST APIs without custom orchestration code.
C. Use Amazon Bedrock Agents for reasoning and conversation management. Use AWS Step Functions and Amazon SQS queues for orchestration. Store the agent state in Amazon DynamoDB to maintain memory between steps.
D. Deploy the reasoning logic as a container on Amazon ECS behind Amazon API Gateway. Use Amazon Aurora to store memory data and identity data.
E. Build a custom RAG pipeline by using Amazon Kendra and Amazon Bedrock. Use AWS Lambda to orchestrate tool invocations. Store the agent state in Amazon S3.

Correct Answer: AB

Question #17 (Topic: Exam A)*Topic 1*

A financial services company is developing a Retrieval Augmented Generation (RAG) application to help investment analysts query complex financial relationships across multiple investment vehicles, market sectors, and regulatory environments. The dataset contains highly interconnected entities that have multi-hop relationships. The analysts must be able to examine the relationships holistically to provide accurate investment guidance. The application must deliver comprehensive answers that capture indirect relationships between financial entities. The application must produce responses in less than 3 seconds.
Which solution will meet these requirements with the LEAST operational overhead?

A. Use Amazon Bedrock Knowledge Bases with Graph RAG and Amazon Neptune Analytics to store the financial data. Analyze the multi-hop relationships between entities and automatically identify related information across documents.
B. Use Amazon Bedrock Knowledge Bases and an Amazon OpenSearch Service vector store to implement custom relationship identification logic that uses AWS Lambda functions to query multiple vector embeddings in sequence.
C. Use an Amazon OpenSearch Serverless vector database with k-nearest neighbor (k-NN) searches. Implement manual relationship mapping in an application layer that runs in an Amazon EC2 Auto Scaling group.
D. Use Amazon DynamoDB to store financial data in a custom indexing system. Use an AWS Lambda function to query relevant records based on input questions. Use Amazon SageMaker AI to generate responses.

Correct Answer: A

Question #18 (Topic: Exam A)*Topic 1*

A healthcare company uses Amazon Bedrock to deploy an application that generates summaries of clinical documents. The application experiences inconsistent response quality with occasional factual hallucinations. Monthly costs exceed the company's projections by 40%. A GenAI developer must implement a near real-time monitoring solution to detect hallucinations, identify abnormal token consumption, and provide early warnings of cost anomalies. The solution must require minimal custom development work and maintenance overhead.
Which solution will meet these requirements?

A. Configure Amazon CloudWatch alarms to monitor InputTokenCount and OutputTokenCount metrics to detect anomalies. Store model invocation logs in an Amazon S3 bucket. Use AWS Glue and Amazon Athena to identify potential hallucinations.
B. Run Amazon Bedrock evaluation jobs that use LLM-based judgments to detect hallucinations. Configure Amazon CloudWatch to track token usage. Create an AWS Lambda function to process CloudWatch metrics. Configure the Lambda function to send usage pattern notifications.
C. Configure Amazon Bedrock to store model invocation logs in an Amazon S3 bucket. Enable text output logging. Configure Amazon Bedrock guardrails to run contextual grounding checks to detect hallucinations. Create Amazon CloudWatch anomaly detection alarms for token usage metrics.
D. Use AWS CloudTrail to log all Amazon Bedrock API calls. Create a custom dashboard in Amazon QuickSight to visualize token usage patterns. Use Amazon SageMaker Model Monitor to detect quality drift in generated summaries.

Correct Answer: C

Question #19 (Topic: Exam A)*Topic 1*

A company is building a generative AI (GenAI) application that produces content based on a variety of internal and external data sources. The company wants to ensure that the generated output is fully traceable. The application must support data source registration and enable metadata tagging to attribute content to its original source. The application must also maintain audit logs of data access and usage throughout the pipeline.
Which solution will meet these requirements?

A. Use AWS Lake Formation to catalog data sources and control access. Apply metadata tags directly in Amazon S3. Use AWS CloudTrail to monitor API activity.
B. Use AWS Glue Data Catalog to register and tag data sources. Use Amazon CloudWatch Logs to monitor access patterns and application behavior.
C. Store data in Amazon S3 and use object tagging for attribution. Use AWS Glue Data Catalog to manage schema information. Use AWS CloudTrail to log access to S3 buckets.
D. Use AWS Glue Data Catalog to register all data sources. Apply metadata tags to attribute data sources. Use AWS CloudTrail to log access and activity across services.

Correct Answer: D

This solution satisfies all three requirements:

- **Data source registration** — AWS Glue Data Catalog serves as a centralized metadata repository where you can register and catalog all internal and external data sources used by the GenAI application.
- **Metadata tagging for attribution** — Glue Data Catalog natively supports metadata tagging on cataloged objects (databases, tables, columns), allowing you to attribute generated content back to its original source.
- **Audit logs across the pipeline** — AWS CloudTrail provides comprehensive logging of API activity *across services*, giving you a full audit trail of data access and usage throughout the entire pipeline.

Question #20 (Topic: Exam A)*Topic 1*

A financial services company needs to build a document analysis system that uses Amazon Bedrock to process quarterly reports. The system must analyze financial data, perform sentiment analysis, and validate compliance across batches of reports. Each batch contains 5 reports. Each report requires multiple foundation model (FM) calls. The solution must finish the analysis within 10 seconds for each batch. Current sequential processing takes 45 seconds for each batch.
Which solution will meet these requirements?

A. Use AWS Lambda functions with provisioned concurrency to process each analysis type sequentially. Configure the Lambda function timeouts to 10 seconds. Configure automatic retries with exponential backoff.
B. Use AWS Step Functions with a Parallel state to invoke separate AWS Lambda functions for each analysis type simultaneously. Configure Amazon Bedrock client timeouts. Use Amazon CloudWatch metrics to track execution time and model inference latency.
C. Create an Amazon SQS queue to buffer analysis requests. Deploy multiple AWS Lambda functions with reserved concurrency. Configure each Lambda function to process different aspects of each report sequentially and then combine the results.
D. Deploy an Amazon ECS cluster that runs containers that process each report sequentially. Use a load balancer to distribute batch workloads. Configure an auto-scaling policy based on CPU utilization to handle demand fluctuations.

Correct Answer: B

Question #21 (Topic: Exam A)*Topic 1*

A company is using Amazon Bedrock to build a customer-facing AI assistant to handle sensitive customer inquiries. The company must use defense-in-depth safety controls to block sophisticated prompt injection attacks. The company must keep audit logs of all safety interventions. The AI assistant must have cross-Region failover capabilities.
Which solution will meet these requirements?

A. Configure Amazon Bedrock guardrails to use content filters to protect against prompt injection attacks. Set the content filters to high. Use a guardrail profile to implement cross-Region guardrail inference. Use Amazon CloudWatch Logs with custom metrics to capture detailed guardrail intervention events.
B. Configure Amazon Bedrock guardrails to use content filters to protect against prompt injection attacks. Set the content filters to high. Use AWS WAF to block suspicious inputs. Use AWS CloudTrail to log API calls for audits.
C. Deploy Amazon Comprehend custom classification to detect prompt injection attacks. Use Amazon API Gateway to validate requests. Use Amazon CloudWatch Logs with custom metrics to capture detailed intervention events.
D. Configure Amazon Bedrock guardrails to use custom content filters to protect against harmful content. Set the content filters to high. Use word filters to protect against known attack patterns. Configure cross-Region guardrail replication to provide failover capabilities. Store logs in AWS CloudTrail for compliance auditing.

Correct Answer: A

Question #22 (Topic: Exam A)*Topic 1*

A company is designing a canary deployment strategy for a payment processing API. The system must support automated gradual traffic shifting between multiple Amazon Bedrock models based on real-time inference metrics, historical traffic patterns, and service health. The solution must be able to gradually increase traffic to new model versions. The system must increase traffic if metrics remain healthy and decrease traffic if the performance degrades below acceptable thresholds.
The company needs to comprehensively monitor inference latency and error rates during the deployment phase. The company must also be able to halt deployments and revert to a previous model version without any manual intervention.
Which solution will meet these requirements?

A. Use Amazon Bedrock with provisioned throughput to host the versions of the model. Configure an Amazon EventBridge rule to invoke an AWS Step Functions workflow when a new model version

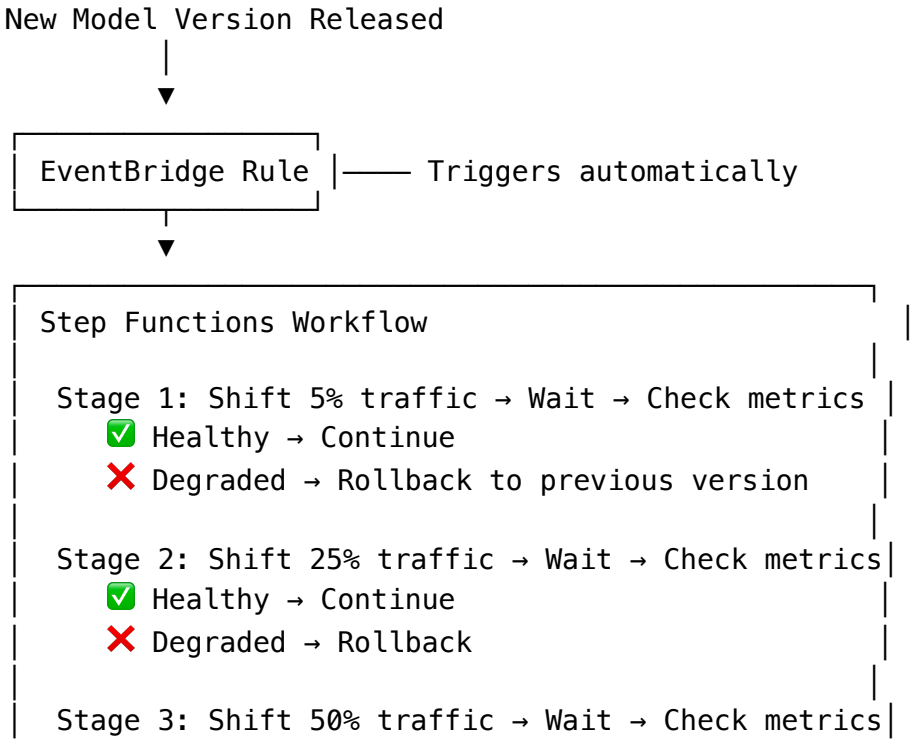
is released. Configure the workflow to shift traffic in stages, wait for a specified time period, and invoke an AWS Lambda function to check Amazon CloudWatch performance metrics. Configure the workflow to increase traffic if the metrics meet thresholds and to trigger a traffic rollback if performance metrics fall below thresholds.

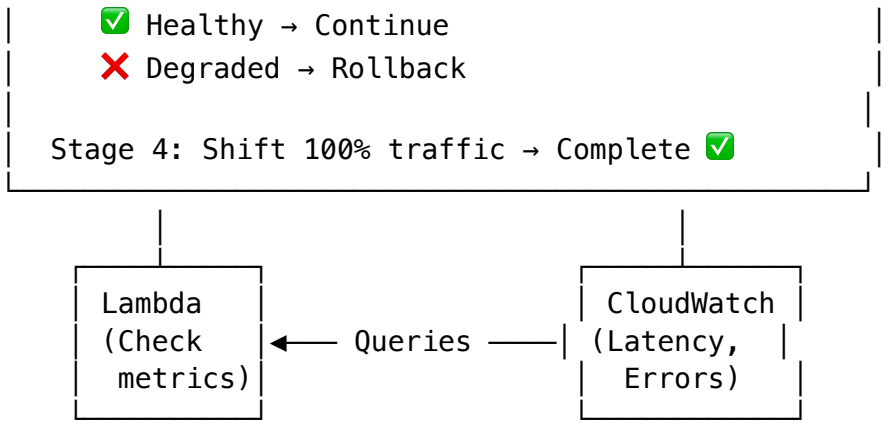
B. Use AWS Lambda functions to invoke various Amazon Bedrock model versions. Use an Amazon API Gateway HTTP API with stage variables and weighted routing to shift traffic gradually to new model versions. Use Amazon CloudWatch to monitor performance metrics. Use external logic to adjust traffic between model versions and to roll back if performance falls below thresholds.

C. Use Amazon SageMaker AI endpoint variants to represent multiple Amazon Bedrock model versions. Use variant weights to shift traffic. Use Amazon CloudWatch to monitor performance metrics. Use SageMaker Model Monitor to trigger AWS Lambda functions to roll back a model deployment if performance drops below a specified threshold. Configure an Amazon EventBridge rule to roll back model deployments if an anomaly is detected.

D. Use Amazon OpenSearch Service to track inference logs. Configure OpenSearch Service to invoke an AWS Systems Manager Automation runbook to update Amazon Bedrock model endpoints to shift traffic based on the inference logs.

Canary Deployment Flow





Question #23 (Topic: Exam A)*Topic 1*

A financial services company uses an AI application to process financial documents by using Amazon Bedrock. During business hours, the application handles approximately 10,000 requests each hour, which requires consistent throughput.

The company uses the CreateProvisionedModelThroughput API to purchase provisioned throughput. Amazon CloudWatch metrics show that the provisioned capacity is unused while on-demand requests are being throttled. The company finds the following code in the application:

python response = bedrock_runtime.invoke_model(modelId="anthropic.claude-v2", body=json.dumps(payload))

The company needs the application to use the provisioned throughput and to resolve the throttling issues.

Which solution will meet these requirements?

A. Increase the number of model units (MUs) in the provisioned throughput configuration.
B. Replace the model ID parameter with the ARN of the provisioned model that the CreateProvisionedModelThroughput API returns.
C. Add exponential backoff retry logic to handle throttling exceptions during peak hours.
D. Modify the application to use the InvokeModelWithResponseStream API instead of the InvokeModel API.

Correct Answer: B

Question #24 (Topic: Exam A)*Topic 1*

A company is building an AI advisory application by using Amazon Bedrock. The application will provide recommendations to customers. The company needs the application to explain its reasoning process and cite specific sources for data. The application must retrieve information from company data sources and show step-by-step reasoning for recommendations. The application must also link data claims to source documents and maintain response latency under 3 seconds.
Which solution will meet these requirements with the LEAST operational overhead?

A. Use Amazon Bedrock Knowledge Bases with source attribution enabled. Use the Anthropic Claude Messages API with RAG to set high-relevance thresholds for source documents. Store reasoning and citations in Amazon S3 for auditing purposes.
B. Use Amazon Bedrock with Anthropic Claude models and extended thinking. Configure a 4,000-token thinking budget. Store reasoning traces and citations in Amazon DynamoDB for auditing purposes.
C. Configure Amazon SageMaker AI with a custom Anthropic Claude model. Use the model's reasoning parameter and AWS Lambda to process responses. Add source citations from a separate Amazon RDS database.
D. Use Amazon Bedrock with Anthropic Claude models and chain-of-thought reasoning. Configure custom retrieval tracking with the Amazon Bedrock Knowledge Bases API. Use Amazon CloudWatch to monitor response latency metrics.

Correct Answer: A

Question #25 (Topic: Exam A)*Topic 1*

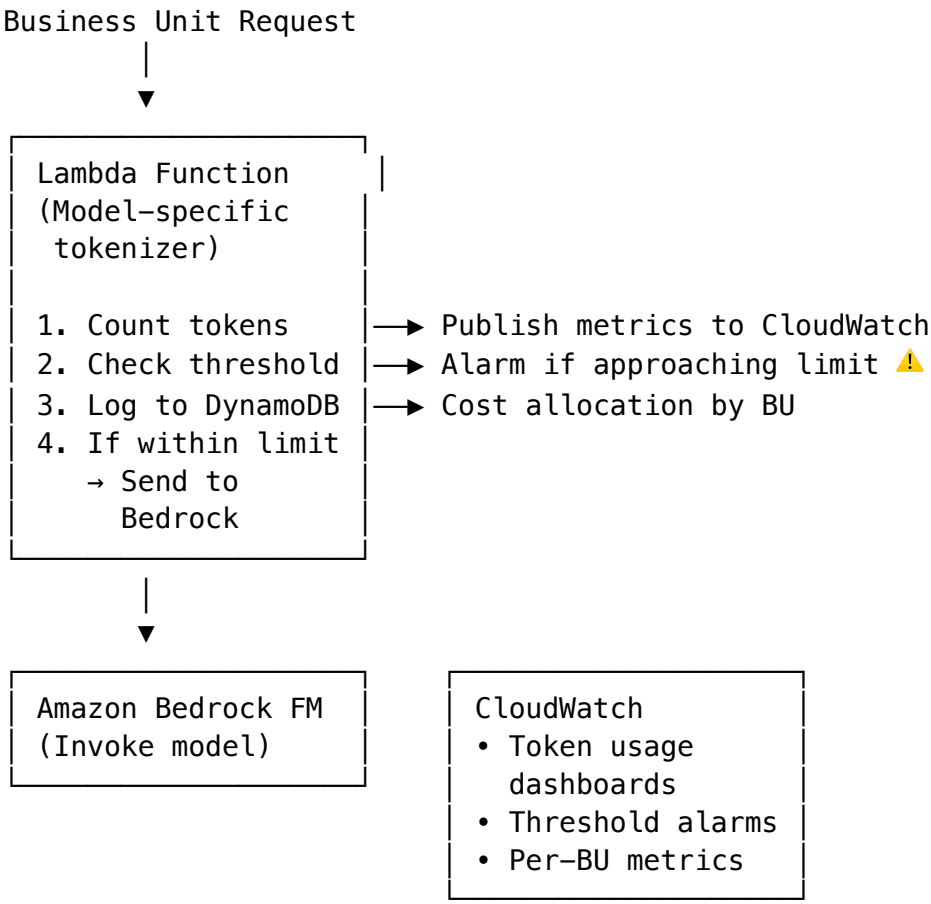
A financial services company uses multiple foundation models (FMs) through Amazon Bedrock for its generative AI (GenAI) applications. To comply with a new regulation for GenAI use with sensitive financial data, the company needs a token management solution.
The token management solution must proactively alert when applications approach model-specific token limits. The solution must also process more than 5,000 requests each minute and maintain token usage metrics to allocate costs across business units.
Which solution will meet these requirements?

A. Develop model-specific tokenizers in an AWS Lambda function. Configure the Lambda function to estimate token usage before sending requests to Amazon Bedrock. Configure the Lambda function to publish metrics to Amazon CloudWatch and trigger alarms when requests approach thresholds. Store detailed token usage in Amazon DynamoDB to report costs.

B. Implement Amazon Bedrock Guardrails with token quota policies. Capture metrics on rejected requests. Configure Amazon EventBridge rules to trigger notifications based on Amazon Bedrock Guardrails metrics. Use Amazon CloudWatch dashboards to visualize token usage trends across models.
C. Deploy an Amazon SQS dead-letter queue for failed requests. Configure an AWS Lambda function to analyze token-related failures. Use Amazon CloudWatch Logs Insights to generate reports on token usage patterns based on error logs from Amazon Bedrock API responses.
D. Use Amazon API Gateway to create a proxy for all Amazon Bedrock API calls. Configure request throttling based on custom usage plans with predefined token quotas. Configure API Gateway to reject requests that will exceed token limits.

Correct Answer: A

Architecture Flow



Question #26 (Topic: Exam A)*Topic 1*

A retail company is developing a customer service application that must process 10,000 daily queries about products, orders, and warranties. The application must be able to respond to queries about 50,000 product documents that are updated every day. The application must integrate with an order management API to check the status of orders and to help process returns. The application must maintain context throughout multi-turn interactions with customers. The company must collect complete audit trails for application responses.
Which solution will meet these requirements with the LEAST operational overhead?

A. Deploy a fine-tuned Amazon Bedrock Anthropic Claude model for each product category. Create AWS Lambda functions to connect each model to the order management API. Store conversation history in Amazon DynamoDB.
B. Create a custom model that uses continued pre-training on Amazon Bedrock to handle all product documentation. Set up an Amazon API Gateway REST API that uses AWS Lambda functions to connect the model to the order management API.
C. Use Amazon SageMaker AI with containers to deploy models. Use Amazon Kendra to search product documents. Use AWS Step Functions to orchestrate calls to the order management API.
D. Use an Amazon Bedrock agent with action groups to integrate with the order management API. Associate an Amazon Bedrock knowledge base with the agent to search product documentation by using Retrieval Augmentation Generation (RAG). Enable trace events to capture audit trails.

Correct Answer: D

Question #27 (Topic: Exam A)*Topic 1*

An ecommerce company is using Amazon Bedrock to build a generative AI (GenAI) application. The application uses AWS Step Functions to orchestrate a multi-agent workflow to produce detailed product descriptions. The workflow consists of three sequential states: a description generator, a technical specifications validator, and a brand voice consistency checker. Each state produces intermediate reasoning traces and outputs that are passed to the next state. The application uses an Amazon S3 bucket for process storage and to store outputs.
During testing, the company discovers that outputs between Step Functions states frequently exceed the 256 KB quota and cause workflow failures.
A GenAI Developer needs to revise the application architecture to efficiently handle the Step Functions 256 KB quota and maintain workflow observability. The revised architecture must preserve the existing multi-agent reasoning and acting (ReAct) pattern.
Which solution will meet these requirements with the LEAST operational overhead?

A. Store intermediate outputs in Amazon DynamoDB. Pass only references between states. Create a Map state that retrieves the complete data from DynamoDB when required for each agent's processing step.
B. Configure an Amazon Bedrock integration to use the S3 bucket URI in the input parameter for large outputs. Use the ResultPath field and the ResultSelector field to route S3 references between the agent steps while maintaining the sequential validation workflow.
C. Use AWS Lambda functions to compress outputs to less than 256 KB before each agent state. Configure each agent task to decompress the outputs before processing and to compress results before passing them to the next state.
D. Configure a separate Step Functions state machine to handle each agent's processing. Use Amazon EventBridge to coordinate the execution flow between state machines. Use S3 references for the outputs as event data.

Correct Answer: B

DynamoDB has a 400 KB item size limit — may still be insufficient for large reasoning traces. Map state is designed for parallel iteration, not sequential data retrieval. Adds an entirely new service (DynamoDB) to manage

Compression/decompression Lambda functions at every state boundary adds 6 additional Lambda invocations (compress + decompress × 3 states). No guarantee compressed output fits under 256 KB. Significant development and maintenance overhead

Question #28 (Topic: Exam A)*Topic 1*

A company provides a service that helps users from around the world discover new restaurants. The service has 50 million monthly active users. The company wants to implement a semantic search solution across a database that contains 20 million restaurants and 200 million reviews. The company currently stores the data in a PostgresQL database.
The solution must support complex natural language queries and return results for at least 95% of queries within 500 ms. The solution must maintain data freshness for restaurant details that update hourly. The solution must also scale cost-effectively during peak usage periods.
Which solution will meet these requirements with the LEAST development effort?

A. Migrate the restaurant data to Amazon OpenSearch Service. Implement keyword-based search rules that use custom analyzers and relevance tuning to find restaurants based on attributes such

as cuisine type, feature, and location. Create Amazon API Gateway HTTP API endpoints to transform user queries into structured search parameters.
B. Migrate the restaurant data to Amazon OpenSearch Service. Use a foundation model (FM) in Amazon Bedrock to generate vector embeddings from restaurant descriptions, reviews, and menu items. When users submit natural language queries, convert the queries to embeddings by using the same FM. Perform k-nearest neighbors (k-NN) searches to find semantically similar results.
C. Keep the restaurant data in PostgresQL and implement a pgvector extension. Use a foundation model (FM) in Amazon Bedrock to generate vector embeddings from restaurant data. Store the vector embeddings directly in PostgreSQL. Create an AWS Lambda function to convert natural language queries to vector representations by using the same FM. Configure the Lambda function to perform similarity searches within the database.
D. Migrate restaurant data to an Amazon Bedrock knowledge base by using a custom ingestion pipeline. Configure the knowledge base to automatically generate embeddings from restaurant information. Use the Amazon Bedrock Retrieve API with built-in vector search capabilities to query the knowledge base directly by using natural language input.

Factor	pgvector (C)	OpenSearch (B)
Designed for	Relational data with vector extension	Full-text + vector search at scale
220M vectors	Performance degrades significantly	Native sharding across nodes
High concurrency	Connection pooling bottlenecks	Distributed query execution
p95 < 500ms	Challenging at this scale	Achievable with HNSW k-NN
Hourly updates	Works but impacts query performance	Near-real-time, non-disruptive

Correct Answer: B

Question #29 (Topic: Exam A)*Topic 1*

A medical company uses Amazon Bedrock to power a clinical documentation summarization system. The system produces inconsistent summaries when handling complex clinical documents. The system performed well on simple clinical documents.
The company needs a solution that diagnoses inconsistencies, compares prompt performance against established metrics, and maintains historical records of prompt versions.
Which solution will meet these requirements?

A. Create multiple prompt variants by using Prompt management in Amazon Bedrock. Manually test the prompts with simple clinical documents. Deploy the highest performing version by using the Amazon Bedrock console.
B. Implement version control for prompts in a code repository with a test suite that contains complex clinical documents and quantifiable evaluation metrics. Use an automated testing framework to compare prompt versions and document performance patterns.
C. Deploy each new prompt version to separate Amazon Bedrock API endpoints. Split production traffic between the endpoints. Configure Amazon CloudWatch to capture response metrics and user feedback for automatic version selection.
D. Create a custom prompt evaluation flow in Amazon Bedrock Flows that applies the same clinical document inputs to different prompt variants. Use Amazon Comprehend Medical to analyze and score the factual accuracy of each version.

Correct Answer: B

Question #30 (Topic: Exam A)*Topic 1*

A company uses Amazon Bedrock to generate technical content for customers. The company has recently experienced a surge in hallucination outputs when the company's model generates summaries of long technical documents. The model outputs include inaccurate or fabricated details. The company's current solution uses a large foundation model (FM) with a basic one-shot prompt that includes the full document in a single input.
The company needs a solution that will reduce hallucinations and meet factual accuracy goals. The solution must process more than 1,000 documents each hour and deliver summaries within 3 seconds for each document.
Which combination of solutions will meet these requirements? (Choose two.)

A. Implement zero-shot chain-of-thought (CoT) instructions that require step-by-step reasoning with explicit fact verification before the model generates each summary.

B. Use Retrieval Augmented Generation (RAG) with an Amazon Bedrock knowledge base. Apply semantic chunking and tuned embeddings to ground summaries in source content.
C. Configure Amazon Bedrock guardrails to block any generated output that matches patterns that are associated with hallucinated content.
D. Increase the temperature parameter in Amazon Bedrock.
E. Prompt the Amazon Bedrock model to summarize each full document in one pass.

Correct Answer: AB

Question #31 (Topic: Exam A)*Topic 1*

A company has a recommendation system. The system's applications run on Amazon EC2 instances. The applications make API calls to Amazon Bedrock foundation models (FMs) to analyze customer behavior and generate personalized product recommendations.
The system is experiencing intermittent issues. Some recommendations do not match customer preferences. The company needs an observability solution to monitor operational metrics and detect patterns of operational performance degradation compared to established baselines. The solution must also generate alerts with correlation data within 10 minutes when FM behavior deviates from expected patterns.
Which solution will meet these requirements?

A. Configure Amazon CloudWatch Container Insights for the application infrastructure. Set up CloudWatch alarms for latency thresholds. Add custom metrics for token counts by using the CloudWatch embedded metric format. Create CloudWatch dashboards to visualize the data.
B. Implement AWS X-Ray to trace requests through the application components. Enable CloudWatch Logs Insights for error pattern detection. Set up AWS CloudTrail to monitor all API calls to Amazon Bedrock. Create custom dashboards in Amazon QuickSight.
C. Enable Amazon CloudWatch Application Insights for the application resources. Create custom metrics for recommendation quality, token usage, and response latency by using the CloudWatch embedded metric format with dimensions for request types and user segments. Configure CloudWatch anomaly detection on the model metrics. Establish log pattern analysis by using CloudWatch Logs Insights.
D. Use Amazon OpenSearch Service with the Observability plugin. Ingest model metrics and logs by using Amazon Kinesis. Create custom Piped Processing Language (PPL) queries to analyze model behavior patterns. Establish operational dashboards to visualize anomalies in real time.

Correct Answer: C

Question #32 (Topic: Exam A)*Topic 1*

An enterprise application uses an Amazon Bedrock foundation model (FM) to process and analyze 50 to 200 pages of technical documents. Users are experiencing inconsistent responses and receiving truncated outputs when processing documents that exceed the FM's context window limits.
Which solution will resolve this problem?

A. Configure fixed-size chunking at 4,000 tokens for each chunk with 20% overlap. Use application-level logic to link multiple chunks sequentially until the FM's maximum context window of 200,000 tokens is reached before making inference calls.
B. Use hierarchical chunking with parent chunks of 8,000 tokens and child chunks of 2,000 tokens. Use Amazon Bedrock Knowledge Bases built-in retrieval to automatically select relevant parent chunks based on query context. Configure overlap tokens to maintain semantic continuity.
C. Use semantic chunking with a breakpoint percentile threshold of 95% and a buffer size of 3 sentences. Use the Amazon Bedrock RetrieveAndGenerate API call to dynamically select the most relevant chunks based on embedding similarity scores.
D. Create a pre-processing AWS Lambda function that analyzes document token count by using the FM's tokenizer. Configure the lambda function to split documents into equal segments that fit within 80% of the context window. Configure the Lambda function to process each segment independently before aggregating the results.

Correct Answer: C

Question #33 (Topic: Exam A)*Topic 1*

A company is developing a generative AI (GenAI) application that analyzes customer service calls in real-time and generates suggested responses for human customer service agents. The application must process 500,000 concurrent calls during peak hours with less than 200 ms end-to-end latency for each suggestion. The company uses existing architecture to transcribe customer call audio streams. The application must not exceed a pre-defined monthly compute budget and must maintain auto scaling capabilities.
Which solution will meet these requirements?

A. Deploy a large, complex reasoning model on Amazon Bedrock. Purchase provisioned throughput and optimize for batch processing.

B. Deploy a low-latency, real-time optimized model on Amazon Bedrock. Purchase provisioned throughput and set up automatic scaling policies.
C. Deploy a large language model (LLM) on an Amazon SageMaker AI real-time endpoint that uses dedicated GPU instances.
D. Deploy a mid-sized language model on an Amazon SageMaker AI serverless endpoint that is optimized for batch processing.

Correct Answer: B

Question #34 (Topic: Exam A)*Topic 1*

An ecommerce company is building an internal platform to develop generative AI applications by using Amazon Bedrock foundation models (FMs). Developers need to select models based on evaluations that are aligned to ecommerce use cases. The platform must display accuracy metrics for text generation and summarization in dashboards. The company has custom ecommerce datasets to use as standardized evaluation inputs.
Which combination of steps will meet these requirements with the LEAST operational overhead? (Choose two.)

A. Import the datasets to an Amazon S3 bucket. Provide appropriate IAM permissions and cross-origin resource sharing (CORS) permissions to give the evaluation jobs access to the datasets.
B. Import the datasets to an Amazon S3 bucket. Provide appropriate IAM permissions and a VPC endpoint configuration to give the evaluation jobs access to the datasets.
C. Configure an AWS Lambda function to create model evaluation jobs on a schedule in the Amazon Bedrock console. Provide the URI of the S3 bucket that contains the datasets as an input. Configure the evaluation jobs to measure the real world knowledge (RWK) score for text generation and BERT Score for summarization. Configure a second Lambda function to check the status of the jobs and publish custom logs to Amazon CloudWatch. Create a custom Amazon CloudWatch Logs Insights dashboard.
D. Use Amazon SageMaker Clarify on a schedule to create model evaluation jobs. Use open source frameworks to create and run standardized evaluations. Publish results to Amazon CloudWatch namespaces. Use the word error rate score for text generation and toxicity for summarization as metrics for accuracy. Configure an AWS Lambda function to check the status of the jobs and publish custom logs to CloudWatch. Create a custom Amazon CloudWatch Logs Insights dashboard.
E. Run an Amazon SageMaker AI notebook job on a schedule by using the fmevals or ragas framework to run evaluations that use the datasets in the S3 bucket. Write Python code in the

notebook that makes direct InvokeModel API calls to the FMs and processes their responses for evaluation. Publish job status and results to Amazon CloudWatch Logs to measure the real world knowledge (RWK) score for text generation and toxicity for summarization as metrics for accuracy. Create a custom CloudWatch Logs Insights dashboard.

Correct Answer: AC

Question #35 (Topic: Exam A)*Topic 1*

An elevator service company has developed an AI assistant application by using Amazon Bedrock. The application generates elevator maintenance recommendations to support the company's elevator technicians. The company uses Amazon Kinesis Data Streams to collect the elevator sensor data.
New regulatory rules require that a human technician must review all AI-generated recommendations. The company needs to establish human oversight workflows to review and approve AI recommendations. The company must store all human technician review decisions for audit purposes.
Which solution will meet these requirements?

A. Create a custom approval workflow by using AWS Lambda functions and Amazon SQS queues for human review of AI recommendations. Store all review decisions in Amazon DynamoDB for audit purposes.
B. Create an AWS Step Functions workflow that has a human approval step that uses the waitForTaskToken API to pause execution. After a human technician completes a review, use an AWS Lambda function to call the SendTaskSuccess API that has the approval decision. Store all review decisions in Amazon DynamoDB.
C. Create an AWS Glue workflow that has a human approval step. After the human technician review, integrate the application with an AWS Lambda function that calls the SendTaskSuccess API. Store all human technician review decisions in Amazon DynamoDB.
D. Configure Amazon EventBridge rules with custom event patterns to route AI recommendations to human technicians for review. Create AWS Glue jobs to process human technician approval queues. Use Amazon ElastiCache to cache all human technician review decisions.

Correct Answer: B

Question #36 (Topic: Exam A)*Topic 1*

A bank is building a generative AI (GenAI) application that uses Amazon Bedrock to assess loan applications by using scanned financial documents. The application must extract structured data from the documents. The application must redact personally identifiable information (PII) before inference. The application must use foundation models (FMs) to generate approvals. The application must route low-confidence document extraction results to human reviewers who are within the same AWS Region as the loan applicant.

The company must ensure that the application complies with strict Regional data residency and auditability requirements. The application must be able to scale to handle 25,000 applications each day and provide 99.9% availability.

Which combination of solutions will meet these requirements? (Choose three.)

A. Deploy Amazon Textract and Amazon Augmented AI (Amazon A2I) within the same Region to extract relevant data from the scanned documents. Route low-confidence pages to human reviewers.

B. Use AWS Lambda functions to detect and redact PII from submitted documents before inference. Apply Amazon Bedrock guardrails to prevent inappropriate or unauthorized content in model outputs. Configure Region-specific IAM roles to enforce data residency requirements and to control access to the extracted data.

C. Use Amazon Kendra and Amazon OpenSearch Service to extract field level values semantically from the uploaded documents before inference.

D. Store uploaded documents in Amazon S3 and apply object metadata. Configure IAM policies to store original documents within the same Region as each applicant. Enable object tagging for future audits.

E. Use AWS Glue Data Quality to validate the structured document data. Use AWS Step Functions to orchestrate a review workflow that includes a prompt engineering step that transforms validated data into optimized prompts before invoking Amazon Bedrock to assess loan applications.

F. Use Amazon SageMaker Clarify to generate fairness and bias reports based on model scoring decisions that Amazon Bedrock makes.

Correct Answer: ABE but the AI and me picked ABD

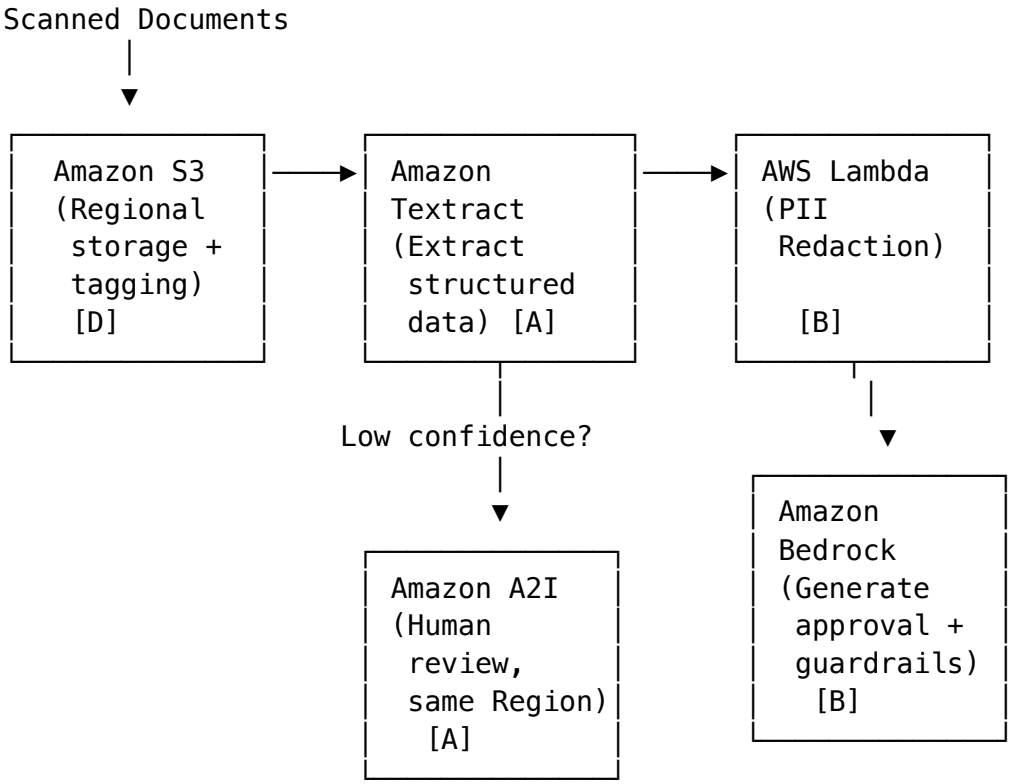
Requirement	How A Addresses It

Extract structured data from scanned documents	Textract is purpose-built for extracting structured data (forms, tables, key-value pairs) from scanned financial documents
Route low-confidence to human reviewers in same Region	A2I natively routes low-confidence extraction results to human review workflows, deployed **within the same Region** as the applicant

Requirement	How B Addresses It
Redact PII before inference	Lambda functions detect and redact PII from documents **before** they're sent to Bedrock — ensuring no PII enters the model
Safe model outputs	Bedrock guardrails prevent inappropriate or unauthorized content in generated loan decisions
Regional data residency	Region-specific IAM roles enforce that data stays within the required Region and controls access to extracted data

Requirement	How D Addresses It
Regional data residency	IAM policies enforce that original documents are stored in the **same Region** as each applicant — strict compliance
Auditability	Object metadata and tagging create a durable, queryable audit trail for every document processed
Durable storage	S3 provides 99.999999999% durability for original loan documents

End-to-End Architecture Flow



Question #37 (Topic: Exam A)*Topic 1*

A software company is using Amazon Q Business to build an AI assistant that allows employees to access company information and personal information by using natural language prompts. The company stores this information in an Amazon S3 bucket.

Each department in the company has a dedicated prefix in the S3 bucket. Each object name includes the S3 prefix of the department that it belongs to. Each department can belong to only a single group in AWS IAM Identity Center. Each employee belongs to a single department.

The company configures Amazon Q Business to access data stored in an S3 bucket as a data source. The company needs to ensure that the AI assistant respects access controls based on the user's IAM Identity Center group membership.

Which solution will meet this requirement with the LEAST operational overhead?

A. Create a JSON file named acl.json in each department folder. In each file, create access control entries that specify the IAM Identity Center group that should have access to that department's data. Indicate the location of the JSON file in the Access Control section of the data source settings.
B. Create a single JSON file named acl.json at the top level of the S3 bucket. Add access control entries that map each department's S3 prefix to its corresponding IAM Identity Center group. Indicate the location of the JSON file in the Access Control section of the data source settings.
C. For each IAM Identity Center group, create a separate permissions set that denies access to all prefixes in the S3 bucket. Add a StringNotEquals condition key to the permissions set for each group that specifies the department each group is associated with. Attach the permissions sets to the Identity Center groups.
D. Create a metadata file named metadata.json at the top level of the S3 bucket. Add an AccessControlList object to the file that specifies the S3 path of each department's prefix. Specify the IAM Identity Center group that should have access to each department's prefix. Reference the file location in the data source metadata settings.

Correct Answer: B

Question #38 (Topic: Exam A)*Topic 1*

A healthcare company is using Amazon Bedrock to build a system to help practitioners make clinical decisions. The system must provide treatment recommendations to physicians based only on approved medical documentation and must cite specific sources. The system must not hallucinate or produce factually incorrect information.
Which solution will meet these requirements with the LEAST operational overhead?

A. Integrate Amazon Bedrock with Amazon Kendra to retrieve approved documents. Implement custom post-processing to compare generated responses against source documents and to include citations.
B. Deploy an Amazon Bedrock knowledge base and connect it to approved clinical source documents. Use the Amazon Bedrock RetrieveAndGenerate API to return citations from the knowledge base.
C. Use Amazon Bedrock and Amazon Comprehend Medical to extract medical entities. Implement verification logic against a medical terminology database.
D. Use an Amazon Bedrock knowledge base with Retrieve API calls and InvokeModel API calls to retrieve approved clinical source documents. Implement verification logic to compare against retrieved sources and to cite sources.

Correct Answer: B

The key differentiator between **B and D**: `RetrieveAndGenerate` is purpose-built to return grounded, cited responses in a single managed API call. Option D manually replicates this with `Retrieve` + `InvokeModel` + custom verification — achieving the same outcome with significantly more operational overhead.

Question #39 (Topic: Exam A)*Topic 1*

A financial services company is developing a real-time generative AI (GenAI) assistant to support human call center agents. The GenAI assistant must transcribe live customer speech, analyze context, and provide incremental suggestions to call center agents while a customer is still speaking. To preserve responsiveness, the GenAI assistant must maintain end-to-end latency under 1 second from speech to initial response display. The architecture must use only managed AWS services and must support bidirectional streaming to ensure that call center agents receive updates in real time.
Which solution will meet these requirements?

A. Use the Amazon Transcribe streaming API to transcribe calls. Pass the text to Amazon Comprehend to perform sentiment analysis. Feed the results to Anthropic Claude on Amazon Bedrock by using the InvokeModel API. Store results in Amazon DynamoDB. Use a WebSocket API to display the results.
B. Use Amazon Transcribe streaming with partial results enabled to deliver fragments of transcribed text before customers finish speaking. Forward text fragments to Amazon Bedrock by using the InvokeModelWithResponseStream API. Stream responses to call center agents through an Amazon API Gateway WebSocket API.
C. Use Amazon Transcribe batch processing to convert calls to text. Pass complete transcripts to Anthropic Claude on Amazon Bedrock by using the ConverseStream API. Return responses through an Amazon Lex chatbot interface that call center agents can access from their work computers.
D. Use the Amazon Transcribe streaming API with an AWS Lambda function to transcribe each audio segment. Configure the Lambda function to call the Amazon Titan Embeddings model on Amazon Bedrock by using the InvokeModel API. Configure the Lambda function to publish results to an Amazon SNS topic. Subscribe the call center agents to the SNS topic.

Correct Answer: B

Question #40 (Topic: Exam A)*Topic 1*

A media company is launching a platform that allows thousands of users every hour to upload images and text content. The platform uses Amazon Bedrock to process the uploaded content to generate creative compositions.
The company needs a solution to ensure that the platform does not process or produce inappropriate content. The platform must not expose personally identifiable information (PII) in the compositions. The solution must integrate with the company's existing Amazon S3 storage workflow.
Which solution will meet these requirements with the LEAST infrastructure management overhead?

A. Enable the Enhanced Monitoring tool. Use an Amazon CloudWatch alarm to filter traffic to the platform. Use Amazon Comprehend PII detection to pre-process the data. Create a CloudWatch alarm to monitor for Amazon Comprehend PII detection events. Create an AWS Step Functions workflow that includes an Amazon Rekognition image moderation step.
B. Use an Amazon API Gateway HTTP API with request validation templates to screen content before storing the uploaded content in Amazon S3. Use Amazon SageMaker AI to build custom content moderation models that process content before sending the processed content to Amazon Bedrock.
C. Create an Amazon Cognito user pool that uses pre-authentication AWS Lambda functions to run content moderation checks. Use Amazon Textract to filter text content and Amazon Rekognition to filter image content before allowing users to upload content to the platform.
D. Create an AWS Step Functions workflow that uses built-in Amazon Bedrock guardrails to filter content. Use Amazon Comprehend PII detection to pre-process the content. Use Amazon Rekognition image moderation.

Correct Answer: D

Question #41 (Topic: Exam A)*Topic 1*

A company has set up Amazon Q Developer Pro licenses for all developers at the company. The company maintains a list of approved resources that developers must use when developing applications. The approved resources include internal libraries, proprietary algorithmic techniques, and sample code with approved styling.
A new team of developers is using Amazon Q Developer to develop a new Java-based application. The company must ensure that the new developer team uses the company's approved resources. The company does not want to make project-level modifications.
Which solution will meet these requirements?

A. Create a Git repository that contains all of the approved internal libraries, algorithms, and code samples. Include this Git repository in the application project locally as part of the workspace. Ensure that the developers use the @workspace context to retrieve suggestions from the Git repository.
B. In the project root folder, create a folder named .amazonq/rules. Add the approved internal libraries, algorithms, and code samples to the folder.
C. Create a folder in the application project named rules. Store the guidelines and code in the folder for Amazon Q Developer to reference product code suggestions.
D. Create an Amazon Q Developer customization that includes the approved data sources. Ensure that the developers use the customization to develop the application.

Correct Answer: D

Question #42 (Topic: Exam A)*Topic 1*

An ecommerce company is using Amazon Bedrock to build a customer service AI assistant. The AI assistant needs to process over 50,000 customer inquiries every day. The AI assistant occasionally experiences traffic spikes of up to 150,000 inquiries every day during promotional events. Analysis shows that 40% of inquiries follow similar patterns that share the same context. A GenAI developer must design a solution that will ensure low latency and consistent performance for the AI assistant during traffic spikes.
Which solution will meet these requirements MOST cost-effectively?

A. Configure latency-optimized inference by setting the latency parameter to optimized in the performance configuration of the request to Amazon Bedrock. Use prompt caching to handle the repetitive inquiries.
B. Purchase provisioned throughput and model units (MUs) that are sized to handle peak traffic loads. Use Amazon ElastiCache (Redis OSS) to cache repetitive inquiries.
C. Use Amazon Bedrock Agents and custom knowledge bases to pre-process customer inquiries. Configure cross-Region inference to distribute traffic.
D. Use AWS Lambda functions to pre-process requests by using a custom prompt routing mechanism. Use Amazon DynamoDB as a caching layer to handle frequently asked questions.

Correct Answer: A

Question #43 (Topic: Exam A)*Topic 1*

A legal research company has a Retrieval Augmented Generation (RAG) application that uses Amazon Bedrock and Amazon OpenSearch Service. The application stores 768-dimensional vector embeddings for 15 million legal documents, including statutes, court rulings, and case summaries.
The company's current chunking strategy segments text into fixed-length blocks of 500 tokens. The current chunking strategy often splits contextually linked information such as legal arguments, court opinions, or statute references across separate chunks. Researchers report that generated outputs frequently omit key context or cite outdated legal information.
Recent application logs show a 40% increase in response times. The p95 latency metric exceeds 2 seconds. The company expects storage needs for the application to grow from 90 GB to 360 GB within a year.
The company needs a solution to improve retrieval relevance and system performance at scale.
Which solution will meet these requirements?

A. Increase the embedding vector dimensionality from 768 to 4,096 without changing the existing chunking or pre-processing strategy.
B. Replace dynamic retrieval with static, pre-written summaries that are stored in Amazon S3. Use Amazon CloudFront to serve the summaries to reduce compute demand and improve predictability.
C. Update the chunking strategy to use semantic boundaries such as complete legal arguments, clauses, or sections rather than fixed token limits. Regenerate vector embeddings to align with the new chunk structure.
D. Migrate from OpenSearch Service to Amazon DynamoDB. Implement keyword-based indexes to enable faster lookups for legal concepts.

Correct Answer: C

Requirement	How C Addresses It
Omitted key context	Semantic chunking keeps legally connected information together — a complete court opinion or statute reference stays in one chunk instead of being split mid-argument
Outdated citations	Properly bounded chunks ensure full context for each legal concept, enabling the model to distinguish current vs. superseded law

Improved retrieval relevance	Semantically coherent chunks produce **higher quality embeddings** — vectors better represent the actual meaning of the legal content
40% latency increase / p95 > 2s	Better embeddings = more relevant results in fewer retrieval passes, reducing unnecessary re-ranking and post-processing overhead
90 GB → 360 GB growth	Semantic chunks are typically more efficient — fewer redundant/fragmented chunks means a cleaner, more scalable index

Question #44 (Topic: Exam A)*Topic 1*

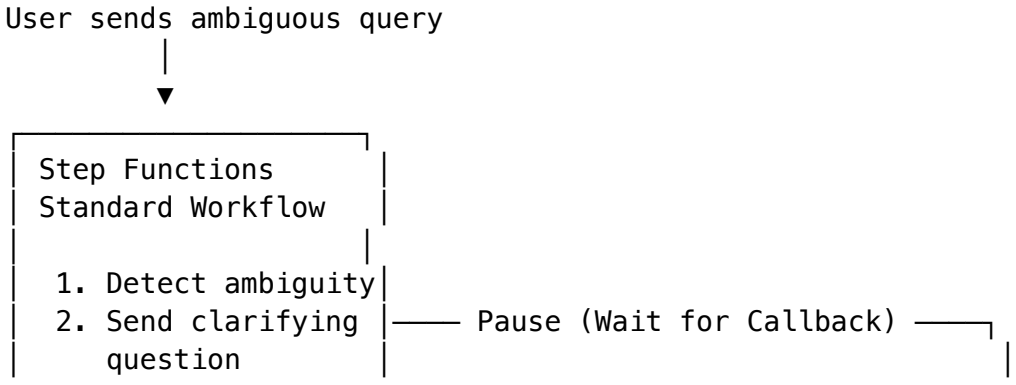
A company is developing a generative AI (GenAI)-powered customer support application that uses Amazon Bedrock foundation models (FMs). The application must maintain conversational context across multiple interactions with the same user. The application must run clarification workflows to handle ambiguous user queries. The company must store encrypted records of each user conversation to use for personalization. The application must be able to handle thousands of concurrent users while responding to each user quickly.
Which solution will meet these requirements?

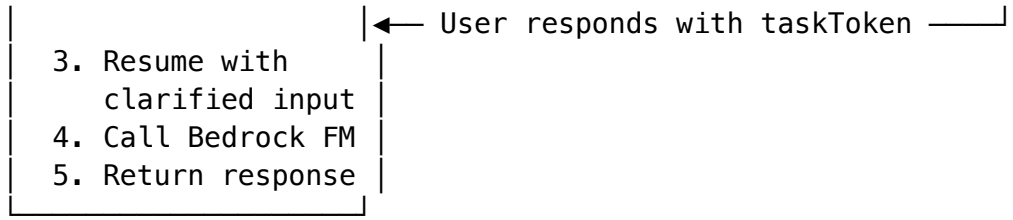
A. Use an AWS Step Functions Express workflow to orchestrate conversation flow. Invoke AWS Lambda functions to run clarification logic. Store conversation history in Amazon RDS and use session IDs as the primary key.
B. Use an AWS Step Functions Standard workflow to orchestrate clarification workflows. Include Wait for a Callback patterns to manage the workflows. Store conversation history in Amazon DynamoDPurchase on-demand capacity and configure server-side encryption.
C. Deploy the application by using an Amazon API Gateway REST API to route user requests to an AWS Lambda function to update and retrieve conversation context. Store conversation history in Amazon S3 and configure server-side encryption. Save each interaction as a separate JSON file.
D. Use AWS Lambda functions to call Amazon Bedrock inference APIs. Use Amazon SQS queues to orchestrate clarification steps. Store conversation history in an Amazon ElastiCache (Redis OSS) cluster. Configure encryption at rest.

Correct Answer: B

Requirement	How B Addresses It
Conversational context	DynamoDB provides **single-digit millisecond** key-value lookups by session/user ID — ideal for retrieving conversation history across interactions
Clarification workflows	Standard workflows support **Wait for Callback** (.waitForTaskToken) — the workflow pauses, sends a clarification prompt to the user, and resumes only when the user responds. Purpose-built for human-in-the-loop patterns
Encrypted conversation records	DynamoDB **server-side encryption** encrypts all data at rest using AWS KMS — meets the encryption requirement natively
Personalization	DynamoDB is a durable, persistent store — conversation history remains available for long-term personalization
Thousands of concurrent users	DynamoDB **on-demand capacity** auto-scales instantly with no capacity planning — handles spiky, high-concurrency workloads
Fast response times	DynamoDB single-digit ms latency + Standard workflows efficiently orchestrate steps

Clarification Workflow Pattern





Why Option C is wrong:

S3 with separate JSON files per interaction is high-latency for conversational context retrieval (S3 GET per turn). No orchestration mechanism for clarification workflows — API Gateway + Lambda alone can't pause and wait for user responses

Question #45 (Topic: Exam A)*Topic 1*

A financial services company needs to pre-process unstructured data such as customer transcripts, financial reports, and documentation. The company stores the unstructured data in Amazon S3 to support an Amazon Bedrock application.
The company must validate data quality, create auditable metadata, monitor data metrics, and customize text chunking to optimize foundation model (FM) performance.
Which solution will meet these requirements with the LEAST development effort?

A. Use Amazon SageMaker Data Wrangler to create a data flow. Configure Amazon CloudWatch metrics and alarms to monitor data quality. Use a custom AWS Lambda function to pre-process the data. Load processed data into Amazon Bedrock.
B. Set up an AWS Glue crawler to catalog data sources. Create AWS Glue ETL jobs to run custom transformation scripts. Use AWS Glue Data Quality to validate and monitor data quality. Load processed data into Amazon Bedrock.
C. Use Amazon Comprehend to extract entities. Create an AWS Lambda function to chunk text. Run Amazon Athena to query and validate data quality. Load processed data into Amazon Bedrock.
D. Create an AWS Step Functions workflow to orchestrate data pre-processing tasks. Run custom code on Amazon EC2 instances to process the data. Use Amazon SageMaker Model Monitor to monitor data quality. Load processed data into Amazon Bedrock.

Correct Answer: B

AWS Glue crawler to catalog → Glue ETL for transformations → Glue Data Quality for validation/monitoring → Load into Bedrock.

Requirement	How B Addresses It
Validate data quality	**AWS Glue Data Quality** is purpose-built for this — define rules, validate datasets, and flag quality issues automatically
Create auditable metadata	**Glue crawler** automatically catalogs data sources, creating a searchable, auditable metadata catalog in the Glue Data Catalog
Monitor data metrics	Glue Data Quality natively tracks and monitors quality metrics over time with built-in dashboards and CloudWatch integration
Customize text chunking	Glue ETL jobs support **custom transformation scripts** (Python/Spark) — full flexibility to implement domain-specific chunking strategies for financial documents
Least development effort	Glue provides an **end-to-end managed pipeline** — cataloging, transformation, and quality validation in one service. No custom infrastructure to manage

Why Option A is wrong: SageMaker Data Wrangler is focused on ML feature engineering, not unstructured text pre-processing. Custom Lambda for pre-processing adds development effort. CloudWatch alone doesn't provide structured data quality validation and auditable metadata

Question #46 (Topic: Exam A)*Topic 1*

A company uses Amazon Bedrock to build a Retrieval Augmented Generation (RAG) system. The RAG system uses an Amazon Bedrock knowledge base that is based on an Amazon S3 bucket as the data source for emergency news video content. The system retrieves transcripts, archived reports, and related documents from the S3 bucket.

The RAG system uses state-of-the-art embedding models and a high-performing retrieval setup. However, users report slow responses and irrelevant results, which cause decreased user satisfaction. The company notices that vector searches are evaluating too many documents across too many content types and over long periods of time.

The company determines that the underlying models will not benefit from additional fine tuning. The company must improve retrieval accuracy by applying smarter constraints. The company

wants a solution that requires minimal changes to the existing architecture.
Which solution will meet these requirements?

A. Enhance embeddings by using a domain-adapted model that is specifically trained on emergency news content for improved vector similarity.
B. Migrate to Amazon OpenSearch Service. Use vector fields and metadata filters to define the scope of results retrieval.
C. Enable metadata-aware filtering within the Amazon Bedrock knowledge base by indexing S3 object metadata.
D. Migrate to an Amazon Q Business index to perform structured metadata filtering and document categorization during retrieval.

Correct Answer: C

Question #47 (Topic: Exam A)*Topic 1*

A financial services company is creating a Retrieval Augmented Generation (RAG) application that uses Amazon Bedrock to generate summaries of market activities. The application relies on a vector database that stores a small proprietary dataset that has a low index count. The application must perform similarity searches. The Amazon Bedrock model's responses must maximize accuracy and maintain high performance.
The company needs to configure the vector database and integrate it with the application.
Which solution will meet these requirements?

A. Launch an Amazon MemoryDB cluster and configure the index by using the Flat algorithm. Configure a horizontal scaling policy based on performance metrics.
B. Launch an Amazon MemoryDB cluster and configure the index by using the Hierarchical Navigable Small World (HNSW) algorithm. Configure a vertical policy based on performance metrics.
C. Launch an Amazon Aurora PostgresSQL cluster and configure the index by using the Inverted File with Flat Compression (IVFFlat) algorithm. Configure the instance class to scale to a larger size when the load increases.
D. Launch an Amazon DocumentDB cluster that has an Inverted File with Flat Compression (IVFFlat) index and a high probe value. Configure connections to the cluster as a replica set Distribute reads to replica instances.

Correct Answer: A

Algorithm	Type	Accuracy	Best For
Flat	Exact Nearest Neighbor	**100%** (no approximation)	Small datasets — scans all vectors
HNSW	Approximate Nearest Neighbor	~95-99% (tunable)	Large datasets — trades accuracy for speed
IVFFlat	Approximate Nearest Neighbor	Variable (probe-dependent)	Medium-large datasets — clusters vectors first

Question #48 (Topic: Exam A)*Topic 1*

A GenAI developer is building a Retrieval Augmented Generation (RAG)-based customer support application that uses Amazon Bedrock foundation models (FMs). The application needs to process 50 GB of historical customer conversations that are stored in an Amazon S3 bucket as JSON files. The application must use the processed data as its retrieval corpus. The application's data processing workflow must extract relevant data from customer support documents, remove customer personally identifiable information (PII), and generate embeddings for vector storage. The processing workflow must be cost-effective and must finish within 4 hours.
Which solution will meet these requirements with the LEAST operational overhead?

A. Use AWS Lambda and Amazon Comprehend to process files in parallel, remove PII, and call Amazon Bedrock APIs to generate vectors. Configure Lambda concurrency limits and memory settings to optimize throughput.
B. Create an AWS Glue ETL job to run PII detection scripts on the data. Use Amazon SageMaker Processing to run the HuggingFaceProcessor to generate embeddings by using a pre-trained model. Store the embeddings in Amazon OpenSearch Service.
C. Deploy an Amazon EMR cluster that runs Apache Spark with user-defined functions (UDFs) that call Amazon Comprehend to detect PII. Use Amazon Bedrock APIs to generate vectors. Store outputs in Amazon Aurora PostgreSQL with the pgvector extension.
D. Implement a data processing pipeline that uses AWS Step Functions to orchestrate a workload that uses Amazon Comprehend to detect PII and Amazon Bedrock to generate embeddings. Directly integrate the workflow with Amazon OpenSearch Serverless to store vectors and provide similarity search capabilities.

Correct Answer: D

Question #49 (Topic: Exam A)*Topic 1*

A financial services company is developing a generative AI (GenAI) application that serves both premium customers and standard customers. The application uses AWS Lambda functions behind an Amazon API Gateway REST API to process requests. The company needs to dynamically switch between AI models based on which customer tier each user belongs to. The company also wants to perform A/B testing for new features without redeploying code. The company needs to validate model parameters like temperature and maximum token limits before applying changes.
Which solution will meet these requirements with the LEAST operational overhead?

A. Create an AWS Systems Manager Parameter Store parameters for each configuration. Use Lambda functions to poll for parameter updates. Use Amazon EventBridge events to trigger redeployments when configurations change.
B. Store model configurations in Amazon DynamoDB tables. Optimize access patterns to retrieve configurations according to customer tier. Configure Lambda functions to query DynamoDB at the beginning of each request to determine which model to use.
C. Use AWS AppConfig to manage model configurations. Use feature flags to perform A/B testing. Define JSON schema validation rules for model parameters. Configure Lambda functions to retrieve configurations by using the AWS AppConfig Agent.
D. Create an Amazon ElastiCache (Redis OSS) cluster to store model configurations. Set short TTL values. Run custom validation logic in Lambda functions. Use Amazon CloudWatch metrics to monitor configuration usage.

Correct Answer: C

Question #50 (Topic: Exam A)*Topic 1*

A company is using Amazon Bedrock and Anthropic Claude 3 Haiku to develop an AI assistant. The AI assistant normally processes 10,000 requests each hour but experiences surges of up 30,000 requests each hour during peak usage periods. The AI assistant must respond within 2 seconds while operating across multiple AWS Regions.
The company observes that during peak usage periods, the AI assistant experiences throughput bottlenecks that cause increased latency and occasional request timeouts. The company must resolve the performance issues.
Which solution will meet this requirement?

A. Purchase provisioned throughput and sufficient model units (MUs) in a single Region. Configure the application to retry failed requests with exponential backoff.
B. Implement token batching to reduce API overhead. Use cross-Region inference profiles to automatically distribute traffic across available Regions.
C. Set up auto scaling AWS Lambda functions in each Region. Implement client-side round-robin request distribution. Purchase one model unit (MU) of provisioned throughput as a backup.
D. Implement batch inference for all requests by using Amazon S3 buckets across multiple Regions. Use Amazon SQS to set up an asynchronous retrieval process.

Correct Answer: B

Question #51 (Topic: Exam A)*Topic 1*

A company uses Amazon Bedrock to develop an AI assistant to provide customer support. Analysis shows that 40% of customer queries use varied phrasing or wording to ask the same questions.
The company wants a solution to reduce redundant model calls. The solution must ensure that semantically equivalent questions receive consistent answers. The solution must ensure low latency.
Which solution will meet these requirements?

A. Deploy an Amazon DynamoDB Accelerator (DAX) cluster as an in-memory cache. Specify the query text as the partition key and the model response text as the sort key. Query the cache by using a filter expression with the LIKE operator.
B. Use Amazon Bedrock to generate embeddings from customer queries. Use Amazon MemoryDB for Valkey to store hash sets of vector embeddings and model responses. Use a RANGE query to find similar queries and their responses.
C. Deploy Amazon OpenSearch Service that has k-nearest neighbor (k-NN) capabilities to store query-response text pairs. Use an approximate k-NN technique to find similar queries and their responses.
D. Create a caching solution by using Amazon DynamoDB to create a global secondary index on the normalized query text. Apply stemming to incoming queries. Query the index of cached customer queries.

Correct Answer: C

Question #52 (Topic: Exam A)*Topic 1*

A company uses AWS Lambda functions to build an AI agent solution. A GenAI developer must set up a Model Context Protocol (MCP) server that accesses user information. The GenAI developer must also configure the AI agent to use the new MCP server. The GenAI developer must ensure that only authorized users can access the MCP server.
Which solution will meet these requirements?

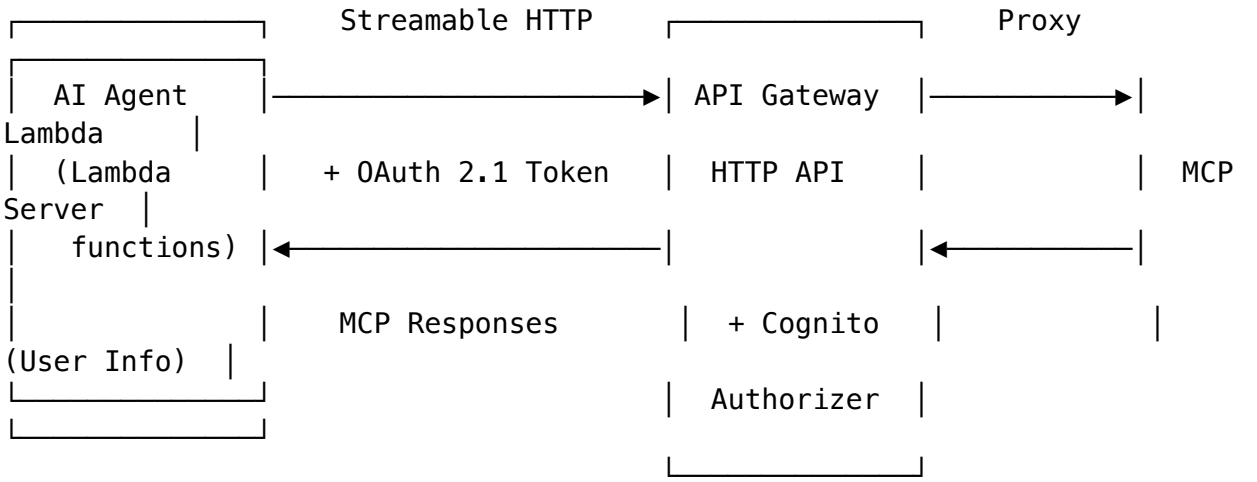
A. Use a Lambda function to host the MCP server. Grant the AI agent Lambda functions permission to invoke the Lambda function that hosts the MCP server. Configure the AI agent's MCP client to invoke the MCP server asynchronously.
B. Use a Lambda function to host the MCP server. Grant the AI agent Lambda functions permission to invoke the Lambda function that hosts the MCP server. Configure the AI agent to use the STDIO transport with the MCP server.
C. Use a Lambda function to host the MCP server. Create an Amazon API Gateway HTTP API that proxies requests to the Lambda function. Configure the AI agent solution to use the Streamable HTTP transport to make requests through the HTTP API. Use Amazon Cognito to enforce OAuth 2.1.
D. Use a Lambda layer to host the MCP server. Add the Lambda layer to the AI agent Lambda functions. Configure the agentic AI solution to use the STDIO transport to send requests to the MCP server. In the AI agent's MCP configuration, specify the Lambda layer ARN as the command. Specify the user credentials as environment variables.

Correct Answer: C

Requirement	How C Addresses It
Host MCP server	Lambda function runs the MCP server logic — serverless, scalable
AI agent accesses MCP server	API Gateway HTTP API provides a stable HTTP endpoint that proxies to the Lambda function
MCP transport protocol	**Streamable HTTP** is the MCP spec's standard transport for remote servers — purpose-built for client-server MCP communication over HTTP

Only authorized users	**Amazon Cognito with OAuth 2.1** is the MCP specification's recommended authorization framework — enforces token-based access control at the API layer

Architecture Flow



Question #53 (Topic: Exam A)*Topic 1*

A company wants to create an annual rewards program for its customers. The rewards that customers earn vary based on different parameters such as the categories of the items ordered and the customers' purchase history.

The company needs a generative AI (GenAI) solution that uses three Amazon Bedrock agents to help customers during online catalog browsing. The agents must use knowledge bases and action groups to handle the search, recommendation, and order modules. The modules must operate sequentially. An AWS Lambda function must calculate estimated rewards for each recommended item. The solution must provide graceful degradation during service disruptions.

Which solution will meet these requirements with the MOST operational efficiency?

A. Define an Amazon API Gateway REST API behind each agent. Create a second Lambda function to orchestrate the calls to the agents and the rewards Lambda function. Configure the second Lambda function with a retry/fallback mechanism.
B. Create an AWS Step Functions state machine with four tasks that run the agents and the rewards Lambda function. Set up retry and catch branches for each of the task steps.
C. Configure each agent with a separate retry/fallback mechanism. Create a second Lambda function to orchestrate the calls to the agents and the rewards Lambda function. Define an Amazon API Gateway REST API behind the second Lambda function.
D. Create a second Lambda function to orchestrate the calls to the agents and the rewards Lambda function. Create an AWS Step Functions state machine with one task that runs the second Lambda function. Set up retry and catch branches for the task step.

Correct Answer: B

Question #54 (Topic: Exam A)*Topic 1*

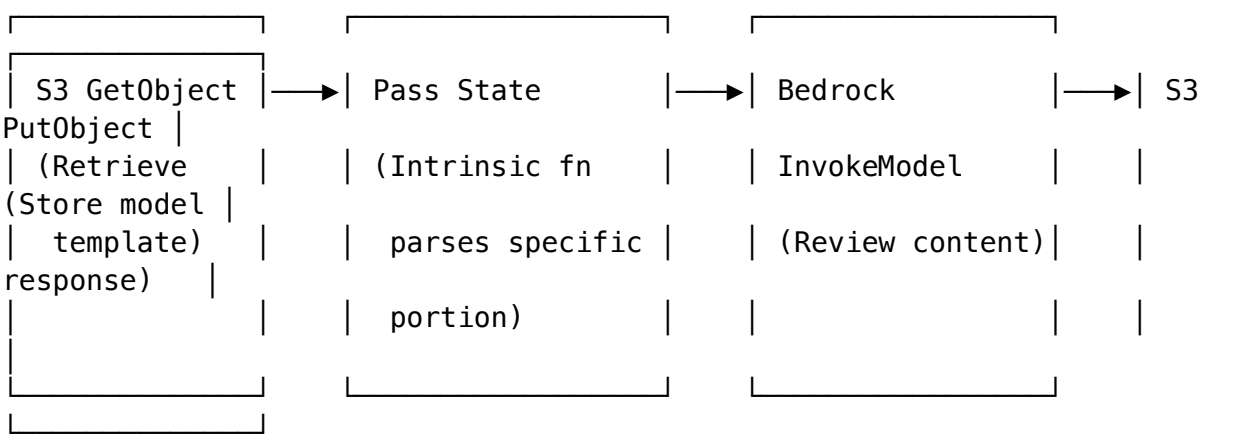
A company is creating a workflow to review customer-facing communications before the company sends the communications. The company uses a pre-defined message template to generate the communications and stores the communications in an Amazon S3 bucket. The workflow needs to capture a specific portion from the template and send it to an Amazon Bedrock model. The workflow must store model responses back to the original S3 bucket.
Which solution will meet these requirements?

A. Create a flow in Amazon Bedrock Flows. Configure S3 action nodes at the beginning and end of the flow to retrieve and store the communications and the model responses. In the middle of the flow, configure an expression to parse each communication. Configure an agent step to send the parsed input to the model for review.
B. Create an AWS Step Functions Express workflow state machine. Use an Amazon S3 integration GetObject step to retrieve the original communications. Use an intrinsic function Pass step to parse the communications and to pass the results to an Amazon Bedrock InvokeModel step. Configure an Amazon S3 integration PutObject step to store the model responses back to the S3 bucket.
C. Create an Amazon Bedrock agent that has an action group. Configure instructions to define how the agent should parse the communications. Configure the action group to retrieve the communications from the S3 bucket, invoke the Amazon Bedrock model, and store the model responses back to the S3 bucket.
D. Create an Amazon Bedrock agent that has a single action group. Configure three AWS Lambda functions in the action group. Configure the functions to retrieve the communications from the S3

bucket, parse the communications and invoke the Amazon Bedrock model, and store the model responses back to the S3 bucket.

Correct Answer: B

Workflow Flow



Requirement	How B Addresses It
Retrieve communications from S3	Native **S3 SDK integration** with GetObject step — no Lambda needed
Capture a specific portion from template	**Intrinsic functions** in a Pass state can parse, extract, and transform JSON data deterministically — perfect for extracting specific template sections
Send to Bedrock model	Native **Bedrock InvokeModel** SDK integration — direct service call, no intermediary
Store responses back to S3	Native **S3 PutObject** integration writes results back to the original bucket

Workflow orchestration	Express workflows handle short-duration, high-volume sequential processing efficiently

`Why option A is not correct:` Bedrock Flows does not have native S3 action nodes for reading/writing objects or expression nodes for deterministic template parsing. It's designed for FM-centric orchestration, not file-level S3 operations

Question #55 (Topic: Exam A)*Topic 1*

A financial technology company is using Amazon Bedrock to build an assessment system for the company's customer service AI assistant. The AI assistant must provide financial recommendations that are factually accurate, compliant with financial regulations, and conversationally appropriate. The company needs to combine automated quality evaluations at scale with targeted human reviews of critical interactions.
What solution will meet these requirements?

A. Configure a pipeline in which financial experts manually score all responses for accuracy, compliance, and conversational quality. Use Amazon SageMaker notebooks to analyze results to identify improvement areas.
B. Configure Amazon Bedrock evaluations that use Anthropic Claude Sonnet as a judge model to assess response accuracy and appropriateness. Configure custom Amazon Bedrock guardrails to check responses for compliance with financial policies. Add Amazon Augmented AI (Amazon A2I) human reviews for flagged critical interactions.
C. Create an Amazon Lex bot to manage the customer service interactions. Configure AWS Lambda functions to check responses against a static compliance database. Configure intents in the bot that call the Lambda functions to check the responses. Add an additional intent to collect end-user reviews.
D. Configure Amazon CloudWatch to monitor response patterns from the AI assistant. Configure CloudWatch alerts for potential compliance violations. Establish a team of human evaluators to review flagged interactions.

Correct Answer: B

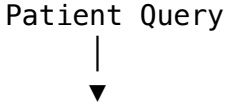
Question #56 (Topic: Exam A)*Topic 1*

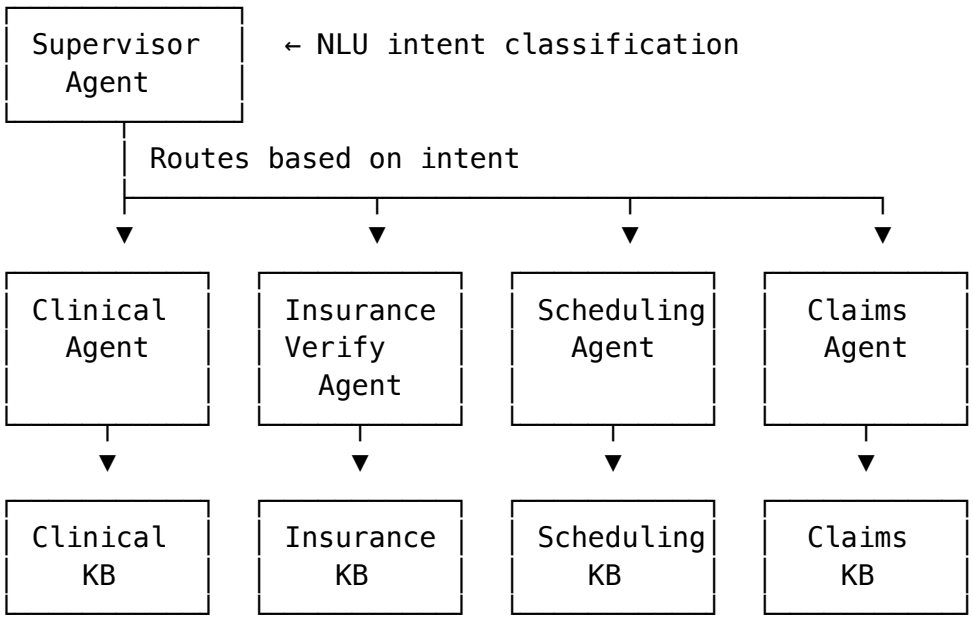
A healthcare company is using Amazon Bedrock to develop a real-time patient care AI assistant to respond to queries for separate departments that handle clinical inquiries, insurance verification, appointment scheduling, and insurance claims. The company wants to use a multi-agent architecture.
The company must ensure that the AI assistant is scalable and can onboard new features for patients. The AI assistant must be able to handle thousands of parallel patient interactions. The company must ensure that patients receive appropriate domain-specific responses to queries.
Which solution will meet these requirements?

A. Isolate data for each agent by using separate knowledge bases. Use IAM filtering to control access to each knowledge base. Deploy a supervisor agent to perform natural language intent classification on patient inquiries. Configure the supervisor agent to route queries to specialized collaborator agents to respond to department-specific queries. Configure each specialized collaborator agent to use Retrieval Augmented Generation (RAG) with the agent's department-specific knowledge base.
B. Create a separate supervisor agent for each department. Configure individual collaborator agents to perform natural language intent classification for each specialty domain within each department. Integrate each collaborator agent with department-specific knowledge bases only. Implement manual handoff processes between the supervisor agents.
C. Isolate data for each department in separate knowledge bases. Use IAM filtering to control access to each knowledge base. Deploy a single general-purpose agent. Configure multiple action groups within the general-purpose agent to perform specific department functions. Implement rule-based routing logic within the general-purpose agent instructions.
D. Implement multiple independent supervisor agents that run in parallel to respond to patient inquiries for each department. Configure multiple collaborator agents for each supervisor agent. Integrate all agents with the same knowledge base. Use external routing logic to merge responses from multiple supervisor agents.

Correct Answer: A

Architecture Flow





Question #57 (Topic: Exam A)*Topic 1*

A company uses an AI assistant application to summarize the company's website content and provide information to customers. The company plans to use Amazon Bedrock to give the application access to a foundation model (FM).

The company needs to deploy the AI assistant application to a development environment and a production environment. The solution must integrate the environments with the FM. The company wants to test the effectiveness of various FMs in each environment. The solution must provide product owners with the ability to easily switch between FMs for testing purposes in each environment.

Which solution will meet these requirements?

A. Create one AWS CDK application. Create multiple pipelines in AWS CodePipeline. Configure each pipeline to have its own settings for each FM. Configure the application to invoke the Amazon Bedrock FMs by using the aws_bedrock.ProvisionedModel.fromProvisionedModelArn() method.

B. Create a separate AWS CDK application for each environment. Configure the applications to invoke the Amazon Bedrock FMs by using the aws_bedrock.FoundationModel.fromFoundationModelId() method. Create a separate pipeline in AWS CodePipeline for each environment.

C. Create one AWS CDK application. Configure the application to invoke the Amazon Bedrock FMs by using the aws_bedrock.FoundationModel.fromFoundationModelId() method. Create a pipeline in AWS CodePipeline pipeline that has a deployment stage for each environment that uses AWS CodeBuild deploy actions.

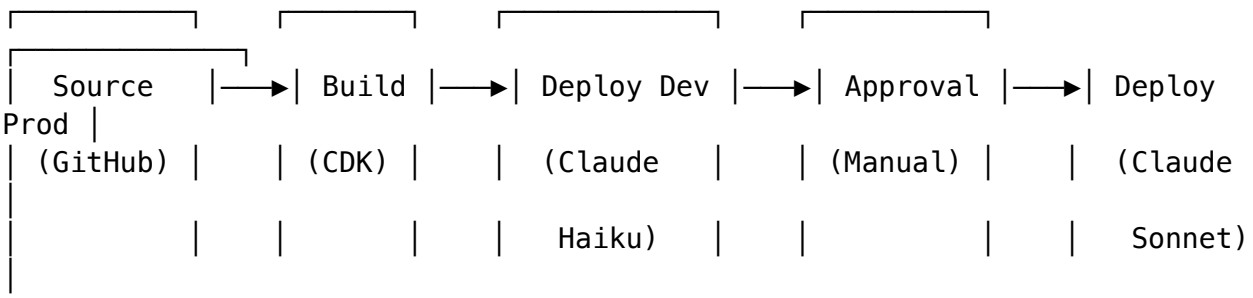
D. Create one AWS CDK application for the production environment. Configure the application to invoke the Amazon Bedrock FMs by using the

aws_bedrock.ProvisionedModel.fromProvisionedModelArn() method. Create a pipeline in AWS CodePipeline. Configure the pipeline to deploy to the production environment by using an AWS CodeBuild deploy action. For the development environment, manually recreate the resources by referring to the production application code.

Correct Answer: C

Requirement	How C Addresses It
Dev and production environments	Single CodePipeline with **separate deployment stages** for each environment — clean, consistent CI/CD
Easily switch between FMs	fromFoundationModelId() references on-demand FMs by a simple **model ID string** (e.g., anthropic.claude-3-sonnet). Switching models is just a parameter change — no infrastructure reprovisioning needed
Test various FMs	Each environment stage can use different model IDs via **environment-specific configuration/context variables**, letting product owners swap FMs without code changes
Single codebase	One CDK app ensures consistency across environments — DRY principle, easy maintenance

Visual Pipeline Flow



Question #58 (Topic: Exam A)*Topic 1*

A hotel company wants to enhance a legacy Java-based property management system (PMS) by adding AI capabilities. The company wants to use Amazon Bedrock Knowledge Bases to provide staff with room availability information and hotel-specific details. The solution must maintain separate access controls for each hotel that the company manages. The solution must provide room availability information in near real time and must maintain consistent performance during peak usage periods.
Which solution will meet these requirements?

A. Deploy a single Amazon Bedrock knowledge base that contains combined data for all hotels. Configure AWS Lambda functions to synchronize data from each hotel's PMS database through direct API connections. Implement AWS CloudTrail logging with hotel-specific filters to audit access logs for each hotel's data.
B. Create an Amazon EventBridge rule for each hotel that is invoked by changes to the PMS database for each hotel. Configure the rule to send updates to a centralized Amazon Bedrock knowledge base in a management AWS account. Configure resource-based policies to enforce hotel-specific access controls for hotel staff.
C. Implement one Amazon Bedrock knowledge base for each hotel in a multi-account structure. Use direct data ingestion to provide real-time room availability information. Schedule regular synchronization for less critical information.
D. Build a centralized Amazon Bedrock agent that uses multiple knowledge bases. Implement AWS IAM Identity Center with hotel-specific permission sets to control hotel staff data access.

Correct Answer: C

Question #59 (Topic: Exam A)*Topic 1*

A company is implementing a serverless inference API by using AWS Lambda. The API will dynamically invoke multiple AI models hosted on Amazon Bedrock. The company needs to design a solution that can switch between model providers without modifying or redeploying Lambda code in real time. The design must include safe rollout of configuration changes and validation and rollback capabilities.
Which solution will meet these requirements?

A. Store the active model provider in AWS Systems Manager Parameter Store. Configure a Lambda function to read the parameter at runtime to determine which model to invoke.
B. Store the active model provider in AWS AppConfig. Configure a Lambda function to read the configuration at runtime to determine which model to invoke.
C. Configure an Amazon API Gateway REST API to route requests to separate Lambda functions. Hardcode each Lambda function to a specific model provider. Switch the integration target manually.
D. Store the active model provider in a JSON file hosted on Amazon S3. Use AWS AppConfig to reference the S3 file as a hosted configuration source. Configure a Lambda function to read the file through AppConfig at runtime to determine which model to invoke.

Correct Answer: B

Question #60 (Topic: Exam A)*Topic 1*

A company is building a generative AI (GenAI) application that uses Amazon Bedrock APIs to process complex customer inquiries. During peak usage periods, the application experiences intermittent API timeouts that cause issues such as broken response chunks and delayed data delivery. The application struggles to ensure that prompts remain within token limits when handling complex customer inquiries of varying lengths. Users have reported truncated inputs and incomplete responses. The company has also observed foundation model (FM) invocation failures. The company needs a retry strategy that automatically handles transient service errors and prevents overwhelming Amazon Bedrock during peak usage periods. The strategy must also adapt to changing service availability and support response streaming and token-aware request handling. Which solution will meet these requirements?

A. Implement a standard retry strategy that uses a 1-second fixed delay between attempts and a 3-retry maximum for all errors. Handle streaming response timeouts by restarting streams. Cap token usage for each session.
B. Implement an adaptive retry strategy that uses exponential backoff with jitter and a circuit breaker pattern that temporarily disables retries when error rates exceed a predefined threshold. Implement a streaming response handler that monitors for chunk delivery timeouts. Configure the handler to buffer successfully received chunks and intelligently resume streaming from the last received chunk when connections are re-established.
C. Use the AWS SDK to configure a retry strategy in standard mode. Wrap Amazon Bedrock API calls in try-catch blocks that handle timeout exceptions. Return cached completions for failed streaming requests. Enforce a global token limit for all users. Add jitter-based retry logic and lightweight token trimming for each request. Resume broken streams by requesting only the

missing chunks from the point of failure. Maintain a small in-memory buffer of the most recent chunks to minimize redundant data transfer.
D. Set Amazon Bedrock client request timeouts to 30 seconds. Implement client-side load shedding. Buffer partial results and stop new requests when the application performance begins to degrade. Set static token usage caps for all requests. Configure exponential backoff retries, dynamic chunk sizing, and context-aware token limits.
Correct Answer: B

Question #61 (Topic: Exam A)*Topic 1*

A bank is developing a generative AI (GenAI)-powered AI assistant that uses Amazon Bedrock to assist the bank's website users with account inquiries and financial guidance. The bank must ensure that the AI assistant does not reveal any personally identifiable information (PII) in customer interactions.
The AI assistant must not send PII in prompts to the GenAI model. The AI assistant must not respond to customer requests to provide investment advice. The bank must collect audit logs of all customer interactions, including any images or documents that are transmitted during customer interactions.
Which solution will meet these requirements with the LEAST operational effort?

A. Use Amazon Macie to detect and redact PII in user inputs and in the model responses. Apply prompt engineering techniques to force the model to avoid investment advice topics. Use AWS CloudTrail to capture conversation logs.
B. Use an AWS Lambda function and Amazon Comprehend to detect and redact PII. Use Amazon Comprehend topic modeling to prevent the AI assistant from discussing investment advice topics. Set up custom metrics in Amazon CloudWatch to capture customer conversations.
C. Configure Amazon Bedrock guardrails to apply a sensitive information policy to detect and filter PII. Set up a topic policy to ensure that the AI assistant avoids investment advice topics. Use the Converse API to log model invocations. Enable delivery and image logging to Amazon S3.
D. Use regex controls to match patterns for PII. Apply prompt engineering techniques to avoid returning PII or investment advice topics to customers. Enable model invocation logging, delivery logging, and image logging to Amazon S3.

Correct Answer: C

Question #62 (Topic: Exam A)*Topic 1*

A financial services company is developing a customer service AI assistant application that uses a foundation model (FM) in Amazon Bedrock. The application must provide transparent responses by documenting reasoning and by citing sources that are used for Retrieval Augmented Generation (RAG). The application must capture comprehensive audit trails for all responses to users. The application must be able to serve up to 10,000 concurrent users and must respond to each customer inquiry within 2 seconds.
Which solution will meet these requirements with the LEAST operational overhead?

A. Enable tracing for Amazon Bedrock agents. Configure structured prompts that direct the FM to provide evidence presentations. Integrate Amazon Bedrock knowledge bases with data sources to enable RAG. Configure the application to reference and cite authoritative content. Deploy the application in a Multi-AZ architecture. Use Amazon API Gateway and AWS Lambda functions to scale the application. Use Amazon CloudFront to provide low-latency delivery.
B. Enable tracing for Amazon Bedrock agents. Integrate a custom RAG pipeline with Amazon OpenSearch Service to retrieve and cite sources. Configure structured prompts to present retrieved evidence. Deploy the application behind an Amazon API Gateway REST API. Use AWS Lambda functions and Amazon CloudFront to scale the application and to provide low latency. Store logs in Amazon S3 and use AWS CloudTrail to capture audit trails.
C. Use Amazon CloudWatch to monitor latency and error rates. Embed model prompts directly in the application backend to cite sources. Store application interactions with users in Amazon RDS for audits.
D. Store generated responses and supporting evidence in an Amazon S3 bucket. Enable versioning on the bucket for audits. Use AWS Glue to catalog retrieved documents. Process the retrieved documents in Amazon Athena to generate periodic compliance reports.

Correct Answer: A

Question #63 (Topic: Exam A)*Topic 1*

A healthcare company is developing a document management system that stores medical research papers in an Amazon S3 bucket. The company needs to build a comprehensive metadata framework that will improve search precision for a generative AI (GenAI) application that analyzes the research papers. The metadata framework must include document timestamps, author information, and research domain classifications.
The solution must maintain a consistent metadata structure across all uploaded documents. The solution must give foundation models (FMs) the ability to understand document context without accessing the full content.
Which solution will meet these requirements?

A. Store document timestamps in Amazon S3 system metadata. Use S3 object tags to implement domain classification. Implement custom user-defined metadata to store author information.
B. Set up S3 Object Lock with legal holds to track document timestamps. Use S3 object tags to store author information. Implement S3 access points for domain classification.
C. Use S3 Inventory reports to track document timestamps. Create S3 access points to implement domain classification. Store author information in S3 Storage Lens dashboards.
D. Use custom user-defined metadata to store author information. Use S3 Object Lock retention periods to track document timestamps. Use S3 Event Notifications to implement domain classification.

Correct Answer: A

Question #64 (Topic: Exam A)*Topic 1*

Example Corp provides a personalized video generation service that millions of enterprise customers use. Customers generate marketing videos by submitting prompts to the company's proprietary generative AI (GenAI) model. To improve output relevance and personalization, Example Corp wants to enhance the prompts by using customer-specific context such as product preferences, customer attributes, and business history.
The customers have strict data governance requirements. The customers must retain full ownership and control over their own data. The customers do not require real-time access. However, semantic accuracy must be high and retrieval latency must remain low to support customer experience use cases.
Example Corp wants to minimize architectural complexity in its integration pattern. Example Corp does not want to deploy and manage services in each customer's environment unless necessary.
Which solution will meet these requirements?

A. Ensure that each customer sets up an Amazon Q Business index that includes the customer's internal data. Ensure that each customer designates Example Corp as a data accessor to allow Example Corp to retrieve relevant content by using a secure API to enrich prompts at runtime.
B. Use federated search with Model Context Protocol (MCP) by deploying real-time MCP servers for each customer. Retrieve data in real time during prompt generation.
C. Ensure that each customer configures an Amazon Bedrock knowledge base. Allow cross-account querying so Example Corp can retrieve structured data for prompt augmentation.
D. Configure Amazon Kendra to crawl customer data sources. Share the resulting indexes across accounts so Example Corp can query each customer's Amazon Kendra index to retrieve augmentation data.

Correct Answer: A

Question #65 (Topic: Exam A)*Topic 1*

A company is building a legal research AI assistant that uses Amazon Bedrock with an Anthropic Claude foundation model (FM). The AI assistant must retrieve highly relevant case law documents to augment the FM's responses. The AI assistant must identify semantic relationships between legal concepts, specific legal terminology, and citations. The AI assistant must perform quickly and return precise results.
Which solution will meet these requirements?

A. Configure an Amazon Bedrock knowledge base to use a default vector search configuration. Use Amazon Bedrock to expand queries to improve retrieval for legal documents based on specific terminology and citations.
B. Use Amazon OpenSearch service to deploy a hybrid search architecture that combines vector search with keyword search. Apply an Amazon Bedrock reranker model to optimize result relevance.
C. Enable the Amazon Kendra query suggestion feature for end users. Use Amazon Bedrock to perform post-processing of search results to identify semantic similarity in the documents and to produce precise results.
D. Use Amazon OpenSearch Service with vector search and Amazon Bedrock Titan embeddings to index and search legal documents. Use custom AWS Lambda functions to merge results with keyword-based filters that are stored in an Amazon RDS database.

Correct Answer: B

Question #66 (Topic: Exam A)*Topic 1*

A company deploys multiple Amazon Bedrock based generative AI (GenAI) applications across multiple business units for customer service, content generation, and document analysis. Some applications show unpredictable token consumption patterns. The company requires a comprehensive observability solution that provides real-time visibility into token usage patterns across multiple models. The observability solution must support custom dashboards for multiple stakeholder groups and provide alerting capabilities for token consumption across all the foundational models that the company's applications use.

Which combination of solutions will meet these requirements with the LEAST operational overhead? (Choose two.)

A. Use Amazon CloudWatch metrics as data sources to create custom Amazon QuickSight dashboards that show token usage trends and usage patterns across FMs.
B. Use Amazon CloudWatch Logs Insights to analyze Amazon Bedrock invocation logs for token consumption patterns and usage attribution by application. Create custom queries to identify high-usage scenarios. Add log widgets to dashboards to enable continuous monitoring.
C. Create custom Amazon CloudWatch dashboards that combine native Amazon Bedrock token and invocation CloudWatch metrics. Set up CloudWatch alarms to monitor token usage thresholds.
D. Create dashboards that show token usage trends and patterns across the company's FMs by using an Amazon Bedrock zero-ETL integration with Amazon Managed Grafana.
E. Implement Amazon EventBridge rules to capture Amazon Bedrock model invocation events. Route token usage data to an Amazon Data Firehose delivery stream that targets Amazon OpenSearch Serverless. Use OpenSearch dashboards to analyze usage patterns.

Correct Answer: BC

Question #67 (Topic: Exam A)*Topic 1*

A company is designing a solution that uses foundation models (FMs) to support multiple AI workloads. Some FMs must be invoked on demand and in real time. Other FMs require consistent high-throughput access for batch processing.
The solution must support hybrid deployment patterns and run workloads across cloud infrastructure and on-premises infrastructure to comply with data residency and compliance requirements.
Which combination of steps will meet these requirements? (Choose two.)

A. Use AWS Lambda to orchestrate low-latency FM inference by invoking FMs hosted on Amazon SageMaker AI asynchronous endpoints.
B. Configure provisioned throughput in Amazon Bedrock to ensure consistent performance for high-volume workloads.
C. Deploy FMs to Amazon SageMaker AI endpoints with support for edge deployment by using Amazon SageMaker Neo. Orchestrate the FMs by using AWS Lambda to support hybrid deployment.
D. Use Amazon Bedrock with auto-scaling to handle unpredictable traffic surges.
E. Use Amazon SageMaker JumpStart to host and invoke the FMs.

Correct Answer: BC

Question #68 (Topic: Exam A)*Topic 1*

A company is planning to deploy multiple generative AI (GenAI) applications to five independent business units that operate in multiple countries in Europe and the Americas. Each application uses Amazon Bedrock Retrieval Augmented Generation (RAG) patterns with business unit-specific knowledge bases that store terabytes of unstructured data.
The company must establish well-architected, standardized components for security controls, observability practices, and deployment patterns across all the GenAI applications. The components must be reusable, versioned, and governed consistently.
Which solution will meet these requirements?

A. Configure Amazon API Gateway REST API endpoints for the GenAI applications. Deploy common security, observability, and RAG patterns based on the AWS Well-Architected Generative AI Lens in standardized AWS CloudFormation templates. Use CloudFormation Guard after the deployment to validate policy compliance in each business unit.
B. Create standardized AWS CloudFormation templates to implement security, observability, and RAG patterns based on the AWS Well-Architected Generative AI Lens. Establish a centralized repository that performs version control. Integrate a CI/CD pipeline with CloudFormation Guard to enforce consistent and repeatable deployments across business units.
C. Use AWS Service Catalog to define standardized portfolios and versioned products for each business unit. Use the portfolios to enforce security, observability, and RAG patterns based on the AWS Well-Architected Generative AI Lens. Require the business units to use the Service Catalog console to deploy resources.
D. Document security controls, observability requirements, and RAG patterns based on the AWS Well-Architected Generative AI Lens in a shared design document. Use Amazon Macie to enforce deployment. Delegate implementation responsibility to each business unit.

Correct Answer: B

Question #69 (Topic: Exam A)*Topic 1*

A company upgraded its Amazon Bedrock powered foundation model (FM) that supports a multilingual customer service assistant. After the upgrade, the assistant exhibited inconsistent behavior across languages. The assistant began generating different responses in some languages when presented with identical questions.

The company needs a solution to detect and address similar problems for future updates. The evaluation must be completed within 45 minutes for all supported languages. The evaluation must process at least 15,000 test conversations in parallel. The evaluation process must be fully automated and integrated into the CI/CD pipeline. The solution must block deployment if quality thresholds are not met.
Which solution will meet these requirements?

A. Create a distributed traffic simulation framework that sends translation-heavy workloads to the assistant in multiple languages simultaneously. Use Amazon CloudWatch metrics to monitor latency, concurrency, and throughput. Run simulations before production releases to identify infrastructure bottlenecks.
B. Deploy the assistant in multiple AWS Regions with Amazon Route 53 latency-based routing and AWS Global Accelerator to improve global performance. Store multilingual conversation logs in Amazon S3. Perform weekly post-deployment audits to review consistency.
C. Create a pre-processing pipeline that normalizes all incoming messages into a consistent format before sending the messages to the assistant. Apply rule-based checks to flag potential hallucinations in the outputs. Focus the evaluation on the normalized text to simplify testing across languages.
D. Set up standardized multilingual test conversations with identical meaning. Run the test conversations in parallel by using Amazon Bedrock model evaluation jobs. Apply similarity and hallucination thresholds. Integrate the process into the CI/CD pipeline to block releases that fail.

Correct Answer: D

Question #70*Topic 1*

A company is developing a generative AI (GenAI) application that uses Amazon Bedrock foundation models (FMs). The application has several custom tool integrations. The application has experienced unexpected token consumption surges despite consistent user traffic.
The company needs a solution that uses Amazon Bedrock model invocation logging to monitor InputTokenCount metrics and OutputTokenCount metrics. The solution must detect unusual patterns in tool usage and identify which specific tool integrations cause abnormal token consumption. The solution must also automatically adjust thresholds as traffic patterns change.
Which solution will meet these requirements?

A. Use Amazon CloudWatch Logs to capture model invocation logs. Create CloudWatch dashboards based on InputTokenCount metrics and OutputTokenCount metrics. Configure static CloudWatch alarms with fixed thresholds for each tool integration.

B. Store model invocation logs in an Amazon S3 bucket. Use AWS Glue to catalog the logs. Analyze token consumption patterns by using scheduled Amazon Athena queries that generate reports on tool usage trends.
C. Use Amazon CloudWatch Logs to capture model invocation logs. Create CloudWatch metric filters to extract tool-specific invocation patterns. Apply CloudWatch anomaly detection alarms that adjust baselines for each tool's metrics.
D. Store model invocation logs in an Amazon S3 bucket. Create an AWS Lambda function to process logs in real time. Manually update Amazon CloudWatch alarm thresholds based on token consumption trends that the Lambda function identifies.

Correct Answer: *C*

Question #71 *Topic 1*

A company is using Amazon Bedrock to develop an AI-powered application that uses a foundation model (FM) that supports cross-Region inference and provisioned throughput. The application must serve users in Europe and North America with consistently low latency. The application must comply with data residency regulations that require European user data to remain within Europe-based AWS Regions.
During testing, the application experiences service degradation when Regional traffic spikes reach service quotas. The company needs a solution that maintains application resilience and minimizes operational complexity.
Which solution will meet these requirements?

A. Deploy separate Amazon Bedrock instances in North American and European Regions. Use a custom routing layer that directs traffic based on user location. Configure Amazon CloudWatch alarms to monitor Regional service usage. Use Amazon SNS to send email alerts to the company when usage approaches specified thresholds.
B. Use Amazon Bedrock cross-Region inference profiles by specifying geographical codes in profile IDs when the application calls the InvokeModel API. Configure separate Amazon API Gateway HTTP APIs to direct European and North American users to the appropriate Regional endpoints.
C. Deploy a multi-Region Amazon API Gateway HTTP API and AWS Lambda functions that implement retry logic to handle throttling. Configure the Lambda functions to call the FM in the nearest secondary Region when the application reaches service quotas in the primary Region. Use intelligent routing to ensure compliance with data residency requirements.

D. Configure provisioned throughput for Amazon Bedrock in multiple Regions. Implement failover logic in the application code to switch between Regions when throttling occurs. Use AWS Global Accelerator to route traffic to the appropriate endpoints based on user location.

Correct Answer: *B*

Requirement	How B solves it
European data residency	Geographic codes like eu.* in the inference profile ID guarantee that **all European inference stays within European Regions** — built into the profile, not custom logic
Low latency (EU + NA)	Separate API Gateway HTTP APIs route European users to eu.* profiles and North American users to us.* profiles — each geography served locally
Traffic spike resilience	Cross-Region inference profiles **automatically distribute load** across multiple Regions within the geography — if eu-west-1 hits quota, traffic routes to eu-central-1 seamlessly
Minimize operational complexity	**Two configurations**: geographic inference profiles + API Gateway routing. No custom failover code, no retry logic, no manual monitoring

Question #72*Topic 1*

An international company is building an AI assistant that uses RAG. The company wants the AI assistant to have near real-time, low-latency performance. The AI assistant must provide service to several geographic areas. The company's customers will use proprietary data with the AI assistant. The proprietary data must not leave the company's immediate geographic area.
Which solution will meet these requirements?

A. Deploy an Amazon Bedrock model with a cross-Region model inference profile. Create Amazon S3 buckets in each AWS Region the company operates in. Store a knowledge base in each

respective S3 bucket. In each Region, configure Amazon Kendra to interact with the respective knowledge base. In each Region, configure an AWS Lambda function that uses Kendra and Amazon Bedrock to process AI assistant prompts.
B. Deploy an Amazon Bedrock model in each AWS Region the company operates in. Configure an Amazon Bedrock cross-Region model inference profile. Configure a vector database that uses Amazon Bedrock Knowledge Bases. Store the knowledge bases in Amazon S3 in each Region the company operates in.
C. Use AWS Outposts to deploy an outpost in each AWS Region the company operates in. Create Amazon S3 buckets to store knowledge bases in each corresponding Region. Deploy Amazon RDS configured as a vector database to each outpost. Deploy an Amazon Bedrock model with a cross-Region inference profile in a central Region.
D. Configure a knowledge base stored in the Amazon S3 Express One Zone storage class in each AWS Local Zone the company operates in. Use Amazon RDS to deploy a vector database in each Local Zone the company operates in. Deploy a large language model (LLM) to Amazon EC2 instances in each Local Zone. Configure the AI assistant to route prompts to the model in the respective Local Zone.

Correct Answer: *B* 📦

Requirement	How B solves it
Data residency	Knowledge bases stored in **S3 in each Region** — proprietary data never leaves its geographic area. Vector databases via Bedrock Knowledge Bases are also Regional
Near real-time, low latency	Bedrock model available in each Region + **cross-Region inference profile** provides geographic routing with automatic failover — users always hit the nearest available model
Multiple geographic areas	Each Region has its own S3 bucket + Knowledge Base + vector database — complete RAG stack per geography
RAG architecture	**Bedrock Knowledge Bases** with vector database is the native, managed RAG solution — retrieval and generation are integrated
Geographic inference boundary	Cross-Region inference profiles (e.g., us.*, eu.*) guarantee inference stays **within the geographic boundary** — proprietary data in prompts doesn't leave the continent

Question #73 *Topic 1*

A company is building a generative AI (GenAI) application that processes financial reports and provides summaries for analysts. The application must run two compute environments. In one environment, AWS Lambda function must use the Python SDK to analyze reports on demand. In the second environment, Amazon EKS containers must use the JavaScript SDK to batch process multiple reports on a schedule. The application must maintain conversational context throughout multi-tum interactions, use the same foundation model (FM) across environments, and ensure consistent authentication.
Which solution will meet these requirements?

A. Use the Amazon Bedrock InvokeModel API with a separate authentication method for each environment. Store conversation states in Amazon DynamoDB. Use custom I/O formatting logic for each programming language.
B. Use the Amazon Bedrock Converse API directly in both environments with a common authentication mechanism that uses IAM roles. Store conversation states in Amazon ElastiCache. Creating programming language-specific wrappers for model parameters.
C. Create a centralized Amazon API Gateway REST API endpoint that handles all model interactions by using the InvokeModel API. Store interaction history in application process memory in each Lambda function or EKS container. Use environment variables to configure model parameters.
D. Use the Amazon Bedrock Converse API and IAM roles for authentication. Pass previous messages in the request messages array to maintain conversational context. Use programming language-specific SDKs to establish consistent API interfaces.

Correct Answer: D

Requirement	How D solves it
Two environments (Python + JavaScript)	AWS **SDKs** provide the same Converse API interface in both languages — boto3 (Python/Lambda) and @aws-sdk/client-bedrock-runtime (JavaScript/EKS). Identical API structure, native language implementation

Conversational context	Converse API's **messages array** natively carries conversation history — pass previous turns in each request. No external state store needed
Same FM across environments	Converse API uses a **unified, model-agnostic interface** — same request/response format regardless of which FM is selected. Switch models without code changes
Consistent authentication	**IAM roles** for both environments — Lambda execution role and EKS pod IAM role (via IRSA/Pod Identity). Same authentication mechanism everywhere

Question #74*Topic 1*

A company is building a serverless application that uses AWS Lambda functions to help students around the world summarize notes. The application uses Anthropic Claude through Amazon Bedrock. The company observed that most of the traffic occurs during evenings in each time zone. Users report experiencing throttling errors during peak usage times in their times zones.
The company needs to resolve the throttling issues by ensuring continuous operation of the application. The solution must maintain application performance quality. The company needs a solution that does not require a fixed hourly cost during low traffic periods.
Which solution will meet these requirements?

A. Create custom Amazon CloudWatch metrics to monitor model errors. Set provisioned throughput to a value that is safely higher than the peak traffic observed.
B. Create custom Amazon CloudWatch metrics to monitor model errors. Set up a failover mechanism to redirect invocations to a backup AWS Region when the errors exceed a specified threshold.
C. Enable invocation logging in Amazon Bedrock. Monitor key metrics such as Invocations, InputTokenCount, OutputTokenCount, and Invocation Throttles. Distribute traffic across cross-Region inference endpoints.
D. Enable invocation logging in Amazon Bedrock. Monitor InvocationLatency, InvocationClientErrors, and InvocationServerErrors metrics. Distribute traffic across multiple versions of the same model.

Correct Answer: *C*

Question #75 *Topic 1*

A company is developing a new AI-powered application that needs to integrate with various specialized tools. These tools currently run as Model Context Protocol (MCP) servers on the local machines of developers and do not maintain states between invocations. The company plans to deploy each MCP server as an AWS Lambda function to support the company's production application.
The solution must be accessible to both internal applications and authorized third-party partners.
The solution must use strict authentication and authorization controls.
Which additional steps will meet these requirements with the LEAST operational overhead?

A. Create a custom Lambda invocation transport by using the Lambda Invoke API. Implement IAM authentication and grant InvokeFunction permissions to authorized users and roles.
B. Expose the Lambda functions through Amazon API Gateway REST API endpoints. Implement API keys for authentication. Configure the applications that need to access the MCP servers to use standard HTTP requests instead of the MCP protocol.
C. Create Lambda function URLs and enable a custom Streamable HTTP transport and SigV4. Implement AWS IAM authentication. Grant InvokeFunctionUrl permissions to authorized users and roles.
D. Expose the Lambda function through Amazon API Gateway HTTP API endpoints with the Streamable HTTP transport. Use Amazon Cognito to implement OAuth authentication. Configure API Gateway to validate OAuth tokens.

Correct Answer: *C*

Requirement	How C solves it
MCP server compatibility	**Streamable HTTP** is the standard MCP transport protocol — Lambda function URLs support streaming responses natively, enabling proper MCP communication

Stateless Lambda deployment	MCP servers "do not maintain states between invocations" — aligns perfectly with Lambda's stateless execution model. Function URLs expose each Lambda directly as an HTTP endpoint
Internal + third-party access	Internal apps use IAM roles natively. Third-party partners are granted **IAM users/roles with InvokeFunctionUrl permissions** — standard cross-account access pattern
Strict auth & authorization	**SigV4 signing** + **IAM authentication** provides the strictest authentication mechanism in AWS. Fine-grained InvokeFunctionUrl permissions control exactly who can invoke each MCP server
Least operational overhead	**No API Gateway**, no Cognito, no additional services — Lambda function URLs are a zero-cost, built-in feature of Lambda

Question #76*Topic 1*

A company has a generative AI (GenAI) application that uses Amazon Bedrock to provide real-time responses to customer queries. The company has noticed intermittent failures with API calls to foundation models (FMs) during peak traffic periods.

The company needs a solution to handle transient errors and provide detailed observability into FM performance. The solution must prevent cascading failures during throttling events and provide distributed tracing across service boundaries to identify latency contributors. The solution must also enable correlation of performance issues with specific FM characteristics.

Which solution will meet these requirements?

A. Implement a custom retry mechanism with a fixed delay of 1 second between retries. Configure Amazon CloudWatch alarms to monitor the application's error rates and latency metrics.

B. Configure the AWS SDK with standard retry mode and exponential backoff with jitter. Use AWS X-Ray tracing with annotations to identify and filter service components.

C. Implement client-side caching of all FM responses. Add custom logging statements in the application code to record API call durations.

D. Configure the AWS SDK with adaptive retry mode. Use AWS CloudTrail distributed tracing to monitor throttling events.

Correct Answer: *B*

Question #77 *Topic 1*

A company is building a video analysis platform on AWS. The platform will analyze a large video archive by using Amazon Rekognition and Amazon Bedrock. The platform must comply with predefined privacy standards. The platform must also use secure model I/O, control foundation model (FM) access patterns, and provide an audit of who accessed what and when.
Which solution will meet these requirements?

A. Configure VPC endpoints for Amazon Bedrock model API calls. Implement Amazon Bedrock Guardrails to filter harmful or unauthorized content in prompts and responses. Use Amazon Bedrock trace events to track all agent and model invocations for auditing purposes. Export the traces to Amazon CloudWatch Logs as an audit record of model usage. Store all prompts and outputs in Amazon S3 with server-side encryption with AWS KMS keys (SSE-KMS).
B. Define access control by using IAM with attribute-based controls to map departments to specific permissions. Configure VPC endpoints for Amazon Bedrock model API calls. Use IAM condition keys to enforce specific GuardrailIdentifier and ModelId values. Configure AWS CloudTrail to capture management and data events for S3 objects and KMS key usage activities. Enable S3 server access logging to record detailed file-level interactions with the video archives. Send all CloudTrail logs to AWS CloudTrail Lake. Set up Amazon CloudWatch alarms to detect and alert on unexpected activity from Amazon Bedrock, Amazon Rekognition, and AWS KMS.
C. Restrict access to services by using VPC endpoint policies. Use AWS Config to track resource changes and compliance with security rules. Use server-side encryption with AWS KMS keys (SSE-KMS) to encrypt data at rest. Store the model's I/O in separate Amazon S3 buckets. Enable S3 server access logging to track file-level interactions.
D. Configure AWS CloudTrail Insights to analyze API call patterns across accounts and detect anomalous activity in Amazon Bedrock, Amazon Rekognition, Amazon S3, and AWS KMS. Deploy Amazon Macie to scan and classify the video archive. Use server-side encryption with AWS KMS keys (SSE-KMS) to encrypt all stored data. Configure CloudTrail to capture KMS API usage events for audit purposes. Configure Amazon EventBridge rules to process CloudTrail Insights anomalies and Macie findings. Use CloudWatch alarms to trigger automated notifications and security responses when potential security issues are detected.

Correct Answer: *B*

Question #78 *Topic 1*

An insurance company uses existing Amazon SageMaker AI infrastructure to support a web-based application that allows customers to predict what their insurance premiums will be. The company stores customer data that is used to train the SageMaker AI model in an Amazon S3 bucket. The dataset is growing rapidly. The company wants a solution to continuously re-train the model. The solution must automatically re-train and re-deploy the model to the application when an employee uploads a new customer data file to the S3 bucket.
Which solution will meet these requirements?

A. Use AWS Glue to run an ETL job on each uploaded file. Configure the ETL job to use the AWS SDK to invoke the Sage Maker AI model endpoint. Use real-time inference with the endpoint to re-deploy the model after it is re-trained on the updated customer dataset.
B. Create an AWS Lambda function and webhook handlers to generate an event when an employee uploads a new file. Configure SageMaker Pipelines to re-deploy the model after it is re-trained on the updated customer dataset. Use Amazon EventBridge to create an event bus. Set the Lambda function event as the source and SageMaker Pipelines as the target.
C. Create an AWS Step Functions Express workflow with AWS SDK integrations to retrieve the customer data from the S3 bucket when an employee uploads a new file to the S3 bucket. Use a SageMaker Data Wrangler flow to export the data from the S3 bucket to SageMaker Autopilot. Use SageMaker Autopilot to re-deploy the model after it has been re-trained on the updated customer dataset.
D. Create an AWS Step Functions Standard workflow. Configure the first state to call an AWS Lambda function to respond when an employee uploads a new file to the S3 bucket. Use a pipeline in SageMaker Pipelines to re-deploy the model after it has been re-trained on the updated customer dataset. Use the next state in the workflow to run the pipeline when the first state receives a response.

Correct Answer: *D*

Question #79 *Topic 1*

A company uses Amazon Bedrock to implement a Retrieval Augmented Generation (RAG)-based system to serve medical information to users. The company needs to compare multiple chunking strategies, evaluate the generation quality of two foundation models (FMs), and enforce quality thresholds for deployment.
Which Amazon Bedrock evaluation configuration will meet these requirements?

A. Create a retrieve-only evaluation job that uses a supported version of Anthropic Claude Sonnet as the evaluator model. Configure metrics for context relevance and context coverage. Define deployment thresholds in a separate CI/CD pipeline.
B. Create a retrieve-and-generate evaluation job that uses custom precision at k metrics and an LLM-as-a-judge metric that uses a scale of 1-5. Include each chunking strategy in the evaluation dataset. Use a supported version of Anthropic Claude Sonnet to evaluate responses from both FMs.
C. Create a separate evaluation job for each chunking strategy and FM combination. Use Amazon Bedrock built-in metrics for correctness and completeness. Manually review scores before deployment approval.
D. Set up a pipeline that uses multiple retrieve-only evaluation jobs to assess retrieval quality. Create separate evaluation jobs for both FMs that use Amazon Nova Pro as the LLM-as-a-judge model. Evaluate based on faithfulness and citation precision metrics.

Correct Answer: *B* 📦

Sure, . Let me break down Option B in detail.

Option B — Component-by-Component Analysis

1. Retrieve-and-Generate Evaluation Job

Aspect	Detail
What it does	Evaluates the **full RAG pipeline** end-to-end — from document retrieval through FM response generation
Why it matters	Retrieval quality alone doesn't guarantee generation quality. A chunking strategy might retrieve perfect passages, but the FM might still hallucinate. Only retrieve-and-generate captures this interaction
Medical context	For medical information, you need confidence that the **final answer** (not just the retrieved chunks) is accurate and complete

2. Custom Precision at K Metrics

Aspect	Detail
What it measures	Of the top K retrieved chunks, how many are actually relevant to the query
How it compares chunking strategies	Strategy A (500-token chunks) might achieve precision@5 = 0.80, while Strategy B (1000-token chunks) might achieve 0.60 — directly quantifiable comparison
Why "custom"	The company defines what "relevant" means for medical content — stricter than generic relevance for a healthcare use case

Example:

Chunking Strategy	Precision@3	Precision@5	Precision@10
500-token fixed	0.67	0.60	0.50
1000-token semantic	0.83	0.78	0.65
Hierarchical	0.90	0.82	0.70

3. LLM-as-a-Judge with 1-5 Scale

Aspect	Detail
How it works	Claude Sonnet reads the original query, retrieved context, and generated response — then scores the response on a 1-5 scale
Why 1-5 scale	Provides granular, quantitative scoring — enables threshold enforcement (e.g., "must average ≥ 4.0 to deploy")
Evaluating 2 FMs	Both FMs answer the same queries with the same retrieved context — Claude judges both, producing directly comparable scores

Scoring example:

Query	FM A Score	FM B Score	Reasoning
"Dosage for metformin?"	5	4	FM A cited exact guidelines; FM B was slightly vague
"Contraindications for aspirin?"	3	5	FM A missed renal impairment; FM B was comprehensive
"Symptoms of Type 2 diabetes?"	4	4	Both accurate and complete

4. Evaluation Dataset Design

Each entry in the evaluation dataset combines a chunking strategy with test queries:

Query: "What are the side effects of lisinopril?"
├— Chunking Strategy A (500-token) → retrieves chunks → FM A generates → scored
├— Chunking Strategy A (500-token) → retrieves chunks → FM B generates → scored
├— Chunking Strategy B (semantic) → retrieves chunks → FM A generates → scored
└— Chunking Strategy B (semantic) → retrieves chunks → FM B generates → scored

5. Quality Threshold Enforcement

The 1-5 scoring directly enables deployment gating:

Metric	Threshold	Strategy A + FM A	Strategy B + FM B
Avg generation score	≥ 4.0	3.8 ❌	4.5 ✅
Precision@5	≥ 0.75	0.60 ❌	0.82 ✅
Faithfulness	≥ 4.0	4.2 ✅	4.7 ✅

Only combinations meeting **all thresholds** are approved for deployment.

Why This Matters for Medical Use Cases

Medical RAG systems have **zero tolerance for inaccuracy**. Option B's configuration ensures:

- **Chunking comparison** → best retrieval strategy selected via precision metrics
- **FM comparison** → best generation model selected via judge scores
- **Threshold enforcement** → only combinations exceeding quality bars reach production
- **Single unified evaluation** → chunking × FM interactions are captured holistically

This is why a retrieve-and-generate evaluation with quantitative scoring is essential — you're making deployment decisions that directly impact patient safety.

Question #80*Topic 1*

A wildlife conservation agency operates zoos globally. The agency uses various sensors, trackers, and audiovisual recorders to monitor animal behavior. The agency wants to launch a generative AI (GenAI) assistant that can ingest multimodal data to study animal behavior.
The GenAI assistant must support natural language queries, avoid speculative behavioral interpretations, and maintain audit logs for ethical research audits.
Which solution will meet these requirements?

A. Ingest raw videos into Amazon Rekognition to detect animal postures and expressions. Use Amazon Data Firehose to stream sensor and GPS data into an Amazon S3 data lake. Prompt an Amazon Bedrock foundation model (FM) by using basic templates that are stored in AWS Systems Manager Parameter Store. Use IAM policies to control access. Use AWS CloudTrail for audit logging.
B. Use Amazon SageMaker Processing and Amazon Transcribe to pre-process multimodal data. Ingest summaries into an Amazon Bedrock Retrieval Augmented Generation (RAG) knowledge base. Apply Amazon Bedrock guardrails to restrict speculative outputs. Use AWS AppConfig to manage prompt templates. Use AWS CloudTrail to log research activity for audits.
C. Use Amazon OpenSearch Serverless to index behavioral logs and telemetry events. Use Amazon Comprehend to extract entities. Use Amazon Bedrock to build a layer to answer questions. Embed study summaries into OpenSearch Serverless documents. Use IAM to control access. Use AWS CloudTrail to log user interactions with the AI assistant.

D. Configure Amazon Q Business to federate data across Amazon S3, Amazon Kinesis, and Amazon SageMaker Feature Store. Configure Amazon EventBridge to invoke data ingestion jobs. Use custom AWS Lambda functions to filter large language model (LLM) outputs for ethical compliance before returning results to users.

Correct Answer: *B*

Question #81 *Topic 1*

A company uses an organization in AWS Organizations with all features enabled to manage multiple AWS accounts. Employees use Amazon Bedrock across multiple accounts. The company must prevent specific topics and proprietary information from being included in prompts to Amazon Bedrock models. The company must ensure that employees can use only approved Amazon Bedrock models. The company centrally manages IAM roles for employees.
Which combination of solutions will meet these requirements? (Choose two.)

A. Create an IAM permissions boundary for each employee's IAM role. Configure the permissions boundary to require an approved Amazon Bedrock guardrail identifier to invoke Amazon Bedrock models. Create an SCP that allows employees to use only approved models.
B. Create an SCP that allows employees to use only approved models. Configure the SCP to require employees to specify a guardrail identifier in calls to invoke an approved model.
C. Create an SCP that prevents an employee from invoking a model if a centrally deployed guardrail identifier is not specified in a call to the model. Create a permissions boundary on each employee's IAM role that allows each employee to invoke only approved models.
D. Use AWS CloudFormation to create a custom Amazon Bedrock guardrail that has a block filtering policy. Use stack sets to deploy the guardrail to each account in the organization.
E. Use AWS CloudFormation to create a custom Amazon Bedrock guardrail that has a mask filtering policy. Use stack sets to deploy the guardrail to each account in the organization.

Correct Answer: *BD*

D → Stack sets deploy guardrails (block policy) to every account

B → SCP ensures employees can ONLY use approved models AND must specify the guardrail ID in every call

Why Option A is wrong: Result: every Bedrock invocation is restricted to approved models and filtered through the deployed guardrail

plits enforcement across two different mechanisms unnecessarily. Permissions boundaries must be managed per IAM role — more overhead than a single SCP that handles both constraints. B consolidates both requirements into one SCP

Question #82 *Topic 1*

A company is designing an API for a generative AI (GenAI) application that uses a foundation model (FM) that is hosted on a managed model service. The API must stream responses to reduce latency, enforce token limits to manage compute resource usage, and implement retry logic to handle model timeouts and partial responses.
Which solution will meet these requirements with the LEAST operational overhead?

A. Integrate an Amazon API Gateway HTTP API with an AWS Lambda function to invoke Amazon Bedrock. Use Lambda response streaming to stream responses. Enforce token limits within the Lambda function. Implement retry logic for model timeouts by using Lambda and API Gateway timeout configurations.
B. Connect an Amazon API Gateway HTTP API directly to Amazon Bedrock. Simulate streaming by using client-side polling. Enforce token limits on the frontend. Configure retry behavior by using API Gateway integration settings.
C. Connect an Amazon API Gateway WebSocket API to an Amazon ECS service that hosts a containerized inference server. Stream responses by using the WebSocket protocol. Enforce token limits within Amazon ECS. Handle model timeouts by using ECS task lifecycle hooks and restart policies.
D. Integrate an Amazon API Gateway REST API with an AWS Lambda function that invokes Amazon Bedrock. Use Lambda response streaming to stream responses. Enforce token limits within the Lambda function. Implement retry logic by using Lambda and API Gateway timeout configurations.

Correct Answer: *D* 📦

Question #83 *Topic 1*

A retail company is using Amazon Bedrock to develop a customer service AI assistant. Analysis shows that 70% of customer inquiries are simple product questions that a smaller model can effectively handle. However, 30% of inquiries are complex return policy questions that require advanced reasoning. The company wants to implement a cost-effective model selection

framework to automatically route customer inquiries to appropriate models based on inquiry complexity. The framework must maintain high customer satisfaction and minimize response latency.
Which solution will meet these requirements with the LEAST implementation effort?

A. Create a multi-stage architecture that uses a small foundation model (FM) to classify the complexity of each inquiry. Route simple inquiries to a smaller, more cost-effective model. Route complex inquiries to a larger, more capable model. Use AWS Lambda functions to handle the routing logic.
B. Use Amazon Bedrock intelligent prompt routing to automatically analyze inquiries. Route simple product inquiries to smaller models, and route complex return policy inquiries to more capable larger models.
C. Implement a single-model solution that uses an Amazon Bedrock mid-sized foundation model (FM) with on-demand pricing. Include special instructions in model prompts to handle both simple and complex inquiries by using the same model.
D. Create separate Amazon Bedrock endpoints for simple and complex inquiries. Implement a rule-based routing system based on keyword detection. Use on-demand pricing for the smaller model and provisioned throughput for the larger model.

Correct Answer: *B*

Question #84*Topic 1*

A specialty coffee company has a mobile app that generates personalized coffee roast profiles by using Amazon Bedrock with a three-stage prompt chain. The prompt chain converts user inputs into structured metadata, retrieves relevant logs for coffee roasts, and generates a personalized roast recommendation for each customer.
Users in multiple AWS Regions report inconsistent roast recommendations for identical inputs, slow inference during the retrieval step, and unsafe recommendations such as brewing at excessively high temperatures. The company must improve the stability of outputs for repeated inputs. The company must also improve app performance and the safety of the app's outputs. The updated solution must ensure 99.5% output consistency for identical inputs and achieve inference latency of less than 1 second. The solution must also block unsafe or hallucinated recommendations by using validated safety controls.
Which solution will meet these requirements?

A. Deploy Amazon Bedrock with provisioned throughput to stabilize inference latency. Apply Amazon Bedrock guardrails that have semantic denial rules to block unsafe outputs. Use Amazon Bedrock Prompt Management to manage prompts by using approval workflows.
B. Use Amazon Bedrock Agents to manage chaining. Log model inputs and outputs to Amazon CloudWatch Logs. Use logs from Amazon CloudWatch to perform A/B testing for prompt versions.
C. Cache prompt results in Amazon ElastiCache. Use AWS Lambda functions to pre-process metadata and to trace end-to-end latency. Use AWS X-Ray to identify and remediate performance bottlenecks.
D. Use Amazon Kendra to improve roast log retrieval accuracy. Store normalized prompt metadata within Amazon DynamoDB. Use AWS Step Functions to orchestrate multistep prompts.

Correct Answer: *A*

Requirement	How A solves it
99.5% output consistency	**Prompt Management with approval workflows** ensures identical, vetted prompts are used consistently across all Regions — no prompt drift or unauthorized changes. **Provisioned throughput** provides dedicated model capacity, eliminating variability from shared infrastructure
Inference latency < 1 second	**Provisioned throughput** guarantees dedicated compute capacity — eliminates queuing delays, cold starts, and throttling that cause slow inference during the retrieval step
Block unsafe recommendations	**Guardrails with semantic denial rules** evaluate outputs against safety policies — detects and blocks dangerous recommendations (e.g., excessive brewing temperatures) before they reach users

Question #85 *Topic 1*

A company is developing a generative AI (GenAI) application by using Amazon Bedrock. The application will analyze patterns and relationships in the company's data. The application will process millions of new data points daily across AWS Regions in Europe, North America, and Asia before storing the data in Amazon S3.

The application must comply with local data protection and storage regulations. Data residency and processing must occur within the same continent. The application must also maintain audit trails of the application's decision-making processes and provide data classification capabilities. Which solution will meet these requirements?

A. Deploy the application in each Region with local IAM policies. Use Amazon Bedrock cross-Region inference to distribute the workload. Use Amazon CloudWatch to log AI decision-making processes and data processing activities. Manually track compliance certifications across Regions.
B. Use SCPs with AWS Organizations to manage location-specific permissions. Use AWS CloudTrail immutable logs to audit the decision-making processes. Import a custom model into Amazon Bedrock and deploy the model to each Region.
C. Use Amazon S3 Object Lock with Region-specific S3 bucket policies. Pre-process the data points within the Region based on geographic origin before sending the data points to Amazon Bedrock. Use Amazon Macie to classify the data. Use AWS CloudTrail immutable logs to audit the decision-making processes.
D. Create separate AWS accounts for each Region with individual compliance frameworks. Use Amazon SageMaker AI with custom monitoring to track model performance and compliance with data residency requirements. Create manual reports for each regulatory jurisdiction.

Correct Answer: *C* 📦

Requirement	How C solves it
Data residency & storage compliance	**Region-specific S3 bucket policies** enforce that data stays in the correct geographic Region. **S3 Object Lock** provides WORM (Write Once Read Many) compliance — data cannot be tampered with or deleted, meeting strict regulatory retention requirements
Processing within same continent	**Pre-processing data within the Region based on geographic origin** ensures that European data is processed in European Regions, Asian data in Asian Regions, etc. — before it ever reaches Bedrock
Audit trails of decision-making	**CloudTrail immutable logs** provide tamper-proof, regulatory-grade audit trails of all application activities and AI decision processes

Data classification capabilities	**Amazon Macie** automatically discovers, classifies, and protects sensitive data — identifying PII, financial data, and other regulated content across the S3 data stores

Question #86*Topic 1*

A financial services company is deploying a generative AI (GenAI) application that uses Amazon Bedrock to assist customer service representatives to provide personalized investment advice to customers. The company must implement a comprehensive governance solution that follows responsible AI practices and meets regulatory requirements.

The solution must detect and prevent hallucinations in recommendations. The solution must have safety controls for customer interactions. The solution must also monitor model behavior drift in real time and maintain audit trails of all prompt-response pairs for regulatory review.

The company must deploy the solution within 60 days. The solution must integrate with the company's existing compliance dashboard and respond to customers within 200 ms.

Which solution will meet these requirements with the LEAST operational overhead?

A. Configure Amazon Bedrock guardrails to apply custom content filters and toxicity detection. Use Amazon Bedrock Model Evaluation to detect hallucinations. Store prompt-response pairs in Amazon DynamoDB to capture audit trails and set a TTL. Integrate Amazon CloudWatch custom metrics with the existing compliance dashboard.
B. Deploy Amazon Bedrock and use AWS PrivateLink to access the application securely. Use AWS Lambda functions to implement custom prompt validation. Store prompt-response pairs in an Amazon S3 bucket and configure S3 Lifecycle policies. Create custom Amazon CloudWatch dashboards to monitor model performance metrics.
C. Use Amazon Bedrock Agents and Amazon Bedrock Knowledge Bases to ground responses. Use Amazon Bedrock Guardrails to enforce content safety. Use Amazon OpenSearch Service to store and index prompt-responses pairs. Integrate OpenSearch Service with Amazon QuickSight to create compliance reports and to detect model behavior drift.
D. Use Amazon SageMaker Model Monitor to detect model behavior drift. Use AWS WAF to filter content. Store customer interactions in an encrypted Amazon RDS database. Use Amazon API Gateway to create custom HTTP APIs to integrate with the compliance dashboard.

Correct Answer: *A*

Aspect	Option A	Option C
Services involved	Guardrails, Model Evaluation, DynamoDB, CloudWatch	Agents, Knowledge Bases, Guardrails, OpenSearch, QuickSight
Infrastructure management	**All fully serverless/managed** — zero cluster management	OpenSearch requires **cluster sizing, node management, scaling policies, index lifecycle**
Setup complexity	Configure guardrails + DynamoDB table + CloudWatch metrics	Set up KB (vector store, data ingestion, chunking), Agents, OpenSearch cluster, QuickSight dashboards
Operational overhead	**Minimal**	**Significantly more**

Question #87*Topic 1*

A company uses AWS Lake Formation to set up a data lake that contains databases and tables for multiple business units across multiple AWS Regions. The company wants to use a foundation model (FM) through Amazon Bedrock to perform fraud detection. The FM must ingest sensitive financial data from the data lake. The data includes some customer personally identifiable information (PM).

The company must design an access control solution that prevents PI I from appearing in a production environment. The FM must access only authorized data subsets that have PH redacted from specific data columns. The company must capture audit trails for all data access.

Which solution will meet these requirements?

A. Create a separate dataset in a separate Amazon S3 bucket for each business unit and Region combination. Configure S3 bucket policies to control access based on IAM roles that are assigned to FM training instances. Use S3 access logs to track data access.

B. Configure the FM to authenticate by using IAM roles and Lake Formation permissions based on LF-Tag expressions. Define business units and Regions as LF-Tags that are assigned to databases and tables. Use AWS CloudTrail to collect comprehensive audit trails of data access.
C. Use direct IAM principal grants on specific databases and tables in Lake Formation. Create a custom application layer that logs access requests and further filters sensitive columns before sending data to the FM.
D. Configure the FM to request temporary credentials from AWS STS. Access the data by using presigned S3 URLs that are generated by an API that applies business unit and Regional filters. Use AWS CloudTrail to collect comprehensive audit trails of data access.

Correct Answer: *B*

Question #88*Topic 1*

A company runs a Retrieval Augmented Generation (RAG) application that uses Amazon Bedrock Knowledge Bases to perform regulatory compliance queries. The application uses the RetrieveAndGenerateStream API. The application retrieves relevant documents from a knowledge base that contains more than 50,000 regulatory documents, legal precedents, and policy updates.

The RAG application is producing suboptimal responses because the initial retrieval often returns semantically similar but contextually irrelevant documents. The poor responses are causing model hallucinations and incorrect regulatory guidance. The company needs to improve the performance of the RAG application so it returns more relevant documents.

Which solution will meet this requirement with the LEAST operational overhead?

A. Deploy an Amazon SageMaker endpoint to run a fine-tuned ranking model. Use an Amazon API Gateway REST API to route requests. Configure the application to make requests through the REST API to rerank the results.
B. Use Amazon Comprehend to classify documents and apply relevance scores. Integrate the RAG application's reranking process with Amazon Textract to run document analysis. Use Amazon Neptune to perform graph-based relevance calculations.
C. Implement a retrieval pipeline that uses the Amazon Bedrock Knowledge Bases Retrieve API to perform initial document retrieval. Call the Amazon Bedrock Rerank API to rerank the results. Invoke the InvokeModelWithResponseStream operation to generate responses.
D. Use the latest Amazon reranker model through the reranking configuration within Amazon Bedrock Knowledge Bases. Use the model to improve document relevance scoring and to reorder results based on contextual assessments.

Question #89 *Topic 1*

A company is developing a customer communication platform that uses an AI assistant powered by an Amazon Bedrock foundation model (FM). The AI assistant summarizes customer messages and generates initial response drafts.

The company wants to use Amazon Comprehend to implement layered content filtering. The layered content filtering must prevent sharing of offensive content, protect customer privacy, and detect potential inappropriate advice solicitation. Inappropriate advice solicitation includes requests for unethical practices, harmful activities, or manipulative behaviors. The solution must maintain acceptable overall response times, so all pre-processing filters must finish before the content reaches the FM.

Which solution will meet these requirements?

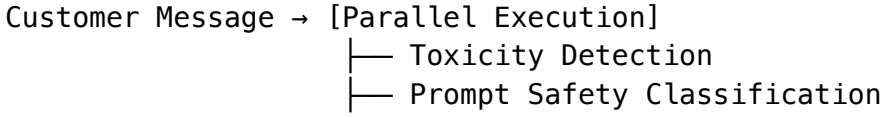
A. Use parallel processing with asynchronous API calls. Use toxicity detection for offensive content. Use prompt safety classification for inappropriate advice solicitation. Use personally identifiable information (PII) detection without redaction.

B. Use custom classification to build an FM that detects offensive content and inappropriate advice solicitation. Apply personally identifiable information (PII) detection as a secondary filter only when messages pass the custom classifier.

C. Deploy a multi-stage process. Configure the process to use prompt safety classification first, then toxicity detection on safe prompts only, and finally personally identifiable information (PII) detection in streaming mode. Route flagged messages through Amazon EventBridge for human review.

D. Use toxicity detection with thresholds configured to 0.5 for all categories. Use parallel processing for both prompt safety classification and personally identifiable information (PII) detection with entity redaction. Apply Amazon CloudWatch alarms to filter metrics.

Correct Answer: *A*



```
  └── PII Detection
→ [All pass?] → FM processing
→ [Any fail?] → Block/Flag
```

Question #90 *Topic 1*

A software as a service (SaaS) company is building a recommendation model that uses Amazon SageMaker AI to support an application that recommends airline cabin upgrades to customers. The company will host SageMaker AI models on Amazon Bedrock by using Amazon Bedrock Custom Model Import. Airline companies will use the application to send customized offers to customers.

The model must examine the travel history of customers to help make more relevant recommendations. The company stores customer travel history data in an Amazon RDS database. The company must ensure that the application delivers consistent, relevant, and accurate results across multiple airlines and customer populations.

Which solution will meet these requirements?

A. Use Amazon Bedrock Knowledge Bases to implement a RAG architecture to analyze customer travel history data to give the application semantic search capabilities. Use the semantic search capabilities to retrieve relevant booking patterns, preferences, and loyalty information to generate personalized cabin upgrade recommendations. Apply Amazon Bedrock guardrails to filter content. Use AWS Step Functions and AWS Lambda functions to orchestrate validation workflows to reduce hallucinations.
B. Implement text-to-SQL transformations with SQL validations to accurately retrieve relevant booking patterns, preferences, and loyalty information from the RDS database. Use the results to generate personalized cabin upgrade recommendations. Apply Amazon Bedrock guardrails to filter content. Use AWS Step Functions and AWS Lambda functions to orchestrate validation workflows to reduce hallucinations.
C. Use Amazon OpenSearch Service to implement vector searches of customer travel history embeddings. Use the vector searches to give the application the ability to perform similarity-based retrieval of booking patterns, preferences, and loyalty information to generate personalized cabin upgrade recommendations. Apply Amazon Bedrock guardrails to filter responses. Use confidence scoring and semantic similarity searches to reduce hallucinations.
D. Implement text-to-SQL transformations with SQL validations to accurately retrieve relevant booking patterns, preferences, and loyalty information from the RDS database. Use the results to

generate personalized cabin upgrade recommendations. Apply Amazon Bedrock guardrails to filter responses. Use confidence scoring and semantic similarity searches to reduce hallucinations.

Correct Answer: *B*

Question #91 *Topic 1*

A company is using Amazon Bedrock to develop a customer support AI assistant. The AI assistant must respond to customer questions about their accounts. The AI assistant must not expose personal information in responses. The company must comply with data residency policies by ensuring that all processing occurs within the same AWS Region where each customer is located.

The company wants to evaluate how effective the AI assistant is at preventing the exposure of personal information before the company makes the AI assistant available to customers.

Which solution will meet these requirements?

A. Configure a cross-Region Amazon Bedrock guardrail to apply sensitive information filters. Set the guardrail to detect mode during development and testing. Switch to block mode for production deployment.
B. Configure an Amazon Bedrock guardrail to apply sensitive information filters. Set the guardrail to mask mode during development and testing. Switch to block mode for production deployment. Deploy a copy of the guardrail to each Region where the company operates.
C. Configure an Amazon Bedrock guardrail to apply content and topic filters. Set the guardrail to detect mode during development, testing, and production. Disable invocation logging for the Amazon Bedrock model.
D. Configure a cross-Region Amazon Bedrock guardrail to apply a set of content and word filters. Set the guardrail to detect mode during development and testing. Switch to mask mode for production deployment.

Requirement	How D solves it
Semantic text search	**Amazon Titan Embeddings** converts journal abstracts into dense vector representations that capture semantic meaning — queries find conceptually similar content, not just keyword matches

Text + metadata search	Aurora PostgreSQL stores vectors **alongside metadata columns** in the same table — a single query can filter by metadata AND rank by vector similarity simultaneously
Match similar abstracts	**pgvector** extension provides native similarity operators (<-> cosine distance, <=> inner product) — directly compares abstract embeddings to find matches
< 1 million files	Aurora PostgreSQL with pgvector comfortably handles this scale — well within its performance envelope
Least operational overhead	**Aurora Serverless** auto-scales capacity (even to zero), eliminates instance management. **Titan Embeddings** is fully managed via API — no model hosting

Correct Answer: *B* 📦

Requirement	How B solves it
Prevent PII exposure	**Sensitive information filters** are the specific Bedrock guardrail type designed to detect personal information (names, SSNs, account numbers, etc.)
Evaluate effectiveness before production	**Mask mode** during development replaces detected PII with placeholders (e.g., {NAME}) — developers can see exactly *where* and *what* the guardrail catches, measure detection accuracy, and identify gaps without exposing real data
Production protection	**Block mode** in production prevents any response containing PII from reaching customers — strictest enforcement
Data residency	Deploying a **copy of the guardrail to each Region** ensures all processing occurs locally where each customer is located — no cross-Region data movement

Option A : Cross-Region directly violates data residency — customer data would traverse Regions. Detect mode only logs PII detections but still passes PII through to the user — doesn't prevent exposure in production

Question #92 *Topic 1*

A university recently digitized a collection of archival documents, academic journals, and manuscripts. The university stores the digital files in an AWS Lake Formation data lake.

The university hires a GenAI developer to build a solution to allow users to search the digital files by using text queries. The solution must return journal abstracts that are semantically similar to a user's query. Users must be able to search the digitized collection based on text and metadata that is associated with the journal abstracts. The metadata of the digitized files does not contain keywords. The solution must match similar abstracts to one another based on the similarity of their text. The data lake contains fewer than 1 million files.

Which solution will meet these requirements with the LEAST operational overhead?

A. Use Amazon Titan Embeddings in Amazon Bedrock to create vector representations of the digitized files. Store embeddings in the OpenSearch Neural Plugin for Amazon OpenSearch Service.
B. Use Amazon Comprehend to extract topics from the digitized files. Store the topics and file metadata in an Amazon Aurora PostgreSQL database. Query the abstract metadata against the data in the Aurora database.
C. Use Amazon SageMaker AI to deploy a sentence-transformer model. Use the model to create vector representations of the digitized files. Store embeddings in an Amazon Aurora PostgreSQL database that has the pgvector extension.
D. Use Amazon Titan Embeddings in Amazon Bedrock to create vector representations of the digitized files. Store embeddings in an Amazon Aurora PostgreSQL Serverless database that has the pgvector extension.

Correct Answer: *D*

Question #93 *Topic 1*

A financial services company wants to use Amazon Bedrock foundation models (FMs) to analyze call center recordings. When calls end, the call center stores recordings as MP3 files in an Amazon S3 bucket. The company needs to generate summaries and sentiment analysis for the recordings in a structured format as soon as new files are created. The recordings average 20 MB in size.

Which combination of solutions will meet these requirements? (Choose two.)

A. Use AWS Step Functions to orchestrate a workflow to process the recordings. Configure steps to invoke Amazon Transcribe to convert audio to text, validate job completion, and to invoke an AWS Lambda function to process the text by using Amazon Bedrock FMs to generate structured analysis output.
B. Use AWS Step Functions to orchestrate a workflow to process the recordings. Configure steps to invoke Amazon Transcribe to convert audio to text, validate job completion, and to directly invoke Amazon Bedrock FMs to generate summaries and sentiment analysis in JSON format.
C. Use AWS Step Functions to orchestrate a workflow to process the recordings. Configure steps to invoke Amazon Transcribe to convert audio to text, validate job completion, and to invoke an AWS Lambda function to create a prompt to invoke Amazon Bedrock FMs to generate structured analysis output.
D. Configure the source S3 bucket to send events to Amazon EventBridge. Create an EventBridge rule to invoke the Step Functions workflow when an object is created in the bucket.
E. Configure the source S3 bucket to send notifications to the Step Functions workflow when an object is created in the bucket.

Correct Answer: *CD*

Requirement	How D solves it
"As soon as new files are created"	S3 event notifications to EventBridge trigger immediately on object creation
Reliable invocation	EventBridge rule matches S3 PutObject events and invokes the Step Functions workflow — this is the supported pattern for S3 → Step Functions

Requirement	How C solves it
Audio → text conversion	Transcribe converts MP3 recordings to text with validated job completion
Structured summaries + sentiment	Lambda constructs a well-engineered **prompt** with the transcript text, then invokes Bedrock FMs to produce structured analysis output
20 MB recordings	Lambda handles reading large transcripts from S3 (Transcribe outputs to S3), avoiding Step Functions' 256 KB state payload limit

Question #94*Topic 1*

A medical device company wants to feed reports of medical procedures that used the company's devices into an AI assistant. To protect patient privacy, the AI assistant must expose patient personally identifiable information (PII) only to surgeons. The AI assistant must redact PII for engineers. The AI assistant must reference only medical reports that are less than 3 years old.

The company stores reports in an Amazon S3 bucket as soon as each report is published. The company has already set up an Amazon Bedrock knowledge base. The AI assistant uses Amazon Cognito to authenticate users.

Which solution will meet these requirements?

A. Enable Amazon Macie PII detection on the S3 bucket. Use an S3 trigger to invoke an AWS Lambda function that redacts PII from the reports. Configure the Lambda function to delete outdated documents from the bucket and to invoke knowledge base syncing.
B. Invoke an AWS Lambda function to sync the S3 bucket and the knowledge base when a new report is uploaded to the bucket. Use a second Lambda function to invoke Amazon Comprehend to detect and redact PII if a user is part of the engineer Cognito user group. Set up an S3 Lifecycle configuration to remove reports that are older than 3 years from the bucket.
C. Set up an S3 Lifecycle configuration on the bucket to remove reports that are older than 3 years. Schedule an AWS Lambda function to run daily syncs between the bucket and the knowledge base.

When users interact with the AI assistant, call the ApplyGuardrail configuration that matches the user's Cognito user group to redact PII from the agent's responses if appropriate.
D. Create a second knowledge base. Set up an S3 Lifecycle configuration on the bucket to remove reports that are older than 3 years. Invoke an AWS Lambda function that syncs the bucket with the original knowledge base when a new report is uploaded to the bucket. Use Amazon Comprehend to detect and redact PII before syncing the bucket with the second knowledge base. When a user interacts with the AI assistant, redirect the model to the appropriate knowledge base depending on the user's Cognito user group.

Correct Answer: *C*

Requirement	How C solves it
Surgeons see PII	Surgeon Cognito group → ApplyGuardrail with a **permissive** configuration (no PII redaction) — full patient data visible
Engineers get PII redacted	Engineer Cognito group → ApplyGuardrail with **sensitive information filters** enabled — PII automatically masked in responses
Only reports < 3 years old	**S3 Lifecycle configuration** automatically deletes reports older than 3 years from the bucket
Knowledge base stays current	**Daily Lambda sync** ensures the KB reflects the bucket's current state — including removals from Lifecycle expiration
Existing architecture	Builds on the existing Bedrock KB and Cognito authentication — minimal new infrastructure

Question #95 *Topic 1*

A large ecommerce company has deployed a foundation model (FM) to generate product descriptions. The company's engineering team monitors technical metrics such as token usage, latency, and error rates by using Amazon CloudWatch. The company's marketing team tracks business metrics such as conversion rates and revenue impact in its own systems.

The company needs a unified observability solution that correlates technical performance with business outcomes. The solution must provide automatic alerts to stakeholders when operational

metrics indicate degradation. The solution must provide comprehensive visibility across both technical and business metrics.

Which solution will meet these requirements?

A. Create CloudWatch dashboards that include technical metrics and imported business metrics. Configure CloudWatch composite alarms that combine technical data and business data. Use Amazon SNS to set up notifications to stakeholders.
B. Use Amazon Managed Grafana to visualize technical metrics from CloudWatch with business metrics from external sources. Configure Amazon Managed Grafana alerts to invoke AWS Lambda functions. Configure the Lambda functions to remediate issues automatically when metrics exceed predefined thresholds.
C. Stream CloudWatch metrics to Amazon S3 by using CloudWatch metric streams. Create Amazon QuickSight dashboards to visualize the combined technical metrics and business metrics. Set up Amazon EventBridge rules to send notifications to stakeholders when metrics exceed predefined thresholds.
D. Configure CloudWatch custom dashboards that integrate operational metrics with imported business metrics. Set up CloudWatch composite alarms with anomaly detection. Use Amazon SNS to create alarm actions to notify stakeholders when correlated metrics indicate performance issues.

Correct Answer: *D* 📦

Requirement	How D solves it
Correlate technical + business metrics	CloudWatch custom dashboards **integrate** operational metrics alongside imported business metrics — single pane of glass showing both dimensions
Automatic degradation alerts	**Anomaly detection** automatically learns baseline patterns and detects deviations — no manual threshold tuning needed for "degradation" detection
Correlated alerting	**Composite alarms** combine multiple individual alarms (e.g., latency spike AND conversion drop) — triggering only when **correlated metrics** together indicate a real issue

Stakeholder notifications	**SNS alarm actions** deliver alerts to appropriate stakeholders when composite alarm conditions are met
Comprehensive visibility	Unified dashboards show technical metrics (token usage, latency, errors) alongside business metrics (conversion, revenue)

Question #96 *Topic 1*

A GenAI developer is evaluating Amazon Bedrock foundation models (FMs) to enhance a Europe-based company's internal business application. The company has a multi-account landing zone in AWS Control Tower. The company uses SCPs to allow its accounts to use only the eu-north-1 Region and the eu-west-1 Region. All customer data must remain in private networks within the approved AWS Regions.

The GenAI developer selects an FM based on analysis and testing and hosts the model in the eu-central-1 Region and the eu-west-3 Region. The GenAI developer must enable access to the FM for the company's employees. The GenAI developer must ensure that requests to the FM are private and remain with the same Regions as the FM.

Which solution will meet these requirements?

A. Deploy an AWS Lambda function that is exposed by a private Amazon API Gateway REST API to a VPC in eu-north-1. Create a VPC endpoint for the selected FM in eu-central-1 and eu-west-3. Extend existing SCPs to allow employees to use the FM. Integrate the REST API with the business application.
B. Deploy the FM on Amazon EC2 instances in eu-north-1. Deploy a private Amazon API Gateway REST API in front of the EC2 instances. Configure an Amazon Bedrock VPC endpoint. Integrate the REST API with the business application.
C. Configure the FM to use cross-Region inference through an eu.amazon.* endpoint to ensure that all calls remain within Europe. Configure an Amazon Bedrock VPC endpoint. Extend existing SCPs to allow employees to use the FM through inference profiles in Europe-based Regions where the FM is available. Use an inference profile to integrate Amazon Bedrock with the business application.
D. Deploy the FM in Amazon SageMaker AI in eu-north-1. Configure a SageMaker AI VPC endpoint. Extend existing SCPs to allow employees to use the SageMaker AI endpoint. Integrate the FM in SageMaker AI with the business application.

Correct Answer: *C*

Question #97 *Topic 1*

A healthcare company is developing an application to process medical queries. The application must answer complex queries with high accuracy by reducing semantic dilution. The application must refer to domain-specific terminology in medical documents to reduce ambiguity in medical terminology. The application must be able to respond to 1,000 queries each minute with response times less than 2 seconds.

Which solution will meet these requirements with the LEAST operational overhead?

A. Use Amazon API Gateway to route incoming queries to an Amazon Bedrock agent. Configure the agent to use an Anthropic Claude model to decompose queries and an Amazon Titan model to expand queries. Create an Amazon Bedrock knowledge base to store the reference medical documents.
B. Configure an Amazon Bedrock knowledge base to store the reference medical documents. Enable query decomposition in the knowledge base. Configure an Amazon Bedrock flow that uses a foundation model (FM) and the knowledge base to support the application.
C. Use Amazon SageMaker AI to host custom ML models for both query decomposition and query expansion. Configure Amazon Bedrock knowledge bases to store the reference medical documents. Encrypt the documents in the knowledge base.
D. Create an Amazon Bedrock agent to orchestrate multiple AWS Lambda functions to decompose queries. Create an Amazon Bedrock knowledge base to store the reference medical documents. Use the agent's built-in knowledge base capabilities. Add deep research and reasoning capabilities to the agent to reduce ambiguity in the medical terminology.

Requirement	How B solves it
Reduce semantic dilution	**Query decomposition** breaks complex medical queries into focused sub-queries — each sub-query targets a specific aspect, preventing diluted results from overly broad searches
Domain-specific terminology	Bedrock **knowledge base** indexes medical documents — retrieval is grounded in actual domain vocabulary, reducing ambiguity in terms like "MI" (myocardial infarction vs. mitral insufficiency)

1,000 queries/min, <2s response	**Bedrock Flows** provide a streamlined, managed orchestration pipeline (FM + KB) — no agent reasoning loops, no Lambda cold starts, no multi-model handoffs adding latency
Least operational overhead	Everything is fully managed: KB handles indexing/retrieval, query decomposition is a native KB feature (toggle on), Flows handle orchestration — **zero custom code**

The architecture:

Query arrives → **Bedrock Flow** receives it

KB **query decomposition** splits complex query into focused sub-queries

Each sub-query retrieves from **medical document KB** with precision

FM synthesizes decomposed results into a comprehensive, accurate answer

Question #98*Topic 1*

A company has a customer service application that uses Amazon Bedrock to generate personalized responses to customer inquiries. The company needs to establish a quality assurance process to evaluate prompt effectiveness and model configurations across updates. The process must automatically compare outputs from multiple prompt templates, detect response quality issues, provide quantitative metrics, and allow human reviewers to give feedback on responses. The process must prevent configurations that do not meet a predefined quality threshold from being deployed.

Which solution will meet these requirements?

A. Create an AWS Lambda function that sends sample customer inquiries to multiple Amazon Bedrock model configurations and stores responses in Amazon S3. Use Amazon QuickSight to visualize response patterns. Manually review outputs daily. Use AWS CodePipeline to deploy the configurations that have quality scores above the specified quality threshold.
B. Use Amazon Bedrock evaluation jobs to compare model outputs by using custom prompt datasets. Configure AWS CodePipeline to run the evaluation jobs when prompt templates change. Configure CodePipeline to deploy the configurations that have quality scores above the specified quality threshold.

C. Set up Amazon CloudWatch alarms to monitor response latency and error rates from Amazon Bedrock. Use Amazon EventBridge rules to notify the company when latency and error rate metrics exceed thresholds. Configure an approval workflow in AWS Systems Manager to perform manual quality checks.
D. Use AWS Lambda functions to create an automated testing framework that samples production traffic and routes duplicate requests to the updated model version. Use Amazon Comprehend sentiment analysis to compare results. Block deployment if sentiment scores decrease.

Correct Answer: *B*

Question #99 *Topic 1*

A retail company runs an application that makes product recommendations to customers on the company's website. The application uses Amazon Bedrock to generate recommendations by dynamically constructing prompts and sending them to foundation models (FMs).

A GenAI developer has deployed an update to the application that instructs the FM to include a specific promotional message when the FM generates a response to prompts. When the developer tests the application, the promotional message does not always appear in the responses. When the promotional message does appear in the responses, it does not always flow with the rest of the text.

The GenAI developer must ensure that the promotional message always appears in the FM responses.

Which solution will meet this requirement?

A. Use an Amazon Bedrock Guardrails filter on the prompt. Set the input filter strength to HIGH.
B. Generate multiple response variants that include the promotional message in different ways. Use a reranker model to select the most coherent version based on relevance to the original prompt.
C. Run the prompt through Amazon Bedrock. Process the response through Amazon Bedrock Agents to add the promotional message. Rerank the results by using the original prompt and the desired message as context.
D. Reinforce the requirement to include the new promotional message within product recommendations by using an output indicator in prompts to the FM.

Correct Answer: *D*

This is fundamentally a **prompt engineering problem** — the FM isn't consistently following the instruction to include the promotional message. An **output indicator** is the standard technique to solve this:

Question #100*Topic 1*

A company is creating a generative AI (GenAI) application that uses Amazon Bedrock foundation models (FMs). The application must use Microsoft Entra ID to authenticate. All FM API calls must stay on private network paths. Access to the application must be limited by department to specific model families. The company also needs a comprehensive audit trail of model interactions.

Which solution will meet these requirements?

A. Configure SAML federation between Microsoft Entra ID and IAM. Create department-specific IAM roles that allow only the required ModelId values. Create AWS PrivateLink interface VPC endpoints for Amazon Bedrock runtime services. Enable AWS CloudTrail to capture Amazon Bedrock API calls. Configure Amazon Bedrock model invocation logging to record detailed model interactions.
B. Create an identity provider (IdP) connection in IAM to authenticate by using Microsoft Entra ID. Assign department permission sets to control access to specific model families. Deploy AWS Lambda functions in private subnets with a NAT gateway for egress to Amazon Bedrock public endpoints. Enable CloudWatch Logs to capture model interactions for auditing purposes.
C. Create a SAML identity provider (IdP) in IAM to authenticate by using Microsoft Entra ID. Use IAM permissions boundaries to limit department roles' access to specific model families. Configure public Amazon Bedrock API endpoints with VPC routing to maintain private network connectivity. Set up AWS CloudTrail with Amazon S3 Lifecycle rules to manage audit logs of model interactions.
D. Configure OpenID Connect (OIDC) federation between Microsoft Entra ID and IAM. Use attribute-based access control to map department attributes to specific model access permissions. Apply SCP policies to restrict access to Amazon Bedrock FM families based on department. Use Microsoft Entra ID's built-in logging capabilities to maintain an audit trail of model interactions.

Correct Answer: *A*

Question #101 *Topic 1*

A healthcare company is developing an AI application to retrieve medical literature to help doctors conduct research. The company wants to deploy the application to multiple countries. The application needs to index 50,000 text-heavy medical documents that contain specialized terminology in English, French, and German. The application must provide high search accuracy for medical terminology. The company wants a solution that requires minimal custom code development and ongoing maintenance.

Which solution will meet these requirements?

A. Use Amazon Titan Text Embeddings to process all documents in their original language without translation. Apply semantic chunking at the medical section level to preserve medical terminology context and to maintain moderate vector dimensions.
B. Use Amazon Titan Text Embeddings to translate all non-English documents to English before processing. Implement document chunking strategies based on logical sections to improve retrieval accuracy.
C. Use Amazon Titan Multimodal Embeddings to convert document pages to images. Generate text captions for the images to create a multimodal knowledge base that handles medical content.
D. Use an Amazon Bedrock knowledge base as the vector store. Use Amazon Kendra to perform multilingual document processing. Configure the vector store to automatically handle language detection and translation for each query.

Correct Answer: *A*

Question #102 *Topic 1*

A publishing company is developing a chat assistant that uses a containerized large language model (LLM) that runs on Amazon SageMaker AI. The architecture consists of an Amazon API Gateway REST API that routes user requests to an AWS Lambda function. The Lambda function invokes a SageMaker AI real-time endpoint that hosts the LLM.

Users report uneven response times. Analytics show that a high number of chats are abandoned after 2 seconds of waiting for the first token. The company wants a solution to ensure that p95 latency is under 800 ms for interactive requests to the chat assistant.

Which combination of solutions will meet this requirement? (Choose two.)

A. Enable model preload upon container startup. Implement dynamic batching to process multiple user requests together in a single inference pass.
B. Select a larger GPU instance type for the SageMaker AI endpoint. Set the minimum number of instances to 0. Continue to perform per-request processing. Lazily load model weights on the first request.
C. Switch to a multi-model endpoint. Use lazy loading without request batching.
D. Set the minimum number of instances to greater than 0. Enable response streaming.
E. Switch to Amazon SageMaker Asynchronous Inference for all requests. Store requests in an Amazon S3 bucket. Set the minimum number of instances to 0.

Correct Answer: *AD* 📦

Problem	How D solves it
Cold start latency	Setting minimum instances **> 0** ensures at least one warm instance is always running — eliminates cold start delays that cause multi-second waits
Time-to-first-token (TTFT)	**Response streaming** sends tokens to the user as they're generated rather than waiting for the full response — directly addresses the "abandoned after 2 seconds waiting for the first token" problem
p95 latency under 800ms	Streaming delivers the first token in milliseconds once inference begins, combined with warm instances means no startup delay

Problem	How A solves it

Model loading latency	**Model preload upon container startup** ensures weights are in GPU memory before any request arrives — no lazy loading delays
Throughput under load	**Dynamic batching** groups concurrent requests into a single inference pass, improving GPU utilization and reducing per-request queuing time during peak traffic
Uneven response times	Batching smooths out latency variance by processing requests efficiently rather than sequentially

Together: **warm instances (D)** + **preloaded models (A)** = zero cold start. **Streaming (D)** = instant first token. **Dynamic batching (A)** = consistent performance under load.

Question #103 *Topic 1*

A company runs a GenAI application that uses multiple foundation models (FMs) from Amazon Bedrock to generate text. The application serves users across three AWS Regions. The application runs on AWS Lambda functions.

The company needs to implement dynamic model selection based on response quality and latency metrics. The company must be able to switch between models without modifying or redeploying the application code. The company wants to gradually roll out new models, starting with 20% of traffic. The company needs a solution that will automatically roll back if error rates exceed 1%.

Which solution will meet these requirements?

A. Use AWS AppConfig to create feature flags for model selection criteria. Set up validation rules for error rates. Implement the AWS AppConfig Agent Lambda extension to retrieve model configurations when the functions run. Define linear deployment strategies to gradually release new models, starting with 20% of traffic.

B. Store model selection configurations in Amazon DynamoDUse global tables to enable multi-Region replication. Configure Amazon EventBridge rules to monitor Amazon CloudWatch metrics for errors and invoke AWS Lambda functions that update the DynamoDB table with model preferences. Add custom code to the Lambda functions to query DynamoDB before each model invocation.
C. Create Lambda function versions for each model configuration. Use Lambda aliases and set up routing configurations to gradually shift traffic between versions, starting with 20% of traffic. Implement AWS CodeDeploy deployments that monitor Amazon CloudWatch metrics for the Lambda function versions to automatically roll back deployments when error rales exceed thresholds.
D. Create Amazon CloudWatch dashboards to monitor response quality and latency metrics. Implement custom logic in the Lambda functions to select models based on metric thresholds. Use environment variables that are stored in AWS Systems Manager Parameter Store to define model selection parameters that can be updated across Regions.

Correct Answer: *A* 📦

Requirement	How A solves it
Dynamic model selection	Feature flags in AppConfig define model selection criteria — switch between models by updating configuration, not code
No code changes or redeployment	The **AppConfig Agent Lambda extension** retrieves configurations at runtime — model preferences are decoupled from application code entirely
Gradual 20% traffic rollout	AppConfig's **linear deployment strategy** natively supports percentage-based rollouts (e.g., start at 20%, increment over time)
Auto-rollback at >1% error rate	AppConfig **validation rules** monitor error rates during deployment and automatically roll back if the threshold is exceeded
Multi-Region	AppConfig is a Regional service that can be configured identically across all three Regions

Question #104 *Topic 1*

A financial services company is developing an AI-powered search assistant application to help investment advisors quickly retrieve investment data. The application runs as an AWS Lambda function. The company is using Amazon Bedrock to develop the application by using an Amazon Bedrock knowledge base that uses Amazon OpenSearch Serverless as its data source. The application agent must manage collections at scale by automatically assigning access permissions to collections and indexes that match a specific pattern.

The company uses Amazon Bedrock tools to test the knowledge base. The knowledge base sync process finishes successfully. However, the test reveals a 400 Bad Authorization error from the BedrockAgentRuntime API and a 403 Forbidden error when the test attempts to access OpenSearch Serverless. The company must resolve the permissions issues.

Which combination of solutions will meet this requirement? (Choose two.)

A. Update the Lambda function execution role to include the bedrock:InvokeAgent permission. Add the aoss:APIAccessAll permission to the Lambda execution role.
B. Create an OpenSearch Serverless data access policy that includes pattern-based resource rules.
C. Configure a VPC endpoint policy for OpenSearch Serverless. Add the endpoint to the Lambda function's VPC configuration.
D. Configure AWS Secrets Manager to store OpenSearch Serverless credentials. Grant the Lambda function access to retrieve the credentials.
E. Enable IAM authentication for the OpenSearch Serverless domain. Add the es:ESHttp* permission to the Lambda function execution role.

Correct Answer: *AB*

Question #105 *Topic 1*

A financial services company is developing a GenAI application that allows business analysts to query sensitive customer data by using natural language. A GenAI developer must implement a content safety framework that performs text-to-SQL transformations while preventing harmful outputs. The solution must detect and block SQL injection attempts in natural language prompts, validate generated SQL queries against predefined schemas, and create an audit trail of query transformations.

Which solution will meet these requirements?

A. Use Amazon Bedrock with an integrated content filtering system. Deploy Amazon GuardDuty with custom SQL threat detection rules. Use AWS Lambda functions to check generated queries against schema definitions in AWS Systems Manager Parameter Store. Use Amazon EventBridge to store records of all transformations in Amazon S3.
B. Implement a multi-stage validation pipeline that uses Amazon Bedrock Guardrails to filter inputs. Configure AWS Lambda functions to use with SQL parsing libraries to validate query syntax. Use the Amazon RDS Data API to verify queries against defined schemas and collect audit logs before the pipeline runs each query.
C. Configure AWS CloudTrail to log all database API calls. Implement IAM roles that have least privilege access to the application. Use AWS WAF to filter malicious inputs to the application.
D. Train a custom Amazon SageMaker AI model to securely transform text to SQL. Attach VPC endpoints to isolate the customer data database. Store query history in Amazon DynamoDB.

Correct Answer: *B*

Question #106*Topic 1*

A company needs a system to automatically generate study materials from multiple content sources. The content sources include document files (PDF files, PowerPoint presentations, and Word documents) and multimedia files (recorded videos). The system must process more than 10,000 content sources daily with peak loads of 500 concurrent uploads. The system must also extract key concepts from document files and multimedia files and create and store contextually accurate summaries. The generated study materials must support real-time collaboration with version control.

Which solution will meet these requirements?

A. Use Amazon Bedrock Data Automation (BDA) with AWS Lambda functions to orchestrate document file processing. Use Amazon Bedrock Knowledge Bases to process all multimedia. Store the content in Amazon DocumentDB with replication. Collaborate by using Amazon SNS topic subscriptions. Track changes by using Amazon Bedrock Agents.
B. Use Amazon Bedrock Data Automation (BDA) with foundation models (FMs) to process document files. Integrate BDA with Amazon Textract for PDF extraction and with Amazon Transcribe for multimedia files. Store the processed content in Amazon S3 with versioning enabled. Store the metadata in Amazon DynamoDCollaborate in real time by using AWS AppSync GraphQL subscriptions with DynamoDB.
C. Use Amazon Bedrock Data Automation (BDA) with Amazon SageMaker AI endpoints to host content extraction and summarization models. Use Amazon Bedrock Guardrails to extract content

from all file types. Store document files in Amazon Neptune for time series analysis. Collaborate by using Amazon Bedrock Chat for real-time messaging.
D. Use Amazon Bedrock Data Automation (BDA) with AWS Lambda functions to process batches of content files. Fine-tune foundation models (FMs) in Amazon Bedrock to classify documents across all content types. Store the processed data in Amazon ElastiCache (Redis OSS) by using Cluster Mode with sharding. Use Amazon Bedrock Prompt Management for version control.

Correct Answer: *B*

Question #107*Topic 1*

A healthcare company wants to develop a proof-of-concept application that uses Amazon Bedrock to automatically summarize medical documents. The company has 3 weeks to validate the application's accuracy. The application must comply with the company's data privacy policies. The application must include metrics to evaluate summarization accuracy and processing time.

Which solution will meet these requirements?

A. Create a dataset that includes 50-100 anonymized patient records. Implement Retrieval Augmented Generation (RAG) with a secure knowledge base. Use a judge model to evaluate accuracy metrics across three foundation models (FMs).
B. Fine-tune a single foundation model (FM) on patient records. Deploy the FM on Amazon Bedrock. Use Amazon Bedrock AgentCore to configure the FM as an agent. Conduct user testing on 500 company staff members.
C. Select the most powerful available AWS foundation model (FM). Create a chat interface by using Converse APIs. Test the application on 50-100 actual patient records by using only qualitative feedback from stakeholders. Use a custom web interface to gather real-world performance metrics.
D. Use the Strands SDK to deploy multiple agents that connect to multiple knowledge bases that contain specialized medical documents. Compare the responses of the agents. Evaluate the integration of the agents with the company's existing systems.

Correct Answer: *A*

Question #108*Topic 1*

A company wants to select a new FM for its AI assistant. A GenAI developer needs to generate evaluation reports to help a data scientist assess the quality and safety of various foundation models (FMs). The data scientist provides the GenAI developer with sample prompts for evaluation. The GenAI developer wants to use Amazon Bedrock to automate report generation and evaluation.

Which solution will meet this requirement?

A. Combine the sample prompts into a single JSON document. Create an Amazon Bedrock knowledge base with the document. Write a prompt that asks the FM to generate a response to each sample prompt. Use the RetrieveAndGenerate API to generate a report for each model.
B. Combine the sample prompts into a single JSONL document. Store the document in an Amazon S3 bucket. Create an Amazon Bedrock evaluation job that uses a judge model. Specify the S3 location as input and a different S3 location as output. Run an evaluation job for each FM and select the FM as the generator.
C. Combine the sample prompts into a single JSONL document. Store the document in an Amazon S3 bucket. Create an Amazon Bedrock evaluation job that uses a judge model. Specify the S3 location as input and Amazon Quick Sight as output. Run an evaluation job for each FM and select the FM as the evaluator.
D. Combine the sample prompts into a single JSON document. Create an Amazon Bedrock knowledge base from the document. Create an Amazon Bedrock evaluation job that uses the retrieval and response generation evaluation type. Specify an Amazon S3 bucket as the output. Run an evaluation job for each FM.

Correct Answer: *B*

Question #109 *Topic 1*

A financial services company is developing a natural language processing (NLP) application that uses an AWS Lambda function to process customer trade inquiries by using Amazon Bedrock foundation models (FMs). The Lambda function must generate comprehensive trade analysis responses that typically exceed 4 MB and require 15-20 seconds to complete. The application must respond to customer inquiries in less than 10 seconds and comply with financial regulatory requirements to retain records for 10 years.

Which solution will meet these requirements?

A. Implement response streaming to display partial responses as the application generates them. Set a 15-second timeout for Lambda function execution time. Enable AWS X-Ray tracing to monitor performance.
B. Use Amazon SQS queues to implement asynchronous invocations. Set a 30-second timeout for Lambda function execution time. Use Amazon CloudWatch to monitor response times.
C. Use Amazon API Gateway to poll the application. Integrate the application with Amazon ElastiCache to cache responses. Set a 30-second timeout for Lambda function execution time. Collect detailed Amazon CloudWatch logs for application performance.
D. Implement response streaming to display partial responses as the application generates them. Set a 30-second timeout for Lambda function execution time. Store complete response records in Amazon S3.

Answer: D

Question #110 *Topic 1*

A university is building an AI-powered application that includes several sub-applications. The sub-applications include AI assistants, assignment graders, and internal analytics applications. The university is defining and testing multiple prompts by using various foundation models (FMs). The university wants to compare variants of each prompt and choose the variant that yield outputs that are best-suited for specified use cases.

The university requires a version control solution for the prompts. The university must be able to test prompt variations and collect audit trails for prompt changes and usage. The solution must also maintain consistency while allowing the prompts to integrate into the main application.

Which combination of solutions will meet these requirements with the LEAST operational overhead? (Choose two.)

A. Use Amazon Bedrock Prompt Management to create versioned prompts. Include parameterized variables for each use case.
B. Store prompts in Amazon S3. Use AWS Step Functions to orchestrate the model interactions and service integrations.
C. Use Amazon Bedrock Flows to create workflows that combine FMs and AWS services.
D. Configure AWS Config to record prompt changes. Use AWS CloudTrail to track prompt usage.
E. Configure Amazon Bedrock intelligent prompt routing.

Answer: AC

Together: **Prompt Management** governs the prompt lifecycle (version, test, audit) → **Flows** operationalizes those prompts into integrated application workflows.

Requirement	How A solves it
Version control	Prompt Management provides **native prompt versioning** — create, iterate, and track prompt versions within the Bedrock console
Test prompt variations	**Parameterized variables** allow testing different prompt variants against the same FM or across different FMs for each use case (assistants, graders, analytics)
Audit trails	Prompt Management natively integrates with CloudTrail — all prompt changes and usage are automatically logged without additional configuration
Consistency	Versioned prompts with parameters ensure all sub-applications reference the same approved prompt version

Requirement	How C solves it

Integration into main application	Flows creates **managed workflows** that chain FMs, prompts, and AWS services into production-ready pipelines for each sub-application
Consistency across sub-applications	Flows connects directly with Prompt Management versioned prompts — all workflow nodes reference the same governed prompt versions
Least operational overhead	Fully managed, visual workflow builder — no custom orchestration code needed

Why C is not correct: Adds redundant operational overhead — Bedrock Prompt Management already integrates with CloudTrail for audit trails natively. AWS Config rules for prompt tracking require custom configuration and maintenance

Question #111 *Topic 1*

A company is building a multicloud generative AI (GenAI)-powered secret resolution application that uses Amazon Bedrock and Agent Squad. The application resolves secrets from multiple sources, including key stores and hardware security modules (HSMs). The application uses AWS Lambda functions to retrieve secrets from the sources. The application uses AWS AppConfig to implement dynamic feature gating. The application supports secret chaining and detects secret drift. The application handles short-lived and expiring secrets. The application also supports prompt flows for templated instructions. The application uses AWS Step Functions to orchestrate agents to resolve the secrets and to manage secret validation and drift detection.

The company finds multiple issues during application testing. The application does not refresh expired secrets in time for agents to use. The application sends alerts for secret drift, but agents still use stale data. Prompt flows within the application reuse outdated templates, which cause cascading failures. The company must resolve the performance issues.

Which solution will meet this requirement?

A. Use Step Functions Map states to run agent workflows in parallel. Pass updated secret metadata through Lambda function outputs. Use AWS AppConfig to version all prompt flows to gate and roll back faulty templates.

B. Use Amazon Bedrock AgentCore only. Configure Amazon Bedrock guardrails to restrict prompt variation. Use an inline JSON schema for a single agent's workflow definition to chain tool calls.
C. Use a centralized Amazon EventBridge pipeline to invoke each event agent. Store intermediate prompts in Amazon DynamoDB. Resolve agent ordering by using TTL-based backoff and retries.
D. Use Amazon EventBridge Pipes to invoke resolvers based on Amazon CloudWatch log patterns. Store response metadata in Amazon DynamoDB. Configure a TTL and versioned writes. Use Amazon Q Developer to dynamically generate fallback prompts that are routed through a Lambda coordinator.

Answer: A

Issue	How A solves it
Expired secrets not refreshed in time	Lambda functions retrieve and pass **updated secret metadata as outputs** at each workflow step — agents always receive fresh credentials because the data flows through the current execution, not a stale cache
Agents using stale data despite drift alerts	Step Functions Map states run agent workflows **in parallel** with freshly resolved secrets passed via Lambda outputs — the execution pipeline itself guarantees current data at each invocation rather than relying on alerts to trigger manual refresh
Outdated templates causing cascading failures	AWS AppConfig **versions all prompt flows** with feature gating — faulty templates can be instantly rolled back to a known-good version, preventing cascading propagation. AppConfig's deployment strategies (percentage-based rollout, automatic rollback on alarms) provide production-safe template management

Question #112*Topic 1*

A financial services company processes more than 10,000 customer inquiries every day through a multi-agent GenAI application that uses Amazon Bedrock AgentCore. The application agents invoke several custom tools. During peak usage periods, users report that the custom tools experience up to 40% failure rates. The tools perform inconsistently for different teams at the company.

A GenAI developer must implement an observability solution that provides end-to-end visibility into agent interactions and tool behavior. The solution must use built-in Amazon Bedrock capabilities and must not require custom instrumentation. The GenAI developer needs a solution that requires minimal performance overhead.

Which solution will meet these requirements?

A. Enable AgentCore Observability and trace collection. Use AWS X-Ray to capture distributed traces for the custom tools. Build Amazon CloudWatch dashboards to visualize metrics for errors, throttling, and latency during peak usage periods.
B. Use Amazon CloudWatch Container Insights to monitor the agents. Configure an AWS Lambda function to poll the Amazon Bedrock API for tool usage metrics. Configure the function to store results in CloudWatch to generate alerts.
C. Build a custom ETL pipeline that uses AWS Lambda functions to process Amazon CloudWatch logs from Amazon Bedrock. Store the processed data in Amazon DynamoDB. Use Amazon Quick Sight to visualize cross-team performance patterns.
D. Enable AgentCore Observability and send trace data to Amazon CloudWatch Logs. Use a custom AWS Lambda function to extract tool performance metrics from the logs. Use Amazon Managed Grafana to visualize trends.

Answer: A

Question #113*Topic 1*

An ecommerce company is using an Anthropic Claude Sonnet model in Amazon Bedrock to generate product recommendations. An AWS Lambda function retrieves customer purchase data from Amazon DynamoDB, product reviews from Amazon S3, and customer profile information from Amazon RDS. Then the function sends the data directly to the Amazon Bedrock model through API calls. Recently, customers who have extensive purchase histories have begun to receive incomplete recommendations.

Amazon CloudWatch logs for the Lambda function show execution timeouts. CloudWatch logs for

Amazon Bedrock API calls show intermittent errors. The company reviews the logs and finds that some requests are failing with context length exceeded errors. Other requests finish but appear to ignore portions of the input data.

The company wants the recommendation system to consider all customer data when the system generates recommendations. The company wants to use Amazon Bedrock Knowledge Bases to improve data organization and retrieval.

Which combination of solutions will meet these requirements? (Choose two.)

A. Implement a chunking strategy that divides the customer data into smaller segments. Configure the model to process each segment separately. Invoke the model a final time to synthesize the individual responses into comprehensive recommendations.
B. Modify the prompt structure to place the most critical information at the beginning and end of the context window. Implement token-counting logic to truncate less important data when the interaction approaches the model's maximum context length.
C. Replace Claude Sonnet with a model that has a larger context window capacity. Increase the Lambda function timeout to accommodate longer processing times for larger inputs.
D. Configure the recommendation system to use the Converse API. Modify the additionalModelRequestFields parameter to increase the maximum token limit beyond the model's default context window size.
E. Implement RAG by using a knowledge base to index the customer data with vector embeddings. Retrieve only the most semantically relevant information for each recommendation request based on the current customer context.

Answers: AE

The combined architecture:

Knowledge Base (E) → indexes all customer data with embeddings → retrieves only semantically relevant information

Chunking (A) → if retrieved data still exceeds context limits → divides into processable segments → synthesizes comprehensive recommendation

Question #114 *Topic 1*

A company developed a multimodal content analysis application by using Amazon Bedrock. The application routes different content types (text, images, and code) to specialized foundation models (FMs).

The application needs to handle multiple types of routing decisions. Simple routing based on file extension must have minimal latency. Complex routing based on content semantics requires analysis before FM selection. The application must provide detailed history and support fallback options when primary FMs fail.

Which solution will meet these requirements?

A. Configure AWS Lambda functions that call Amazon Bedrock FMs for all routing logic. Use conditional statements to determine the appropriate FM based on content type and semantics.
B. Create a hybrid solution. Handle simple routing based on file extensions in application code. Handle complex content-based routing by using an AWS Step Functions state machine with JSONata for content analysis and the InvokeModel API for specialized FMs.
C. Deploy separate AWS Step Functions workflows for each content type with routing logic in AWS Lambda functions. Use Amazon EventBridge to coordinate between workflows when fallback to alternate FMs is required.
D. Use Amazon SQS with different SQS queues for each content type. Configure AWS Lambda consumers that analyze content and invoke appropriate FMs based on message attributes by using Amazon Bedrock with an AWS SDK.

Answer: B

Question #115 *Topic 1*

A company is developing a customer support chat assistant that uses an Amazon Bedrock foundation model (FM). The company wants to update to a newer FM version but needs to implement a validation system to detect semantic drift in responses. The company wants to ensure that performance and functionality for end users remains consistent.

The company needs a solution to compare responses between current and new FM versions for 500 test cases. The solution must detect changes in response meaning, generate quantitative

similarity scores, complete validations, and log detailed results for historical comparison.

Which solution will meet these requirements with the LEAST operational complexity?

A. Configure Amazon CloudWatch Synthetics canaries that use custom JavaScript code to send identical prompts to both new and existing FM versions. Store responses in Amazon S3. Perform manual reviews to identify semantic differences between FM versions.
B. Configure an AWS Step Functions workflow that sends test prompts to both new and existing FM versions. Use Amazon Bedrock embedding models to calculate cosine similarity scores. Store the results in Amazon DynamoDB with a composite key schema.
C. Use Amazon Bedrock model evaluation jobs to compare the new FM version against the current version using the 500 test cases. Configure the evaluation to calculate semantic similarity metrics. Store results in Amazon S3 for historical comparison.
D. Build a custom solution by using Amazon SageMaker AI to train a classifier model on historical responses to detect anomalies when the solution compares responses from the new FM version to previous patterns.

Answer: C

Question #116*Topic 1*

A media company is building an AI-powered content moderation system by using Amazon Bedrock. The system first classifies text by using a small, low-latency model. Then the system escalates requests that have a confidence score below 0.65 to a larger, more expensive model.

The system must respond in near real time for high-confidence results. The system must process low-confidence requests asynchronously. The system must scale to meet sudden spikes in demand. The company wants to optimize costs for the system by invoking the larger model only when required. The company wants to use decoupled components to achieve high resiliency for the system.

Which solution will meet these requirements?

A. Use Amazon API Gateway to invoke the small model synchronously. If the small model's confidence score is below 0.65, synchronously call the larger model. Use provisioned concurrency to handle traffic spikes.

B. Use an AWS Step Functions workflow that has parallel branches to run both the small model and the large model for every request. Choose the large model result when confidence score values differ.
C. Send requests to an Amazon SQS queue. Use AWS Fargate to process messages. Invoke the small model first. If the confidence score is below 0.65, place the request in a second SQS queue to process asynchronously by using the large model.
D. Deploy both models on Amazon EC2 instances and enable auto scaling. Use a custom application heuristic to route requests to the appropriate instance based on phrase length and keyword rules.

Answer : C

Requirement	How C solves it
Near real-time for high-confidence	Fargate processes messages from the first queue, invokes the small model, and returns results immediately for scores ≥ 0.65
Async for low-confidence	Requests with confidence < 0.65 are placed on a **second SQS queue** — true asynchronous processing with the larger model
Scale for sudden spikes	SQS absorbs traffic bursts as a buffer; Fargate auto-scales task count based on queue depth
Cost optimization	The large model is invoked **only** for requests routed to the second queue — requests with high confidence never touch it
Decoupled components / high resiliency	Two SQS queues completely decouple: ingest → small model processing → large model processing. If any component fails, messages persist in the queue and are retried

Why Option A is wrong:

The larger model is called synchronously, directly violating the asynchronous processing requirement for low-confidence requests. Tightly coupled — API Gateway → small model → large

model is a synchronous chain with no decoupling. A failure in the large model blocks the entire request

Question #117 *Topic 1*

A social messaging company is building an AI chat assistant by using Amazon Bedrock. The company must ensure that every inference complies with an approved safety policy. The company wants to block harmful prompts before model invocations, filter streamed model outputs in real time, and route flagged cases for human review.

Which solution will meet these requirements?

A. Configure an AWS Step Functions workflow. Configure a step to use an AWS Lambda function to pre-check inputs by using the ApplyGuardrail API. Use an InvokeModelWithResponseStream API step that has the guardrail attached. Configure a second Lambda function step to post-check outputs by using the ApplyGuardrail API. Route flagged items to an Amazon SQS queue. Enforce guardrail use by using a bedrock:GuardrailIdentifier IAM condition.
B. Configure an AWS Step Functions workflow. Configure a step to call the ApplyGuardrail API before inference. Then call the InvokeModel API without streaming and use the guardrail. Store the results in Amazon S3. Use the client UI to hide problematic tokens.
C. Configure an AWS Step Functions workflow. Configure a step to use the InvokeModelWithResponseStream API that has the guardrail attached for in-stream filtering. Run an AWS Lambda post-check step by using the ApplyGuardrail API to check flagged cases. Do not perform pre-inference.
D. Configure an AWS Step Functions workflow that includes steps to perform pre-checks and post-checks by using the ApplyGuardrail API and the InvokeModelWithResponseStream API. Attach the guardrail to the check steps. Use process controls such as code reviews instead of IAM enforcement to ensure that guardrails are always applied.

Answer: A

Question #118 *Topic 1*

A financial services company uses Amazon Bedrock to analyze customer data that is stored in an Amazon S3 bucket. The data includes personally identifiable information (PII). The company must mask PII from foundation model (FM) responses.

Which solution will meet this requirement with the LEAST operational effort?

A. Create a guardrail in Amazon Bedrock to filter PII content. Define the PII type and set the guardrail action to MASK. Configure Amazon Bedrock to apply the filter to each FM response.
B. Use Amazon Comprehend to detect PII entities in the S3 data before invoking Amazon Bed-rock. Configure an AWS Lambda function to call the Amazon Comprehend DetectPiiEntities API to mask detected PII. Store the processed data back to the original S3 bucket.
C. Use Amazon Macie to scan the S3 bucket for PII data. Configure an AWS Lambda function to store PII in a second S3 bucket. Use an Amazon EventBridge rule to invoke the Lambda func-tion.
D. Configure an AWS Lambda function to search for PII data. Implement a step in the Lambda function code to store the PII in a second S3 bucket and non-PII data into a third S3 bucket.

Answer: A

Question #119*Topic 1*

A healthcare company is implementing a clinical knowledge base application on Amazon Bed-rock that provides medical information to doctors. During testing, a quality team discovers that the generative AI (GenAI) model occasionally fabricates medical treatment recommendations that do not originate from the approved clinical guidelines.

The quality team needs to implement a solution to detect the hallucinations before releasing the application. The solution must analyze model responses against verified medical information. The solution must identify semantic inconsistencies when users ask similar questions in different ways.

Which solution will meet these requirements?

A. Deploy a real-time monitoring system that uses Amazon CloudWatch Logs Insights to analyze response patterns and to flag factual inaccuracies based on predefined keywords. Implement AWS Lambda functions to compare responses with known answers from an Amazon Dyna-moDB table.
B. Create a reference dataset with validated question-answer pairs from clinical guidelines. Im-plement output diffing by using Amazon Bedrock Guardrails for response consistency analysis. Use Amazon Bedrock evaluation to detect factual inaccuracies.
C. Configure Amazon Bedrock Guardrails with custom rules to detect and block potentially hal-lucinated content by identifying specific patterns in the GenAI responses. Use Amazon Sage-Maker Feature Store to maintain a repository of verified clinical guidelines.

D. Create an Amazon Bedrock automatic model evaluation job with a custom prompt dataset. Set up anomaly detection alarms to identify factual inaccuracies. Implement Amazon SNS notifications when inaccuracies are detected.

Answer: B

Question #120 *Topic 1*

A company deploys a GenAI application that uses an Amazon Bedrock foundation model (FM) and custom prompts. The company must implement an automated quality assurance system to verify FM outputs against expected responses and detect regressions over time.

The solution must compare outputs against more than 500 golden examples by using semantic similarity scoring. The solution must support regularly scheduled test rules. The solution must be able to scale to more than 2,000 test cases within 6 months without the need to make architecture changes. The company wants to generate visualizations for quality metrics. The company must receive alerts if the solution detects quality degradation.

Which solution will meet these requirements?

A. Use Amazon SageMaker Model Monitor to track data drift in the FM. Configure Amazon CloudWatch alarms for drift metrics. Use SageMaker AI processing jobs to run scheduled evaluations of FM outputs.
B. Create AWS Lambda functions that invoke the FM on test cases. Compare results against expected outputs. Configure Amazon EventBridge scheduled rules to run the test executions. Publish custom metrics to Amazon CloudWatch to generate metric visualizations and alarms.
C. Use AWS Step Functions workflows to run test cases and validate responses. Store the test results in Amazon DynamoDB. Use Amazon CloudWatch dashboards to visualize the results.
D. Configure Amazon CloudWatch Synthetics canaries to invoke the FM APIs. Compare re-sponses with baseline scripts. Create CloudWatch alarms based on canary success rates.

Correct Answer is B:

Question #121 *Topic 1*

A financial services company needs to build a zero-touch promotion workflow for a multilingual AI assistant that automatically moves prompt updates from a development environment to a production environment. The workflow must promote prompt updates only if the updates meet several criteria.

The updates must improve accuracy in all five languages that the AI assistant supports. The updates must show lower variance across similar queries than older prompt versions. The updates must pass all regulatory compliance checks for content related to investment risks and fee disclosures.

The workflow must run inside the company's existing CI/CD pipeline, support prompt version comparisons, generate auditable evaluation reports, and remain cost-effective over time.

Which solution will meet these requirements?

A. Use Amazon SageMaker Canvas to visually compare prompt outputs. Create a dashboard to display metrics. Export findings to a document collaboration system to review for regulatory compliance. Schedule weekly meetings between prompt engineers and compliance officers to discuss improvements.
B. Implement prompt version control by using Amazon S3 object versions. Configure AWS Lambda functions to manually score responses. Store evaluation metrics in an Amazon Dyna-moDB table. Build a custom Amazon Quick Sight dashboard to visualize accuracy across lan-guages and product types.
C. Use Amazon Bedrock Prompt Management to implement prompt version control and to per-form A/B testing for prompts. Use Amazon Bedrock Evaluations and LLM-based judges to per-form automated evaluations for accuracy and consistency metrics. Create a flow in Amazon Bed-rock to streamline testing across languages. Integrate the flow with the existing CI/CD pipeline.
D. Build a custom evaluation framework that uses AWS Amplify as the frontend and AWS Lambda functions as the backend. Implement crowd-sourcing features in which team members vote on response quality. Use Amazon Comprehend to analyze sentiments in responses and to evaluate for consistency across languages.

The answer is C

Question #122*Topic 1*

A financial company is building an AI-powered recommendation system. The system uses real-time market data and verified internal documents. The system must include evaluation capabilities that collect explicit user ratings and detect problematic responses in real time. The system must include evaluation capabilities that track model performance over time and support A/B

testing of different prompt or model variants.

The system must orchestrate multi-step workflows. The workflows must retrieve trusted knowledge, validate inputs, and generate recommendations. The system must securely store all user feedback, flagged events, and audit logs in a compliance-ready immutable repository. The repository must be write- once, tamper-evident storage that is suitable for regulated audit retention.

Which architecture will meet these requirements?

A. Deploy a fine-tuned FM on Amazon SageMaker JumpStart. Use Amazon Bedrock AgentCore to orchestrate input validation and retrieval workflows. Store user feedback and flagged queries in Amazon DynamoDB. Run scheduled Amazon Athena queries on AWS CloudTrail logs for anomaly detection. Store generated outputs in encrypted Amazon S3 buckets. Enable S3 Object Lock.
B. Configure Amazon API Gateway to invoke an Amazon Bedrock FM that retrieves from Am-azon Kendra. Store all responses in encrypted Amazon S3 buckets. Enable S3 Object Lock. Use Amazon Macie to scan stored logs for problematic content.
C. Use Amazon Bedrock AgentCore to orchestrate retrieval from curated Amazon Kendra in-dexes. Configure an AWS Lambda function to sanitize input. Configure a function to check post-generation fact consistency. Configure a function to split traffic for A/B testing. Store user feedback in Amazon DynamoDB. Write audit logs through AWS CloudTrail. Store the logs in Amazon S3. Enable S3 Object Lock. Use Amazon SageMaker Model Monitor to track data drift and output anomalies continuously.
D. Build an Amazon Lex chat-based AI assistant with aliases that route a percentage of queries to two Amazon SageMaker AI endpoints for recommendations. Log interactions to Amazon CloudWatch Logs. Conduct quarterly manual audits for hallucination and prompt-injection detection. Store feedback in Amazon RDS with encryption.

The answer is C:

Requirement	How C solves it
Multi-step workflow orchestration	Bedrock AgentCore orchestrates: input sanitization → knowledge retrieval → fact consistency check → recommendation generation

Trusted knowledge retrieval	Curated **Amazon Kendra indexes** provide retrieval from verified internal documents and real-time market data sources
Input validation	Dedicated Lambda function sanitizes and validates inputs before FM processing
Real-time problematic response detection	Post-generation Lambda checks **fact consistency** on every response; **SageMaker Model Monitor** continuously tracks anomalies
Explicit user ratings / feedback	DynamoDB stores user feedback with low-latency writes
A/B testing	Lambda function splits traffic between prompt/model variants — a clean, controllable approach
Track performance over time	SageMaker Model Monitor provides **continuous data drift and output anomaly tracking** with historical dashboards
Immutable, compliance-ready storage	CloudTrail audit logs → S3 with **Object Lock** = write-once, tamper-evident (WORM) storage meeting regulated audit retention requirements

Question #123 *Topic 1*

A retail company has a product recommendation system that uses a foundation model (FM) in Amazon Bedrock. The system processes and searches millions of product embeddings that are updated hourly. The company regularly adds new products.

During peak shopping events, the current solution experiences high latency and occasional failures when the system performs similarity searches across the product catalog. The company must resolve the performance issues.

Which solution will meet this requirement?

A. Implement Amazon OpenSearch Service with a multi-node architecture. Configure vector-specific index settings and memory circuit breakers. Establish dedicated vector operation queues to handle surge capacity during peak usage events.
B. Deploy an Amazon DynamoDB table that uses sparse vector compression. Implement parallel query executors to handle search operations. Configure DynamoDB Streams and AWS Lambda functions to maintain the vector index in real time.
C. Set up an Amazon Bedrock knowledge base that uses custom vector partitioning. Implement hierarchical metadata structures. Integrate the knowledge base with Amazon EventBridge to provide synchronized catalog updates and to generate vectors.
D. Deploy an Amazon Neptune cluster with specialized vector endpoints. Configure custom similarity computation functions. Use the Neptune bulk loader API to manage vector updates across the product catalog.

The answer is A:

Requirement	How A solves it
Millions of embeddings	OpenSearch's **k-NN plugin** (HNSW/Faiss/Lucene engines) natively indexes and searches millions of vectors efficiently
Hourly updates + new products	Multi-node architecture supports **index refresh** operations and dynamic document ingestion without downtime
Peak event latency/failures	**Memory circuit breakers** prevent out-of-memory crashes during traffic surges. Dedicated **vector operation queues** (thread pools) isolate similarity search from other operations, preventing resource contention
Surge capacity	Multi-node horizontal scaling distributes vector search load across data nodes, and dedicated queues ensure vector operations get priority during peaks

Why C is wrong

Designed for RAG-based Q&A, not high-throughput product recommendation engines processing millions of embeddings. "Custom vector partitioning" and "hierarchical metadata structures" are not standard KB configuration options. Doesn't address the core performance/scaling problem

Question #124 *Topic 1*

A healthcare company is developing an application that processes sensitive patient data and generates treatment summaries by using a foundation model (FM). The application must maintain an audit trail of all prompts and completions. The application must securely handle protected health information (PHI) data throughout the processing lifecycle. The company must track all prompt-completion pairs with original patient data, but the company must redact any PHI from stored records. The company must enforce healthcare-specific data retention policies.

Which solution will meet these requirements?

A. Store prompt-completion pairs in Amazon S3 and enable default server-side encryption. De-ploy an AWS Lambda function that scans records by using Amazon Comprehend Medical. Con-figure the Lambda function to delete PHI data after a required retention period.
B. Use Amazon Bedrock Prompt Management and Amazon Bedrock Flows to detect PHI. Con-figure Amazon Bedrock Guardrails to use sensitive information filters to mask PHI automatical-ly. Store logs in Amazon S3 and configure appropriate retention settings.
C. Store prompt-completion pairs in Amazon S3. Configure S3 Object Lock in compliance mode. Apply tag-based S3 Lifecycle policies to PHI data. Configure an AWS Lambda function to re-dact patient information after the application processes the data.
D. Store patient data in an Amazon Bedrock knowledge base. Use Amazon Comprehend Medi-cal to identify PHI. Configure an AWS Lambda function to manage data retention according to healthcare policies.

Answer : B

Question #125 *Topic 1*

A financial services company runs an AI application that uses Amazon Bedrock foundation models (FMs) to generate personalized investment recommendations based on customer docu-ments. The AI application processes customer portfolios, financial statements, and regulatory documentation

to create personalized recommendations.

The company is experiencing inconsistent model performance because of poor data quality. Some documents contain personally identifiable information (PII) that the company must redact. Other documents contain formatting inconsistencies that confuse the FMs. The company needs a solution that ensures consistent, high-quality inputs to optimize FM. The solution must comply with financial regulations and PII handling policies.

Which solution will meet these requirements with the LEAST operational overhead?

A. Use Amazon Macie to automatically detect and redact PII. Use AWS Lambda functions and Amazon Textract to enforce standard document formatting. Create dynamic prompts that adapt to individual document schemas to improve model consistency.
B. Use Amazon Comprehend to detect PII by using custom entity recognition. Use AWS Step Functions to orchestrate document preprocessing workflows. Use prompt templates with conditional logic to handle schema variations effectively.
C. Use Amazon Bedrock Guardrails to filter out PII during inference. Use pre-processing AWS Lambda functions to standardized document formatting. Use few-shot prompting techniques and standardized examples to improve model performance across document types.
D. Use Amazon Textract to analyze document structures and extract text. Use Amazon Comprehend to detect PII and analyze sentiment. Design prompt engineering strategies that use structured metadata to improve FM reasoning and reduce hallucinations.

Answer is C

Requirement	How C solves it
PII redaction	Bedrock Guardrails **sensitive information filters** automatically detect and mask/block PII in both inputs and outputs — no separate service pipeline needed
Formatting inconsistencies	Lambda preprocessing functions standardize document formats before inference — straightforward, serverless, and scalable
Consistent FM performance	Few-shot prompting with standardized examples teaches the model how to handle different document types consistently — a technique, not additional infrastructure

Regulatory compliance	Guardrails natively log all PII handling decisions for audit trails, meeting financial regulation requirements out of the box
Least operational overhead	Guardrails are configured once in the Bedrock console — no model training, no multi-service orchestration, no custom ML pipelines

Question #126 *Topic 1*

A global company uses an organization in AWS Organizations to manage 25 AWS accounts. The company's products have 50,000 daily user interactions. The company is using Amazon Bedrock AgentCore to build an AI assistant.

The AI assistant must access customer data that is stored in multiple AWS accounts across several business units. The AI assistant needs to perform actions on behalf of specific users by querying Amazon DynamoDB tables that contain customer information in each business unit's AWS account. Each business unit maintains its own AWS account that hosts its own DynamoDB ta-bles.

The company must enforce strict data access security boundaries and must collect full audit trails that show which user's credentials are used to perform each action. The solution must allow the AgentCore application to access resources across multiple accounts while preserving user identity context. The solution must support the company's plan to expand to more than 40 AWS accounts in the next year.

Which solution will meet these requirements with the LEAST operational overhead?

A. Configure the AI assistant to use an outbound credential provider that runs AssumeRole operations to dynamically obtain temporary credentials so the AI assistant can access resources across accounts. Create an IAM role in each business unit account that has a trust relationship to the AI assistant's execution role and permissions to access the DynamoDB tables. Ensure that AssumeRole calls include session tags to identify the end user.
B. Create a separate AI assistant in each business unit account. Deploy an Amazon API Gateway REST API that runs request routing logic. Create AWS Lambda authorizers to direct user queries to the appropriate AI assistant based on the customer's business unit identifier from the request context.
C. Use AWS Resource Access Manager (AWS RAM) to share the DynamoDB tables from each business unit account to a central account where the AI assistant runs. Configure AWS RAM

permissions to allow cross-account access to the DynamoDB tables. Grant the AgentCore execution role permissions to access the shared resources directly.
D. Configure SCPs for the organization to explicitly allow the AI assistant's execution role principal to perform dynamodb:GetItem and dynamodb:Query actions across all member accounts. Configure the AI assistant to reference cross-account DynamoDB table ARNs directly in their resource definitions.

Answer is A

Question #127*Topic 1*

A hospital is building an AI application to help medical clinicians to make treatment decisions. The application uses Amazon Bedrock to analyze patient case histories and suggest diagnoses. The application must maintain sub-500 ms response times to integrate with the hospital's existing real-time clinical workflow. To comply with privacy regulations, the application must log all per-sonally identifiable information (PII) handling decisions for audits. The application must detect and remove PII from responses with at least 99% accuracy.

After initial deployment, clinicians report that diagnostic summaries from the application occasionally include patient names and medical record numbers that were not present in the original case history inputs. An investigation reveals that Amazon Comprehend Medical successfully detects and removes PII from inputs with 95% accuracy, and the application replaces all detected entities with tokens before it sends inputs to Amazon Bedrock. However, the application continues to generate patient-identifying information in approximately 3-5% of outputs.

The company needs a solution to prevent the application from displaying PII in outputs while meeting all other operational requirements.

Which solution will meet these requirements?

A. Configure Amazon Bedrock guardrails with sensitive information filters to detect and block PII in model outputs.
B. Implement a secondary PII detection layer by using regular expressions and custom entity recognition to detect identifiers that Amazon Comprehend Medical misses before sending inputs to Amazon Bedrock.
C. Remove detailed medical context from case histories during pre-processing to prevent the model from generating patient-specific information based on clinical pattern associations.

D. Enable session isolation in Amazon Bedrock API calls. Clear conversation history between requests to prevent patient information from persisting across multiple case analyses.

Answer is A

Question #128*Topic 1*

A company is designing a GenAI solution to help children between the ages of 8 and 14 understand world history. The system must support concurrent use for up to 30 children in a classroom and provide responses in less than 1 second. The company is using Amazon Bedrock and Amazon Nova Pro with a knowledge base that contains history textbooks that are written for students between the ages of 6 and 21. The company uses an Amazon Bedrock flow to ensure that the solution uses the knowledge base to generate every response. The company sets the prompt response temperature to 0.7.

Early testing of the solution results in some responses that are historically inaccurate, violent, and not appropriate for children. A GenAI developer must prevent the solution from responding to prompts inaccurately and inappropriately.

Which solution will meet this requirement?

A. Increase the inference temperature in the flow. Add a node to the flow after the initial response that calls the Amazon Nova model to remove any violence from the response. Return the result of the model call.
B. Configure Amazon Bedrock guardrails for the model responses by using content filters. Set the content filter strength high and set the guardrail action to block. Add the guardrail to the flow. Reduce the inference temperature in the flow.
C. Use Amazon Bedrock Data Automation with a custom output that removes all violent content from the knowledge base. Increase the inference temperature in the flow.
D. Configure Amazon Bedrock guardrails for the prompts by using content filters. Set the content filter strength to high and set the guardrail action to block. Add the guardrail to the flow. Reduce the inference temperature in the flow.

Answer is B

Question #129 *Topic 1*

A legal research company is developing an AI-powered chat assistant that uses Amazon Bedrock with a knowledge base that contains millions of legal documents. When attorneys search for legal case precedents, the initial retrieval returns many documents. The company wants the chat assistant to generate all responses within 3 seconds.

The foundation model (FM) that the chat assistant uses has a context window limitation that allows effective processing for only 10 documents for each query. Currently, the retrieval configuration fetches 50 documents by using semantic search. Attorneys report that highly relevant precedents frequently appear in positions 15-40 of the results.

The company needs a solution to improve the relevance of query results. The solution must meet the company's latency and context window constraints.

Which solution will meet these requirements?

A. Maintain the 50-document retrieval but increase the FM's temperature parameter. Implement prompt engineering techniques that instruct the model to identify the most relevant documents from the full result set before the model generates responses.
B. Enable reranking on the knowledge base by using a reranker model with language-specific configurations. Configure the chat assistant to initially retrieve 50 documents. Use the reranker model's cross-encoder architecture to evaluate query-document pairs. Return only the top 10 highest-scored documents.
C. Configure three separate knowledge bases. Configure one knowledge base to use 500 token chunks, another to use 1,000 token chunks, and the third to use 2,000 token chunks. Implement a pre-processing action group by using an AWS Lambda function that retrieves 50 documents from each knowledge base, merges the 150 total results, and uses term frequency - inverse document frequency (TF-IDF) scoring to select the top 10 documents.
D. Implement a post-processing action group by using an AWS Lambda function that retrieves 50 documents and calculates cosine similarity scores between the query embedding and each document embedding by using Amazon Titan Embeddings. Configure the action group to sort by similarity score and return the top 10 documents.

Answer is B:

Requirement	How B solves it
Relevant precedents at positions 15-40	Initial retrieval still fetches 50 documents (capturing those buried results), then the **cross-encoder reranker** evaluates each query-document pair holistically to re-score and surface the truly relevant ones to the top
Context window (10 docs)	After reranking, only the **top 10 highest-scored** documents are passed to the FM — perfectly fitting the constraint
3-second latency	Reranking adds minimal overhead (typically hundreds of milliseconds). Amazon Bedrock knowledge bases support reranking natively, keeping the pipeline well within the SLA
Improved relevance	Cross-encoders are fundamentally more accurate than bi-encoders (used in initial semantic search) because they process the **query and document together** rather than comparing pre-computed embeddings independently

Question #130*Topic 1*

A multinational healthcare company is building a GenAI solution to analyze patient medical records and clinical trial data to identify treatment patterns and generate research insights. Some of the countries where the company operates require the company to store patient data on premises within each country.

The solution must provide low-latency, AI-assisted diagnostics for remote clinics around the world. The solution must use cloud-based foundation models (FMs) to perform advanced, large-scale analytics. The company needs a solution that ensures data residency, delivers sub-100 ms inference at clinics, and supports consistent development workflows across hybrid environments.

Which solution will meet these requirements with the LEAST operational overhead?

A. Use AWS Global Accelerator to reduce latency for remote clinics. Replicate on-premises patient data to Amazon S3 by using AWS DataSync. Process all model requests through a centralized inference endpoint that is hosted in one AWS Region.
B. Deploy open source FMs on Amazon EC2 instances in each country to comply with data residency requirements. Store patient data locally on Amazon EBS volumes. Use AWS Systems Manager to manage model updates across countries. Manually export aggregated research insights as CSV files to Amazon S3 buckets in each AWS Region to perform cross-country analysis.
C. Deploy FMs in multiple AWS Regions on Amazon EC2 GPU instances. Replicate patient data from on-premises systems to the cloud. Centralize model training and inference.
D. Use AWS Outposts to process and store sensitive on-premises patient data in countries that have data residency requirements. Deploy AWS Wavelength at remote clinics to perform real-time inference.

Answer: D

Question #131 *Topic 1*

A global company runs an AI-powered document-processing pipeline that uses Amazon Bedrock to summarize text documents. The pipeline occasionally times out during peak traffic periods. The timeouts cause service degradation.

The company needs a solution to detect when error rates exceed 10% within any 60-second window. The solution must switch to an alternate process path when the error rate threshold is exceeded. The solution must automatically resume normal Amazon Bedrock processing when the pipeline recovers. The company must ensure that the pipeline processes all documents within 15 minutes.

Which solution will meet these requirements?

A. Configure an AWS Step Functions workflow that implements a circuit breaker pattern by tracking Amazon Bedrock invocation failures in Amazon DynamoDB. Configure the workflow to route processing jobs between normal and alternate processing paths based on the current failure state.
B. Configure an AWS Step Functions workflow that uses an AWS Lambda function to poll Amazon CloudWatch metrics for Amazon Bedrock. Store error rate metrics in Amazon DynamoDWhen the error rate exceeds 10%, instruct company staff to manually update the workflow to divert traffic to the alternate processing path.

C. Configure an AWS Step Functions state machine that includes a Task state to implement Retry/Catch logic for Amazon Bedrock calls. Configure an Amazon CloudWatch alarm that triggers an Amazon EventBridge rule to direct traffic from the state machine to the alternate processing path when the error rate exceeds 10%.
D. Deploy multiple identical AWS Step Functions state machines that use multiple Amazon Bedrock models behind an Amazon API Gateway API. Use a custom health check AWS Lambda function to route requests to whichever state machine responds fastest.

Answer A:

Requirement	How A solves it
Detect >10% errors in 60s	DynamoDB tracks invocation success/failure counts with TTL-based records, enabling precise error rate calculation within any sliding window
Switch to alternate path	Circuit breaker transitions from **Closed → Open** state when the threshold is breached, and Step Functions routes to the alternate processing path
Automatic recovery	Circuit breaker's **Half-Open** state periodically tests Bedrock availability and automatically transitions back to **Closed** (normal) when the service recovers — no manual intervention
15-minute processing SLA	Step Functions (Express Workflows) supports up to 5-minute executions, and Standard Workflows can orchestrate the full pipeline well within 15 minutes

Option C is Retry/Catch handles individual invocation failures, not aggregate error rate monitoring. EventBridge cannot dynamically re-route traffic within a running Step Functions execution. CloudWatch alarm evaluation periods introduce latency that may miss the precise 60-second window requirement

Question #132*Topic 1*

A media company needs to automatically generate searchable metadata for a video library that contains 10,000 hours of content. Each video includes visual scenes, spoken dialogue, background music, and on-screen text.

The company needs a solution to extract scene descriptions, speaker transcripts with timestamps, text overlay content, and emotional context from each video. The output must synchronize all metadata types with one another and create a searchable index. Users must be able to search based on any metadata type and jump directly to a specific moment in each video.

What solution will meet these requirements?

A. Create an Amazon SageMaker AI processing job that extracts video frames at regular intervals. Use Amazon Transcribe to process audio and perform speech-to-text conversions. Use an Anthropic Claude model in Amazon Bedrock to process frames individually to produce multimodal scene analyses. Store outputs in separate Amazon S3 buckets that are organized by content type.
B. Create an Amazon SageMaker AI processing job that extracts video frames at fixed intervals and attaches timestamps. Use Amazon Transcribe with time markers enabled to process audio. Use an Anthropic Claude model in Amazon Bedrock to analyze frame-transcript pairs. Store outputs in a unified JSON structure that preserves temporal relationships.
C. Create an Amazon SageMaker AI parallel processing job that is orchestrated by a SageMaker AI pipeline. Configure multiple steps in the pipeline to extract video segments, process audio streams by using Amazon Transcribe, analyze video frames by using Amazon Rekognition to detect text in scenes, use Amazon Comprehend to perform sentiment analysis, and store unified metadata in Amazon DynamoDB with time-based partitioning.
D. Create an Amazon SageMaker AI processing job that configures Anthropic Claude Opus in Amazon Bedrock to produce a comprehensive multimodal analysis for each video. Configure the processing job to extract metadata by using a single inference request and to index results in Amazon OpenSearch Service to provide temporal search capabilities.

Answer B

Claude's multimodal capabilities generate rich, natural-language scene descriptions from video frames — something no single-purpose service can do

Incorrect Option C Rekognition detects text/labels but cannot generate rich scene descriptions (only a multimodal LLM can). Each service (Rekognition, Comprehend, Transcribe) processes independently, making cross-metadata synchronization difficult. Comprehend only analyzes text-based sentiment, missing visual emotional cues

Question #133*Topic 1*

A financial services company is developing a GenAI application that needs to analyze data across multiple AWS accounts by using a foundation model (FM). The data and application are in Amazon S3 buckets in three separate accounts. Account A contains transaction data. Account B contains customer profiles, including personally identifiable information (PII). Account C hosts the GenAI application. The company plans to add data sources from seven more business units within the next few months.

The company must have strict control over PII data. The company must ensure that the FM accesses only necessary data fields.

The company needs a solution to establish secure cross-account access for the GenAI application.

Which solution will meet these requirements?

A. Use AWS Identity Center to create a permission set for the FM service account. Configure resource-based policies on each S3 bucket to allow specific access to Account C. Deploy a custom AWS Lambda authorizer to filter PII data before passing it to the FM.
B. Configure Amazon Bedrock cross-account IAM roles. Create role-based access policies in each source account that allow access only for specific S3 bucket paths. Apply tagbased access controls on S3 objects to classify and control access to PII data.
C. Set up AWS Lake Formation in each source account. Register the S3 buckets with Lake Formation. Create IAM roles in each source account that have permissions to access specific S3 bucket paths. Configure Lake Formation permissions to allow access only to specific data columns by using Lake Formation tag-based access controls. Use Amazon SageMaker AI to deploy the FM with an execution role in Account C that can assume the cross-account IAM roles to access the data through Lake Formation.
D. Configure resource-based policies on S3 buckets in each source account to allow access from the GenAI application in Account C. Configure identity-based policies for the FM's service role that allow access only to specific data columns by using AWS Lake Formation tag-based access controls. Set up a VPC endpoint in each account to provide secure cross-account access.

Answer: C

Question #134*Topic 1*

A logistics company is building an agentic GenAI-powered solution to automate freight optimization. The solution must retrieve data in real time from multiple internal and external systems. The solution must include a human-in-the-loop approval step before the optimization process is finished. The solution must support modular growth as the number of integrations and amount of logic increases.

Which solution will meet these requirements with the LEAST operational overhead?

A. Use an Amazon SageMaker AI endpoint that hosts a large language model (LLM) that directly calls all internal databases and external APIs. Use a custom web application that provides a UI to implement the human-in-the-loop review step.
B. Build a hierarchical system by using the Strands Agents SDK and Amazon Bedrock AgentCore. Configure a coordinating agent to delegate tasks to multiple specialized agents. Use MCP to facilitate inter-agent messaging. Use AWS Step Functions to implement a human-in-the-loop approval step.
C. Use AWS Glue to aggregate operational data into Amazon S3. Use Amazon Athena to query the data. Invoke an AWS Lambda function to generate route assignments. Use Amazon SNS to send notifications to supervisor agents.
D. Use a single Amazon Bedrock AgentCore agent with AWS Lambda based tools to integrate with all internal and external systems. Use AWS Step Functions to orchestrate the approval workflow.

Answer B:

Question #135*Topic 1*

A financial services company is building a fraud detection system by using Amazon Bedrock. The system will monitor activity in multiple stock trading applications that run in the United States and Europe. The system must process 1,000 transactions every second with sub-500 ms response times. The system must also maintain high availability during connectivity disruptions.

The company must ensure that data for European customers is processed only in AWS Regions that are based in Europe.

Which solution will meet these requirements?

A. Configure AWS Lambda functions and Amazon EKS applications to use the InvokeModel API with a global inference profile. Deploy an automated failover system that uses Amazon Route 53 health checks. Create a dedicated European inference profile and enable Geographic cross-Region inference for European applications. Use Amazon CloudWatch alarms to monitor utilization metrics.
B. Configure all applications to use the InvokeModel API with provisioned throughput for an Anthropic Claude model in each Region separately. Set up a custom Application Load Balancer to distribute traffic based on Regional capacity and response times. Implement a Regional failover mechanism that uses Amazon EventBridge rules to handle connectivity disruptions.
C. Configure all applications to use the InvokeModelWithResponseStream API with on-demand throughput. Deploy an Amazon API Gateway REST API with Regional endpoints in each location where the company operates to route requests to the closest Amazon Bedrock endpoint. Create separate IAM roles for applications that run in the United States and Europe. Grant the IAM roles Region-specific permissions.
D. Configure applications that run in the United States to use provisioned throughput with the InvokeModel API. Configure European applications to use a Europe-specific geographic inference profile to ensure data sovereignty. Configure automatic scaling for provisioned capacity based on utilization metrics. Use Amazon EventBridge and AWS Lambda functions to implement cross-Region failover mechanisms.

Answer in examtopics is A but Quick and I picked D

Question #136*Topic 1*

A company is implementing a 2-week initiative to incorporate GenAI into several hundred existing applications to provide context-driven natural language help for users. To simplify the process to choose an appropriate model for each application, a GenAI Developer is building a system to automatically compare foundation models (FMs) based on general text generation accuracy. The system runs a job that performs a series of steps.

First, the system creates a service role and attaches an IAM permissions policy that allows Amazon Bedrock to invoke the FMs and grants access to Amazon S3. Then the system requests and receives access to all the FMs that need to be tested. Then the system creates S3 buckets to store the evaluation results. Then the system creates evaluation configurations by using the TREX built-in dataset. Finally, the system specifies the Accuracy metric and Generation evaluation type in each evaluation configuration.

When the GenAI developer attempts to run the system, it fails with an AccessDenied exception. The GenAI developer must resolve the issue.

Which solution will meet this requirement?

A. Create a trust policy that defines Amazon Bedrock as the service principal. Attach the trust policy to the service role.
B. Configure cross-origin resource sharing (CORS) permissions that allow GET and DELETE actions on the S3 buckets.
C. Change the test dataset from TREX to Gigaword in the evaluation configurations.
D. Change the evaluation type to Summarization and the metric type to Correctness in the evaluation configurations.

The correct answer is **A. Create a trust policy that defines Amazon Bedrock as the service principal. Attach the trust policy to the service role.**

Here's why:

The scenario describes creating a service role and attaching an IAM **permissions policy** (what the role *can do*), but it never mentions creating a **trust policy** (who *can assume* the role). Every IAM service role requires both:

- **Permissions policy** — grants actions like invoking FMs and accessing S3 ✅ *(already done)*
- **Trust policy** — defines which AWS service (e.g., `bedrock.amazonaws.com`) is allowed to assume the role ❌ *(missing)*

Without the trust policy, Amazon Bedrock cannot assume the service role, which directly causes the `AccessDenied` exception.

Why the other options are wrong

Table

Option	Why it's incorrect
B – Configure CORS on S3	CORS governs browser-based cross-origin requests; it's irrelevant to a server-side Bedrock evaluation job and wouldn't cause an AccessDenied on role assumption
C – Change TREX to Gigaword	TREX is a valid built-in evaluation dataset for text generation; switching datasets doesn't resolve IAM permission issues

D – Change to Summarization/Correctness	Changing the evaluation type/metric is a configuration choice, not a fix for an access control error